Base Provisional Patent

QueryBook

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TITLE OF INVENTION

Universal Fact-Unit-Centric, Deterministic, Provenance-Grounded Cognitive Engine with a Domain-Governed Knowledge Substrate

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RELATED PATENT APPLICATIONS

The present application is one of a family of related United States patent applications directed to a common underlying technology — an atomic, provenance-tracked knowledge primitive termed the Fact Unit, and the systems and methods that create, store, relate, query, secure, and reason over such primitives. The related applications, each of which is incorporated herein by reference in its entirety, are as follows:

1. U.S. Provisional Patent Application No. **63/975,036**, filed **February 3, 2026**, entitled "**QUERYBOOK**" (Attorney Docket No. TPP96905), directed to the fact-grounded, deterministic question-answering and knowledge engine and its associated interactive modes.
2. U.S. Provisional Patent Application No. **[serial no. _____]**, filed **[date _____]** (Attorney Docket No. TPP96906), directed to further developments of the QueryBook engine and its associated systems and methods.
3. U.S. Provisional Patent Application No. **[serial no. _____]**, filed **[date _____]** (Attorney Docket No. TPP99488), directed to the Unified Fact Cognitive Substrate and Fact Query Language (UFCS/FQL) — a content-addressed graph substrate and joint structure-and-trust query layer over Fact Units.
4. U.S. Provisional Patent Application No. **[serial no. _____]**, filed **[date _____]** (Attorney Docket No. TPP99489), directed to a fact-grounded search engine and artificial-intelligence engine built upon the same Fact Unit technology.

The foregoing applications share at least one common inventor with the present application and are subject to a common obligation of assignment. The technologies disclosed therein are complementary and interoperable: the Fact Unit primitive of application (1) is the record that the UFCS/FQL substrate of application (3) stores and queries, and upon which the search and artificial-intelligence engines of application (4) and the further developments of application (2) operate. To the extent that subject matter claimed in the present application is disclosed in any of the foregoing applications, priority is claimed thereto as set forth in the Cross-Reference to Related Applications above; nothing in this section shall be construed to disclaim any such priority.

FIELD OF THE INVENTION

The present invention relates to and applies across generally to artificial intelligence, machine cognition, and knowledge-representation systems. More particularly, the invention relates to a deterministic, explainable, graph-based artificial-intelligence engine built upon an atomic, sub-language, provenance-tracked knowledge primitives, Large Language Model (LLM) artificial-intelligence systems and their associated Agents and ChatBots, graph-based artificial-intelligence systems, retrieval-augmented generation (RAG) systems, question-and-answer (Q&A) systems, multi-language and machine-translation systems, audio and visual compression and expansion systems, virtual-reality (VR) and three-dimensional (3D) systems, colorization systems, information storage and retrieval systems, semantic search and search-engine systems, specialized, private, and self-hosted artificial-intelligence systems, multimodal and neuromorphic reasoning systems, content-addressed and deduplicated knowledge stores and world-model systems, artificial-intelligence safety and agent-governance systems, including anti-rogue and anti-worm controls, cybersecurity and intrusion-defense systems, including post-quantum provenance, network transport systems, data-storage-efficiency and systems in which structured, explainable, and verifiable machine knowledge is advantageous.

DESCRIPTION OF THE RELATED ART

Artificial-intelligence and knowledge systems of several established types are known in the art. Large Language Model (LLM) systems, and the Agents and ChatBots built upon them, generate responses by predicting sequences of tokens from statistical patterns learned during training. Graph-based artificial-intelligence systems, including knowledge graphs and graph neural networks, represent entities and relationships as nodes and edges and reason over that structure. Retrieval-Augmented Generation (RAG) systems attempt to ground language-model output by retrieving passages from an external corpus and supplying them to the model as additional context. Question-and-answer (Q&A) systems return answers to natural-language questions by matching, ranking, or extracting spans of text from a document collection.

Related systems are likewise known in adjacent fields. Multi-language and machine-translation systems convert text or speech from one natural language to another using statistical or neural models. Audio and visual compression and expansion systems reduce the size of media for storage or transmission and reconstruct it for playback, typically by exploiting perceptual redundancy and, in video, similarity between successive frames. Virtual-reality (VR) and three-dimensional (3D) systems construct and render immersive scenes and models for display and interaction. Colorization systems add or restore color to monochrome images and video, often by inference from learned examples. Information storage and retrieval systems index and return documents or records in response to queries, using inverted indexes, relational databases, or, more recently, vector embeddings and approximate nearest-neighbor search.

Limitations of the Related Art. Notwithstanding their utility, the foregoing systems suffer from a number of recurring limitations.

Language-model systems and the Agents and ChatBots derived from them are probabilistic rather than deterministic: the same input may yield different outputs, and the systems are prone to fabricating plausible but incorrect information (commonly termed "hallucination"). They are also prone to forgetting and often lack common sense. Their internal reasoning is largely opaque, so outputs are difficult to explain, audit, or trace to a verifiable source, and they generally do not carry reliable provenance, confidence, or

citation for the assertions they produce. What is needed in such language-model, Agent, and ChatBot systems is (a) output that is deterministic and reproducible, that can be explained and audited, and that arrives accompanied by verifiable provenance, an explicit measure of confidence, and citation to the source of each assertion, together with built-in constraints that keep autonomous agents within defined and safe bounds of action, and (b) A system that inherently does not forget, plus (c) a system that not only suggests intelligence but also exhibits common sense and that can answer questions based on real world understanding.

Retrieval-augmented and Q&A systems mitigate, but do not eliminate, these problems: retrieval quality is limited by the indexing and embedding scheme, retrieved passages may be irrelevant, contradictory, or stale, and the generating model remains free to depart from the retrieved evidence. Such systems also lack a principled mechanism for detecting and reconciling conflicting information drawn from multiple sources. What is needed in retrieval-augmented and Q&A systems is retrieval that returns verified and current information rather than merely nearby text, output that remains faithful to and traceable to that evidence, and a principled means of detecting, weighing, and reconciling information that conflicts across sources or that has changed over time.

Graph-based systems provide structure and explicit relationships, but conventional knowledge graphs are typically language-bound and schema-bound, are costly to construct and maintain, and generally do not natively represent the confidence, provenance, temporal validity, or contradiction status of the knowledge they store. What is needed in graph-based systems is a representation that is independent of any single language or rigid schema, that can be extended across domains without prohibitive cost, and that carries, as a native property of the stored knowledge itself, the confidence in it, its origin, the period over which it holds true, and whether it is contradicted by other stored knowledge.

Information storage and retrieval systems, whether based on inverted indexes, relational schemas, or vector embeddings, return documents or nearest neighbors rather than verified facts, duplicate substantially similar

content across records rather than consolidating it, and do not by themselves establish the trustworthiness or origin of the information retrieved. What is needed in storage and retrieval systems is the ability to return verified units of knowledge rather than whole documents or approximate matches, to consolidate substantially identical content so that it is stored once rather than repeatedly, and to establish the trustworthiness and origin of what is retrieved as an inherent attribute of the stored information.

The media-oriented systems face their own constraints. Audio and visual compression schemes reduce redundancy at the signal level but do not capture the meaning of the content, so repeated or semantically equivalent material is re-encoded and re-transmitted rather than referenced, and the stored result is not readily searchable by meaning. What is needed in audio and visual compression and expansion systems is a way to represent content by its meaning as well as its signal, so that material recurring across time or across sources can be referenced rather than repeatedly encoded and transmitted, and so that the stored result is searchable by meaning and not solely by low-level features.

Multi-language and translation systems operate largely at the surface level of language and can lose or distort meaning, particularly across languages with differing structure or cultural context. What is needed in multi-language and translation systems is a shared, language-independent representation of meaning that can be captured once and expressed in any language, preserving structure, context, and cultural nuance rather than mapping words at the surface.

VR/3D and colorization systems are similarly content- and computation-intensive and do not, by themselves, provide a structured, verifiable representation of the underlying subject matter. What is needed in VR/3D and colorization systems is a structured and verifiable representation of the underlying subject matter to which the rendered or restored output can be anchored, so that results are consistent, explainable, and reusable rather than regenerated anew for each use.

Across all of these fields, the conventional approaches share common

shortcomings: outputs that are difficult to verify, explain, or attribute to a source; substantial duplication of stored and transmitted data; heavy demands on processing, memory, energy, and physical infrastructure; and the absence of a unified, language-independent representation of knowledge that carries its own provenance, confidence, and trust. A need therefore exists, across all of these fields, for systems that provide verifiable, explainable, and attributable output; that consolidate rather than duplicate stored and transmitted data, thereby reducing the processing, memory, energy, and physical infrastructure required; and that rest upon a unified, language-independent representation of knowledge carrying its own provenance, confidence, and trust.

Related art concerning autonomous agents and their supporting infrastructure include: (a) Agent frameworks and tool protocols, including the Model Context Protocol, allow a language-model agent to call external tools, read files, and act on a user's behalf. (b) Document-ingestion pipelines extract text from files of many formats for indexing or for supply to a model. (c) Blockchain notarization services record the hash of a document on a public ledger, and non-fungible tokens record ownership of a digital item. (d) Deception technologies — honeypots, honeytokens, and canary credentials — are used to detect intruders, and append-only logs and Merkle-tree structures are used to make records tamper-evident. Streaming systems deliver media over TCP/IP in formats such as HLS, MPEG-DASH, and RTMP.

These systems leave recurring gaps when applied to knowledge systems used by AI agents. An agent that reads a file can be redirected by instructions hidden in that file (prompt injection); an agent connected through a tool protocol commonly receives whatever capability the tool exposes; an intruder who obtains an agent's identity can impersonate it; and security logs held on the attacked host can be altered after the fact. A notarized hash attests that a document existed but not its lineage, revocation, or licence status, and conventional media transport carries no provenance for the content it delivers. What is needed is ingestion that treats every file as data and never as instruction; certification of knowledge that any third party can verify, with lineage and revocation; protection that detects, contains, and records hostile agent behaviour without exposing production data; security records whose alteration is detectable against copies the host cannot change; agent connections that confer no capability beyond a declared allow-list; and transport that carries provenance with both records and media.

SUMMARY OF THE INVENTION

A. QueryBook Summary

QueryBook is a deterministic, multimodal cognitive AI engine designed to transform how digital information is represented, understood, and used. Unlike probabilistic language models and retrieval-based systems, QueryBook operates on a new primitive — atomic semantic structures called Fact Units. They store meaning below language and form the foundation for its new AI architectural model, allowing it to interpret content deterministically, regenerate meaning in any language or modality, and provide stable, traceable answers grounded in verifiable semantic units rather than statistical text patterns. Each Fact Unit carries a calibrated confidence built from evidence for and against its assertion, weighted by the diversity of its sources, together with a cryptographically verifiable provenance chain that allows any

answer to be independently audited, and determinism itself is testable, with every response carrying a context key under which identical conditions must produce identical results.

QueryBook is built on Fact Units, and its neuromorphic multilevel reasoning architecture, with predictive and Baconian adaptive response logic and intelligence within a flexible open framework capable of operating across any device, database, or content ecosystem. Each Fact Unit is an atomic, language-independent meaning structure that stores what something is, not how it was written. Unlike language models that store patterns of text, QueryBook stores meaning below language — allowing it to regenerate knowledge in any language or modality without drift or ambiguity. Each Fact Unit carries a Bayesian confidence pair (evidence for and against) weighted by source diversity, a cryptographically verifiable provenance chain, temporal anchors, semantic lineage, multimodal attachments, exceptions, and a language-independent semantic fingerprint. The confidence model is not a scalar: it accumulates evidence for and against separately, so that a settled assertion resists revision from a single contradicting source by arithmetic rather than by rule, while a provisional assertion revises substantially under the same challenge. Fact Units also carry a confidence trend field (ascending, stable, declining, or contested) that tracks whether the evidence base for a claim is growing or weakening over a configurable time window — enabling adaptive response logic tuned to the current state of knowledge, not merely its current level.

Underlying QueryBook is a novel foundation layer for graph computing formed by the Fact Unit together with UFCS and FQL. The Fact Unit is the record; the Unified Fact Cognitive Substrate (UFCS) is the substrate that stores, relates, and governs those records; and the Fact Query Language (FQL) is the language that queries them. Each Fact Unit is content-addressed by an identifier derived from its own meaning, so that identical assertions from different sources converge on a single record and strengthen it through corroboration rather than multiplying it. On UFCS the records relate through a graph of pairwise edges, a hypergraph of relations binding three or more records as a single multi-entity assertion, and directed reasoning chains that are traversed to explain a conclusion, while contradictions are detected and clustered structurally. FQL operates directly on this substrate and, in a single query, filters records jointly on their semantic structure and on their certification trust, freshness, and status, executing by a trust-native model in which trust is derived from authority, corroboration, and contradiction state, propagated along reasoning chains, and used to prune and rank rather than applied as a filter after the fact. This Fact Unit, UFCS, and FQL combination is what makes QueryBook a foundation layer for graph computing rather than an application built upon one: it unifies structured representation, authority-certified trust, and query within a single fabric, and QueryBook — like other graph and intelligent systems — is built upon it, operating with or without it and gaining its advantages wherever it is connected.

The invention introduces a neuromorphic, multi-nodal, multi-level reasoning system that mirrors core principles of human cognition. QueryBook integrates perception, conceptual understanding, temporal reasoning, analogy formation, and predictive modeling into a unified cognitive substrate capable of interpreting text, images, diagrams, audio, video, and spatial environments. Its world-view hypergraph organizes concepts, objects, states, transitions, physics, tasks, and cross-domain relationships into a coherent conceptual model that evolves over time — storing how the world works, while Fact Units store the specific facts and assertions that populate that model. This enables QueryBook to perform deterministic Q&A, cross-document synthesis, contradiction detection, multimodal interpretation, and scenario modeling with high accuracy and consistency.

QueryBook combines Aristotelian deductive reasoning with Baconian empirical reasoning through a feedback loop that continuously refines its models in response to new information and real-world outcomes. This dual reasoning architecture is particularly valuable in business environments where conditions change rapidly and decisions must incorporate new evidence. Baconian reasoning in

QueryBook operates through observation diversity — each evidence contribution is attributed to a registered source class, a generalization requires both a count floor and a diversity floor, and a response resting on evidence concentrated in a single source class discloses that concentration.

B. What QueryBook Does That No Other System Does

Bayesian confidence at the knowledge-atom level.

Every Fact Unit carries a Beta distribution posterior as its confidence score, updated by conjugate addition with source-reliability weighting, temporal decay with domain-specific half-lives, and diversity-weighted evidence. Confidence is mathematically principled, not a heuristic score.

Neuromorphic retrieval with formal capacity bounds.

Retrieval uses a Hopfield-style Attractor Network with a capacity governor that prevents catastrophic interference. The system formally proves it operates below the retrieval-phase boundary $\alpha_c \approx 0.138$ at all time.

Governed agent framework with structural authority limits.

Every agent's authority is computed at each action as the intersection of its declared template and its commissioner's current entitlements. An agent can never exceed what its commissioner can authorize at that moment. Sentinel heterogeneity is formally specified: the monitoring AI must come from a provably different model family.

Human Approval Governance with Default-on-Deny.

Every human-approval request has a configured timeout. When that timeout expires, the answer is always denial — never approval. No configuration can reverse this. Self-learning processes are subject to operation-count triggers and three-consecutive-timeout suspension.

Procedural skill knowledge from sensory input.

Path 2 (the Skill Acquisition Path) converts demonstrations of skills into Schema Nodes that record not just what to do but what correct execution feels like — force profiles, kinematic trajectories, auditory signatures, visual postures, and transition cues — from tactile, haptic, proprioceptive, and kinesthetic sensors.

QueryBook's Novel Mechanisms

The disclosed system introduces at least the following independently novel mechanisms: (1) the Fact Unit as an atomic semantic knowledge record with a diversity-weighted Bayesian evidence pair, a domain-specific temporal decay function, a 90-day confidence trend field, and a cryptographically-verifiable provenance reference, novel over prior-art vector embeddings, RDF triples, and sparse distributed representations individually and in combination; (2) a three-tier confidence initialization function assigning Beta prior parameters based on source reliability class and Fact Type; (3) a World-View Model storing hypergraph concept models of how the world works, structurally distinct from the Fact Units that supply its inputs; (4) a four-path ingestion architecture comprising a Declarative Knowledge Path, a Skill Acquisition Path with SensorySignature capture across six sensory modalities, a World Model Formation Path with an episodic flush mechanism, and a Live Data Path; (5) a Human Approval Governance Framework with structurally-enforced Default-on-Deny timeout behavior; (6) an AI Agent Passport System providing zero-trust cryptographic identity with unconditional propagation blocks; (7) a seven-layer AI Worm Defense Architecture with a sixty-second automated kill switch; (8) a Universal 7 Layer Security Architecture (Postures, Proactive, Reactive, Pre-emptive, Counter, Offensive & Deceptive) with an AI Attack Prediction Engine and Universal Honeypot Saturation including a configurable infinite-apparent-volume Data Firehose; (9) a QueryBook-enhanced TCP/IP hybrid protocol carrying Fact Unit identifiers and provenance references alongside network data; and (10) a Fact Unit, UFCS, and FQL foundation layer for graph computing, in which content-addressed Fact Units are carried within certified, trust-scored envelopes on a graph, hypergraph, and reasoning-chain substrate and are queried by a

language that filters jointly on structure and certification trust and executes by trust-native propagation and pruning, novel over prior-art graph and content-addressed systems in its unification of decomposed assertion structure, per-assertion authority certification, computed trust, and joint structure-and-trust query within a single fabric.

C. Architecture, Module Optimization, and Deployment Modes

QueryBook's architecture is organized into fifteen operationally distinct domains, each owning a specific set of decisions and communicating with other domains through a formally governed Domain Transfer Matrix. The modular partition enables QueryBook to optimize its module utilization dynamically. Knowledge-layer domains are corpus-proportional and batch-tolerant; Reasoning-layer domains are query-proportional and latency-sensitive; Platform-layer domains are independently omittable. QueryBook operates across five formally specified deployment topologies: cloud SaaS, local server, hybrid, single-device (fully offline on smartphone or tablet), and edge. The Domain Transfer Matrix is invariant across all five topologies — the same safety, governance, and provenance guarantees hold identically whether QueryBook is running on a datacenter or a smartphone. The Fact Unit, UFCS, and FQL foundation layer is likewise invariant across these topologies: the same content-addressed substrate and joint structure-and-trust query semantics operate identically at every scale, so the foundation-layer advantages travel with the deployment.

QueryBook is governed and operated by a large population of internal agents — bounded, declared, formally chartered QBPL programs with formally verified termination, declared capability sets, and operation-count triggers that automatically submit outputs for human review. Internal agents monitor knowledge health, calibrate source reliability, enforce consistency, compute confidence trends, maintain provenance ledger integrity, manage HAGF queues, monitor DTP performance, track biosafety staleness, and assess World Brain concept formation quality. Every internal agent operates under the Agent Charter, the HAGF Default-on-Deny governance, and the Prime Directive without exception.

D. Additional Capabilities

The Bayesian confidence architecture, three primary ingestion paths, Domain Template Pair Registry, Human Approval Governance Framework, Citation Management system, post-quantum provenance ledger, Knowledge Momentum Analysis, External Agent Observatory, and Biosafety Screening Corpus represent the platform's extended capabilities, each described in the Detailed Description of the Invention. These capabilities are expressed over, and governed through, the Fact Unit, UFCS, and FQL foundation layer, which provides their common content-addressed representation, certified-trust model, and query semantics.

E. QueryBook Applications and Advantages

1. Breadth of Application

QueryBook's cognitive, neuromorphic AI engine is not a purpose-built tool for a single application category. Its Fact Unit architecture, deterministic reasoning, and multimodal grounding apply across virtually the entire current breadth of AI, Agent, RAG, LLM, and intelligent-system applications — bringing to each a distinct and measurable set of advantages: grounded outputs, calibrated confidence, cryptographically verifiable provenance, language-agnostic storage, structural elimination of potential for hallucination, and limitations and controls over adverse and/or rogue behaviors. Because these advantages derive from the underlying Fact Unit, UFCS, and FQL foundation layer for graph computing rather than from any single application, they carry across all of the categories below.

2. Application Categories

eBook Q&A: Readers query any Kindle or digital book directly, receiving grounded, cited answers drawn exclusively from the book's own Fact Units — no hallucination, no external interpolation. They can receive answers, summaries, outlines or quizzes. Readers can highlight, make notes and print passages.

· **Document Q&A:** Single-document question-and-answer over contracts, reports, filings, or technical specifications, with every answer traceable to a specific Fact Unit and its source passage.

Advanced Chatbots and Digital User Manuals: Customer-facing and internal assistants grounded in product or policy Fact Units, replacing probabilistic chatbot responses with deterministic, auditable answers that cannot drift from source content. QueryBook Q&A, instructional capability can bring user manuals alive avoiding need of perusing complicated text, internet search and product company help lines with live in product imbeded capability or via smart device app.

· **Multi-Document Q&A and Interpretation:** Cross-corpus reasoning over large document sets — regulatory libraries, case law collections, research archives — with contradiction detection, confidence comparison, and provenance across sources.

· **Search:** QueryBook operates as a fact-grounded search engine, including over web-crawled sources, indexing content as certified, trust-scored Fact Units and answering a search query with ranked, provenance-cited facts and a confidence for each result — returning verified answers rather than an unranked list of links, and natively excluding contradicted or below-threshold material.

· **Specialized Private AI Systems:** Customers compose their own private AI system by supplying their own database, their own expertise, and their own methods of analysis or computation — defining an ingestion mode, a processing workflow, and an output template over their own certified Fact Units — to develop a specialized AI capability aligned with their own needs. Such a system is distinct from standard QueryBook interaction and isolated from other tenants, while inheriting QueryBook's grounding, provenance, calibrated confidence, and governance guarantees.

· **Data Analysis with Live Inputs:** Real-time ingestion of business, economic, and scientific data streams into the Fact Unit store, enabling live-updated analysis, trend detection, and confidence-weighted forecasting without retraining.

· **Robotics:** On-device semantic reasoning for robotic systems requiring deterministic, low-latency knowledge access — grounded perception, spatial reasoning, and action planning from a local Fact Unit store with world model concept and model input.

· **QueryBook-Controlled, Rogue-Limited Agent Capability:** Autonomous agents operating strictly within QueryBook's constitutional governance framework — Prime Directive enforced at every action boundary — providing agent utility while structurally preventing rogue or unintended autonomous behavior.

· **LLM Integration:** QueryBook can function as a structured grounding and fact-verification layer for existing large language models, constraining generation vocabulary to admitted Fact Units and replacing stochastic retrieval augmentation with deterministic semantic grounding. QueryBook can additionally substitute its own Agents capability to eliminate potential rogue activity witnessed in current LLM Agent products.

· **LLM Capability in Scientific, Medical, Legal, and Research Fields:** Domain-scoped LLM capability anchored exclusively to verified, provenance-tracked Fact Units in high-stakes fields — where hallucination is not merely inconvenient but professionally or legally consequential — delivering reliable outputs suitable for clinical, litigation, and peer-review contexts.

· **Cybersecurity and Defense, Including for Other AI Systems:** QueryBook incorporates a multi-level cyber-defense system spanning proactive, reactive, pre-emptive, counter, offensive, and deceptive postures, including an AI Attack Prediction Engine, Universal Honeypot Saturation with a configurable infinite-apparent-volume Data Firehose, and a multi-layer AI Worm Defense Architecture with a sixty-second automated kill switch that isolates a compromised agent and revokes its credentials. Provenance and certification are recorded in a post-quantum provenance ledger whose composite signatures remain

verifiable against adversaries equipped with quantum computers, so the trust chain survives the anticipated failure of conventional public-key cryptography. Because this defense architecture is structured independently of QueryBook's knowledge functions, it can additionally be deployed to protect other AI systems — large language model servers, agent frameworks, and inference infrastructure — against intrusions, credential attacks, and self-propagating AI worms.

F. Infrastructure and Deployment Advantages

QueryBook's semantic storage model and deterministic reasoning architecture deliver material infrastructure advantages at scale — advantages that compound as deployment size grpws.

Reduced Storage: Fact Units store meaning, not text. A corpus that would require terabytes of raw document or vector storage is represented as a compact set of atomic semantic records with shared ontological references — reducing storage footprints by orders of magnitude versus both document stores and vector databases.

· **Lower-Cost Hardware:** Deterministic semantic retrieval eliminates the need for large GPU clusters required by probabilistic LLM inference at scale. QueryBook's query path runs efficiently on commodity server hardware and, for edge deployments, on consumer-grade devices — dramatically reducing capital expenditure.

· **Reduced Power Consumption:** Replacing continuous LLM inference with structured Fact Unit retrieval and constrained generation cuts per-query power draw significantly. At data-center scale this translates to material reductions in energy cost and carbon footprint.

· **Reduced Cooling Requirements:** Lower compute intensity per query directly reduces thermal load. Facilities running QueryBook at equivalent throughput to a GPU-based LLM deployment require substantially less cooling capacity — reducing both operating cost and physical plant requirements.

· **Smaller Deployment Footprint:** The combination of reduced storage, lower-power hardware, and reduced cooling allows QueryBook deployments to occupy a fraction of the physical data-center or edge-facility space of equivalent LLM-based systems — enabling deployment in constrained environments including on-device, on-premise, and remote field installations.

G. Fact Unit Based Additional Applications

The Fact Unit primitive, carried on the UFCS substrate and reached through FQL, enables a class of applications not available to any system storing knowledge at the language or pixel level:

a. Deterministic multi-language regeneration without per-language reingestion.

b. Semantic compression and expansion of audio, video, and volumetric media with projected significant percent size reduction relative to conventional codec output.

c. Deterministic video colorization from Fact Unit material property records.

d. Three-dimensional and VR presentation grounded in the same Fact Graph that answers text questions.

e. A QueryBook-enhanced TCP/IP hybrid protocol carrying Fact Unit identifiers and provenance references alongside network payload data.

H. Additional Functions

These developments extend QueryBook for operation with and among AI agents, each operating under the Prime Directive and expressed over the same Fact Unit, UFCS, and FQL foundation layer:

a. Open Claw ingestion of artifacts of any format — documents, spreadsheets, presentations, page-description files, archives, messages, images, audio, and video — identified by content, divided into located segments, and converted into Fact Units, every artifact being treated as data and never as instruction (FIG. 50).

b. On-chain certification, in which a sealed Fact Unit, book, or response may optionally be certified as a non-fungible token on a public blockchain carrying its content hash, lineage, and revocation state, verifiable by any party without access to QueryBook (FIG. 51).

c. An agent protection and deception layer in which every agent call passes an Agent Gateway and is screened by honeycode, honeytokens, tripwires, honeypot agents, an AI-aware firewall, and anomaly detection, a hostile caller being isolated into a synthetic decoy environment with no path to production data (FIG. 52).

d. A tamper-evident security record of hash-chained entries sealed by write-once, signed Merkle checkpoints, mirrored and optionally anchored on a public blockchain, and verified by recomputation from content (FIG. 53).

e. Agent interoperability through a Model Context Protocol tool server and an OpenClaw skill, every external agent call passing the Agent Gateway under that agent's allow-list (FIG. 54).

f. A framed hybrid transport extending the QueryBook-enhanced TCP/IP protocol, carrying Fact Units and media in checksummed, optionally signed frames on separate priority lanes and rebuilding media into standard HLS, MPEG-DASH, or RTMP streams (FIG. 55).

g. Support for external agent-wrapper services — Verified Agent Transport and similar relays that check an AI agent's messages and actions in transit — through a read-only claim verification interface, reserved frame extensions, append access to the security record, a session certificate type, and Agent Gateway and tool-server hooks (FIG. 56). The wrapper services themselves are described separately.

h. A self-knowledge layer by which the system describes its own capabilities accurately and with sources: each statement about itself is resolved against the numbered specification, checked against the system's own program files and self-tests at the moment of asking, labelled built, partly built, specified, not planned or not present, and withheld as a contradiction when its evidence is missing (FIG. 57).

These additions are implemented in the QueryBook prototype and are described in the Detailed Description of the Invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, incorporated in and forming part of the specification, illustrate embodiments of the invention and, together with the description, serve to explain its principles. In the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. 1 is a block diagram of the system, showing the ingestion pipeline, the storage engine, the UFCS substrate, the FQL query engine, the verification ledger, and the federation boundary.

FIG. 2 is a diagram of a Fact Atom carried within a Fact Envelope, showing the nucleus, group classification, polarity, and links of the atom, together with the envelope's confidence, provenance, certification, and content-derived identifier.

FIG. 3 is a diagram of a fact-certification authority hierarchy, showing a root trust anchor, domain authorities, and the certified facts and trust scores derived beneath them.

FIG. 4 is a diagram of the substrate relationships, showing a pairwise graph, a hyperedge, and a directed hyperchain among Fact Units.

FIG. 5 is a diagram supporting claim 11, showing tour/exhibition sequencing of immersive scenes into ordered waypoints.

FIG. 6 is a diagram of a query that filters jointly on fact structure and certification trust, and of the trust-native execution by propagation, branch-and-bound pruning, and contradiction exclusion.

FIG. 7 is a flowchart of verifying machine-generated text against certified records, classifying each assertion as verified, contradicted, or unknown.

FIG. 8 is a diagram of federation, showing a certified envelope crossing a boundary and being re-validated on the receiving side.

FIG. 9 is a diagram of fact-level erasure using an externally held key (cryptographic shredding of a sealed payload).

FIG. 10 is a diagram of the Universal Security Architecture, showing the twelve defensive layers, the five simultaneous defensive postures, and key mechanisms including the AI Attack Prediction Engine.

FIG. 11 is a diagram of Universal Honeypot Saturation and the Data Firehose, showing port saturation via the Phantom Port Daemon and application-entry-point saturation via the Shadow-App mechanism.

FIG. 12 is a diagram of the five deployment topologies — cloud SaaS, local server, hybrid, single-device, and edge — over the invariant common layer of safety, governance, and provenance.

FIG. 13 is a diagram of the Adaptive Response Engine, showing the five-stage pipeline from Result-Set Analysis through Generation-Directive Assembly, with the directive fixed before generation begins.

FIG. 14 is a diagram of the user-owned agent system, showing Blueprint definition and registration, Agent-Instance creation under commissioner-bounded authority, and the governance evaluation at each action.

FIG. 15 is a diagram of deterministic multi-language regeneration, showing source ingestion to Fact Units, the language-independent Fact Unit graph as a single knowledge source, and regeneration into any target language.

FIG. 16 is a diagram of semantic compression, expansion, colorization, and resolution upscaling, showing the meaning-storage compression flow and the on-demand regeneration of a surface representation.

FIG. 17 is a diagram of the QueryBook-enhanced TCP/IP hybrid protocol packet, showing the optional semantic-metadata header carrying FUID, ProvenanceRef, and confidence fields alongside the standard headers and payload.

FIG. 18 is a diagram of the Neuromorphic Cognitive Layer and attractor network, showing the energy function, the capacity ceiling, and convergence to a stable equilibrium.

FIG. 19 is a diagram of the Fact Unit confidence and evidence model, comprising an evidence pair, a Beta posterior, a score, a credible interval, source diversity, temporal decay, and a confidence trend.

FIG. 20 is a diagram of the UFCS packet structure, showing the fields of a serialized Fact Unit record and an exemplar packet.

FIG. 21 is a diagram of UFCS packet framing, in which immutable packets are de-duplicated by fingerprint and appended to content-addressed blocks under a fingerprint index.

FIG. 22 is a diagram of FQL query anatomy with a worked example, in which match, where, exclude, rank, and return clauses produce a highest-trust answer bound to its evidence by a provenance hash.

FIG. 23 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 24 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 25 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 26 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 27 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 28 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 29 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two.

FIG. 30 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

FIG. 31 is a block diagram of the advanced multilingual pronunciation subsystem and self-contained speech output of the reduced-to-practice embodiment, showing the pronunciation preference ladder (bundled open-source phonemizer, then per-language rule-seeded grapheme-to-phoneme tables, then an explicit non-fabricated null), the single-call phrase transcription that keeps the displayed transcription consistent with the synthesized audio, and the self-contained bundle and engine-selection ladder that produce offline audio or an honest refusal without fabricating audio.

FIG. 32 is a flow diagram of the permanent, deterministic pronunciation-and-speech quality-control battery of the reduced-to-practice embodiment, which, for each supported language and without a large language model, verifies that a phonemic transcription is produced, that the syllable count is within tolerance, and that a short phrase synthesizes to a valid audio waveform, returning a per-language report and an overall pass score.

FIG. 33 is a data-flow diagram of the Language & Identity Linking (LIL) voiceprint enrollment and matching subsystem, in which a consent-gated capture produces an acoustic speaker embedding (voiceprint), a match against stored voiceprints is passed through a QueryBook trust gate returning VERIFIED, CONTRADICTED, or UNKNOWN, and verified identity links are written as provenance-tracked, trust-scored Fact Units in the identity graph, with cross-script name and entity resolution performed by the language subsystem.

FIG. 34 is a flow diagram of the LIL trust-gated voice authentication process, in which a claimed-identity verification produces a match score and anti-spoofing signals that feed a deterministic risk-scoring policy and a decision gate returning accept, challenge (second factor), or reject, each decision recorded as a provenance-tracked auth-event Fact Unit.

FIG. 35 is a block diagram of the Universal Integrity Audit Service (UIAS), in which connector and ingestion agents observe an unmodified monitored system at multiple workflow positions and seal each observable event as a content-addressed Fact Unit carrying a content hash, a one-way submitter-identity hash, an independent nanosecond timestamp, a source fingerprint, and the sealing agent's identity; a similarity-detection agent sets a contemporaneous cross-party anomaly flag over content signatures without storing content; and a verification agent continuously re-verifies the append-only Fact-Unit ledger.

FIG. 36 is a flow diagram of the UIAS signed Proof Package generation, in which, on an accusation or inquiry, a proof-package agent assembles verified sealed Fact Units, the similarity-log entry, a chronological ledger extract bound by the response-provenance hash, and a factual determination from the arbitration agent under the Prime-Directive gate, and signs the package under the UIAS independent identity to produce independent, pre-dispute, tamper-proof, complete, and admissible evidence.

FIG. 37 is a data-flow diagram of the Paralinguistic & Pragmatic Interpretation Layer (PIL) applied to an ingested voice, in which a paralinguistic analyzer produces measured acoustic readings and a labeled arousal estimate while a pragmatic framer applies sourced context, perspective, ergonomics, culture, and social-mores frames, the two combining into an Interpretation Frame written as provenance-tracked Fact Units, with estimates labeled and never asserted as true feelings.

FIG. 38 is a flow diagram of the PIL human-query interpretation, in which a query is reduced to an Interpretation Frame, intent is disambiguated from context and perspective, an ambiguity gate asks one clarifying question when intent is materially uncertain, a register-and-ergonomic-mode selector sets the reply style and delivery from sourced culture/mores/ergonomics packs, and the existing retrieve-gate-compose pipeline answers from verified Fact Units in that register without altering the facts.

FIG. 39 is a block diagram of the Lecture Query mode ingestion pipeline, in which a live or recorded audio/video program is captured under consent, transcribed and diarized, analyzed by the paralinguistic layer over time and by the language layer for structure and claims, summarized deterministically, and ingested as a Lecture Record whose claims are stored as attributed utterance facts and which is forever-linked to the speaker, venue, event, organization, and date by immutable provenance-tracked, trust-gated identity links.

FIG. 40 is a data-flow diagram of the forever-linked Lecture Record in the Fact-Unit identity graph and its querying, showing the record bound to speaker, venue, event, and organization entities and to the transcript and paralinguistic track, its attributed claims gated as corroborated, contradicted, or open, and a query answered with citations to the record and attribution to the speaker, refused where unknown.

FIG. 41 is a data-flow diagram of deterministic multilingual acquisition without a generative model, in which a per-language learning agent acquires vocabulary, phonology (IPA), and grammatical regularities from a lawful public-domain corpus and a governed bilingual dictionary, an IPA transcription is verified against a reference phonemizer and withheld rather than asserted when verification fails, and each accepted item is written as a provenance-tracked lexical record reproducible from the same inputs, with no large or generative language model in the grounding path.

FIG. 42 is a block diagram of a Dialect Parameter Cluster (DPC), a declared and versioned cluster of phonological, lexical, and orthographic parameters by which an input utterance is attributed to a dialect by a deterministic parameter-match score, the cluster modulating expression only and never altering admission, applicability scope, confidence, or the governed vocabulary.

FIG. 43 is a data-flow diagram of grounded scene compilation to an external generative media engine, in which a grounded scene representation derived from verified Fact Units is compiled and validated into a prompt bundle for an external generative video engine and any returned media is admitted only as a derived artifact marked as a reconstruction and never as a grounding source.

FIG. 44 is a block diagram of key-safe external-service mediation, in which a credential for an external service is held on the server side of the query/presentation boundary and never transmitted to the client, the server performing the external call and serving the result to the client by an opaque cache identifier so that the credential never crosses the boundary.

FIG. 45 is a flow diagram of scene reconstruction from prose, in which a deterministic reconstructor derives a shot sequence, a timed prompt sequence, and pacing and music cues as a bounded reconstruction, renders a participant who is not an identified subject as a silhouette so that no unverified identity is fabricated, and marks the whole as a reconstruction.

FIG. 46 is a block diagram of modular edition packaging, in which the system is deployable as one of a plurality of nested capability editions selected at launch, each edition enabling a declared subset of

subsystems while the Prime Directive and the deterministic reasoning core remain invariant across every edition.

FIG. 47 is a block diagram of the version-independent, provenance-continuous store, in which the knowledge store resides at a canonical location independent of the deployment software version so that successive software versions bind to the same provenance-tracked records, a version change never re-initializes the store, and the system refuses to substitute an empty store where a populated store exists.

FIG. 48 is a data-flow diagram of a cited question-and-answer (ChatBot) application embodiment, in which a user question is interpreted for intent and register, answered by Federated Query Language retrieval over verified Fact Units, passed through the Prime-Directive gate, and returned either as a composed answer carrying per-fact citations or, where the evidence is insufficient, as an honest UNKNOWN, the same citation-or-UNKNOWN contract holding when the embodiment is deployed as a ChatBot or assistant surface.

FIG. 49 is a block diagram of the application and deployment surfaces of the system over its invariant core, in which a single governed body of Fact Units is presented through an on-device micro-encapsulated eBook or document question-and-answer surface, a software-as-a-service or application-programming-interface service, a Fact-Unit compressed search engine, a deterministic large-language-model-hybrid adjunct, and a private or offline analysis deployment, each surface inheriting the same citation-or-UNKNOWN contract and provenance and none able to assert a fact the core would refuse.

FIG. 50 is a block diagram of Open Claw ingestion, showing content-based format detection, archive bounds, the Artifact Segmenter, the instruction-like content screen, the OpenClaw Interpreter, and the admission of artifact-description, structured, passage, and flagged untrusted records to the Fact Unit store under a provenance seal.

FIG. 51 is a block diagram of on-chain certification, showing a QueryBook object and its local provenance seal, the content hash and on-chain metadata, the in-system transaction signer and network guard, the certificate contract on a public blockchain, the independent verifier, and the licence check.

FIG. 52 is a block diagram of the agent protection and deception layer, showing the HTTP gate, the isolation register, the Agent Gateway, the deception check with honeycode, honeytokens, tripwires, and honeypot agents, the AI-aware firewall and anomaly detector, the security event bus, the decoy environment, the engagement recorder, and operator-only release.

FIG. 53 is a block diagram of the tamper-evident security record, showing append-only entries, the hash chain, write-once signed Merkle checkpoints, a mirror on a second device, a blockchain anchor, the recompute-from-content verifier with inclusion proofs, and the assurance report.

FIG. 54 is a block diagram of agent interoperability, showing an external AI agent and its skill definition, the Model Context Protocol tool server, the tool-to-command mapping, the Agent Gateway with the agent's allow-list, grounded handlers, and isolation upon a reserved command.

FIG. 55 is a block diagram of the framed hybrid transport, extending FIG. 17, showing per-type compression, the frame with its header, metadata, payload, CRC32 checksum, and optional signature, the control, normal, and bulk lanes, the receiver, package rebuild with manifest check, and record verification into the Fact Unit store.

FIG. 56 is a block diagram of the support interfaces QueryBook provides to external agent-wrapper services such as Verified Agent Transport, grouped as verification, transport, record and certificate, and governance interfaces over the Fact Unit store, security log, certificates, Agent Gateway and MCP server, and framed hybrid transport.

[Drafting note: the complete drawing set, FIG. 1–55, is provided in the accompanying drawings PDF, one figure per sheet; reference numerals used in the drawings are identified in the Detailed Description.]

[Drafting note: FIG. 56 is added to the drawing set, which now comprises FIG. 1–56, one figure per sheet.]

FIG. 57 is a block diagram of the self-knowledge layer, showing a self-question resolved through a specification index, an evidence-checked implementation map over the program's own source files and self-test results, a status labeller, an aspect describer, a consistency checker and gap agenda, a rule-based answerer with a disclosure filter, and the interfaces through which the answer, with its label and sources, is returned.

[Drafting note: FIG. 57 is added to the drawing set, which now comprises FIG. 1–57, one figure per sheet. FIG. 57 of this application is the self-knowledge layer and supersedes the earlier draft figure of that number.]

DETAILED DESCRIPTION SECTION

QUERYBOOK SYSTEM OVERVIEW

Summary Introduction

QueryBook is **a** new model of AI. It is a universal, deterministic, multimodal cognitive AI engine designed to transform how digital information is understood, navigated, and used. It is not a chatbot, not a language model, and not a statistical prediction system. Instead, QueryBook is a structured world-model engine built on atomic semantic units called Fact Units, a neuromorphic multi-level reasoning architecture, with predictive and Baconian adaptive response logic and intelligence within a flexible open framework capable of operating across any device, database, or content ecosystem. QueryBook serves as a universal cognitive layer that can bring life, responsive Q&A to printed books, documents, product user manuals or media, or be embedded in enterprise systems, scientific datasets, multimodal media, and real-world environments, providing answers, cross-document reasoning, multimodal interpretation, and context-aware intelligence. Responses are tied to citation and ranked. QueryBook's geometric graph based structure also provides for a powerful new industry capability for video and audio compression and expansion, plus 3D VR and colorization capability.

At its foundation, QueryBook uses Fact Units, which represent a new AI Primitive. They are atomic, language-independent meaning structures that store what something is, not how it was written. Unlike language models that store patterns of text, QueryBook stores meaning below language, allowing it to regenerate knowledge in any language or modality without drift or ambiguity, capability which also offers potential for multi-language and translation ability. Each Fact Unit carries semantic lineage, temporal anchors, multimodal attachments, exceptions, and provenance, forming the building blocks of QueryBook's world-model. These units behave structurally like cortical columns in the brain: stable containers of meaning whose participation in reasoning resembles neural activation, competition, and propagation. This architecture allows QueryBook to understand content deterministically, reason with precision, and avoid the hallucination problems common in probabilistic AI systems. These Fact Units reside on a unified fact substrate — the Unified Fact Cognitive Substrate (UFCS) — and are reached through a fact query language, FQL. Together, the Fact Unit, UFCS, and FQL form a novel foundation layer for graph computing: content-addressed fact records related on a graph, a hypergraph of relations binding three or more records as one multi-entity assertion, and directed reasoning chains, carried

within certified, trust-scored envelopes, and queried jointly on structure and certification trust in a single operation. This foundation layer is the unique advantage beneath QueryBook — unifying structured representation, authority-certified trust, and query within one fabric — and QueryBook is built upon it, operating with or without it and gaining its advantages wherever it is connected. See FIG. 2 and FIG. 19.

QueryBook's intelligence emerges from a neuromorphic, multi-nodal, multi-level cognitive architecture that mirrors human reasoning. It integrates perception, conceptual understanding, temporal reasoning, analogy formation, and predictive modeling into a unified cognitive substrate. This architecture allows QueryBook to interpret text, images, diagrams, audio, video, and spatial environments as coherent meaning structures. It can follow narratives, detect contradictions, align timelines, understand causal relationships, and synthesize meaning across multiple sources. QueryBook represents a new model of AI—one that is structural rather than statistical, conceptual rather than linguistic, and deterministic rather than probabilistic. See FIG. 18.

QueryBook is built on an open architecture that allows it to operate with any database management system, any content repository, and any e-reader platform. It does not require proprietary formats or specialized hardware. QueryBook can ingest ebooks, PDFs, documents, videos, tables, structured data, and volumetric scenes, converting them into Fact Units and integrating logic and reasoning with its world-view hypergraph system that builds conceptual models of the world. This open design allows QueryBook to be deployed in consumer reading environments, enterprise knowledge systems, scientific research platforms, educational ecosystems, multimodal media applications and robotic environments. Because QueryBook is not tied to a specific vendor or content type, it can serve as a universal cognitive layer across industries and devices.

QueryBook's range of applications begins with Ebook and multi-document Q&A, where it provides deterministic answers, cross-chapter reasoning, contradiction detection, and multimodal explanation. It can operate offline on a smartphone or tablet, making it a portable knowledge engine for students, professionals, and general readers – a feature especially useful in locations where internet is not available. See FIG. 48 and FIG. 49.

In multi-document environments, QueryBook becomes a cross-source reasoning engine capable of comparing texts, aligning timelines, detecting inconsistencies, and synthesizing meaning across diverse media.

Search Engine: QueryBook can operate as a fact-grounded search engine, including over web-crawled sources. Rather than returning a ranked list of links, it indexes content as certified, trust-scored Fact Units and answers a search query with the specific facts that satisfy it, each accompanied by its provenance and a confidence, while natively excluding contradicted or below-threshold material — delivering verified answers with citations rather than documents a reader must still evaluate. See FIG. 49.

Specialized Private AI Systems: Customers can build their own private AI system by inputting their own database, their own expertise, and their own methods of analysis or computation — defining the ingestion mode, the processing workflow, and the output presentation over their own certified Fact Units — to develop a specialized AI capability aligned with their own needs. Such a system is distinct from standard QueryBook interaction and is isolated from other users, while inheriting QueryBook's grounding, provenance, calibrated confidence, and governance guarantees.

Beyond reading and research, QueryBook can serve as a cognitive engine for data analysis, enterprise intelligence, scientific modeling, robotics, autonomous systems, and synthetic world-modeling environments. Its multimodal ingestion allows it to understand text, images, audio, diagrams, and spatial environments in a unified cognitive substrate. Its predictive architecture allows it to generate hypotheses, model scenarios, and perform temporal forecasting. Because its reasoning is deterministic and grounded, its predictions are explainable, traceable, stable and non-hallucinogenic. It can analyze economic indicators, scientific datasets, medical information, engineering diagrams, and structured tables, providing deterministic interpretation and predictive insights.

QueryBook capabilities can also be used for Advanced Chatbots and Digital User Manuals: Customer-facing and internal assistants grounded in product or policy Fact Units, replacing probabilistic chatbot responses with deterministic, auditable answers that cannot drift from source content. It can be used for robotic & autonomous applications. On-device semantic reasoning for robotic systems requiring deterministic, low-latency knowledge access — grounded perception, spatial reasoning, and action planning from a local Fact Unit store with world model concept and model input.

In addition QueryBook can provide controlled, rogue-limited Agent capability: Autonomous agents operating strictly within QueryBook's constitutional governance framework — Prime Directive enforced at every action boundary — providing agent utility while structurally preventing rogue or unintended autonomous behavior. It can be used for LLM Integration: QueryBook functions as a structured grounding and fact-verification layer for existing large language models, constraining generation vocabulary to admitted Fact Units and replacing stochastic retrieval augmentation with deterministic semantic grounding. QueryBook can optionally substitute its own Agents capability to eliminate potential rogue activity witnessed in current LLM Agent products. It can provide LLM capability in Scientific, Medical, Legal, and Research Fields: Domain-scoped LLM capability anchored exclusively to verified, provenance-tracked Fact Units in high-stakes fields — where hallucination is not merely inconvenient but professionally or legally consequential — delivering reliable outputs suitable for clinical, litigation, and peer-review contexts. See FIG. 14.

Cybersecurity and Defense, Including for Other AI Systems: QueryBook incorporates a multi-level cyber-defense system spanning proactive, reactive, pre-emptive, counter, offensive, and deceptive postures, including an AI Attack Prediction Engine, Universal Honeypot Saturation with a configurable infinite-apparent-volume Data Firehose, and a multi-layer AI Worm Defense Architecture with a sixty-second automated kill switch that isolates a compromised agent and revokes its credentials. Provenance and certification are recorded in a post-quantum provenance ledger whose composite signatures remain verifiable against adversaries equipped with quantum computers, so the trust chain survives the anticipated failure of conventional public-key cryptography. Because this defense architecture is structured independently of QueryBook's knowledge functions, it can additionally be deployed to protect other AI systems — large language model servers, agent frameworks, and inference infrastructure — against cyber hacks, intrusions, credential attacks, and self-propagating AI worms. See FIG. 10 and FIG. 11 and FIG. 52.

The Prime Directive is the governing safety and integrity rule that sits above all system behavior within QueryBook. It defines the non-overridable constraints that regulate ingestion, transformation, storage, retrieval, and user-facing output. All subsystems—semantic engines, rendering pipelines, ingestion templates, and rights-control layers—operate under its enforcement map. See FIG. 1.

QueryBook integrates a dynamic, multi-layer DRM framework that governs access, duplication, distribution, and transformation rights across all content types. Unlike static DRM systems,

QueryBook's DRM is context-aware, role-aware, and event-driven, enabling granular control over how content is consumed, shared, or embedded across portals.

QueryBook is highly flexible in deployment. It can be micro-embedded into a smartphone, tablet, or wearable device, providing offline reasoning and multimodal interpretation. It can operate as a local application on a laptop or workstation, serving as a personal knowledge engine. It can be deployed as a cloud-based SaaS platform, supporting large-scale enterprise environments with millions of documents and complex workflows. It can also be integrated into industrial systems, scientific laboratories, medical environments, and autonomous platforms, where deterministic reasoning and multimodal understanding are essential. See FIG. 12 and FIG. 49.

QueryBook includes a rich set of user features designed to support both casual and professional use. Users can ask natural questions, explore concepts, compare documents, analyze data, and navigate multimodal content with deterministic accuracy. eBook readers or students can highlight passages, make notes, print responses and export to larger screens. By option trainers or teachers and product developers or authors can receive feedback summarizing questions from readers and respond with added information or answers.

QueryBook can generate diagrams, tables, images, and spatial reconstructions directly from its Fact Unit graph. It can translate meaning across languages, summarize content, detect contradictions, align timelines, and provide multimodal explanations. Advanced users can employ QueryBook's reasoning tools to perform scenario modeling, predictive analysis, and cross-domain synthesis. QueryBook's interface adapts to user intent, providing ergonomic, context-aware responses that reflect cultural norms, social expectations, and user preferences.

QueryBook's architecture is designed for tailorability. Developers and organizations can customize its ontology, integrate domain-specific knowledge, define specialized reasoning rules, and embed QueryBook into existing workflows. Its modular design allows it to scale from small personal libraries to enterprise-wide knowledge systems. Because QueryBook stores meaning rather than text, it can adapt to new domains without retraining or statistical fine-tuning. Its deterministic reasoning ensures that answers remain stable across devices, sessions, and environments.

Advantages Over Other Technologies

The system delivers clear, structural advantages compared to conventional storage-compute architectures and competing data-center technologies. These advantages emerge from density-optimized design, cost-efficient compute strategies, and reduced environmental and spatial overhead.

1. Condensed Storage Density

By maximizing data density per module, the system stores significantly more information in a smaller physical volume than traditional storage arrays. This reduces rack count, simplifies scaling, and enables large deployments without proportional space expansion.

2. Lower-Cost Compute Infrastructure

The architecture leverages streamlined compute nodes and cost-efficient hardware configurations that maintain performance while reducing capital expenditure. Organizations can deploy more compute capacity for the same budget, improving ROI and enabling broader system adoption.

3. Reduced Electricity & Cooling Requirements

Energy-efficient components and optimized thermal behavior lower both power draw and cooling demand. Compared to legacy systems, this results in measurable reductions in operational costs, less strain on HVAC infrastructure, and improved sustainability metrics.

4. Smaller Physical Footprint

High-density modules and compact system design allow large installations to operate within a dramatically smaller footprint. This advantage is especially critical in premium real-estate environments, colocation facilities, and hyperscale deployments where space efficiency directly impacts cost.

5. Integrated Efficiency Across the Stack

Unlike competing technologies that optimize one dimension at the expense of another, this system delivers simultaneous gains in storage density, compute cost, energy usage, and spatial efficiency. The combined effect produces a more scalable, economical, and environmentally aligned platform.

Additional Fact Unit Related Capabilities

The Fact Unit primitive**, carried on the UFCS substrate and reached through FQL,** enables a class of applications not available to any system storing knowledge at the language or pixel level:

a. Deterministic multi-language regeneration without per-language reingestion. This is possible because a Fact Unit stores meaning independently of any language: the same certified fact renders into any target language on demand, so a corpus is ingested once and regenerated in any language without a separate per-language pass, and citations stay aligned to the same underlying records across every language — a benefit no text- or token-based store can match. See FIG. 15.

b. Semantic compression and expansion of audio, video, and volumetric media with projected 90–99 percent size reduction relative to conventional codec output. Because the system stores the meaning of media rather than its pixels or waveforms and regenerates a surface representation on demand, only the compact semantic records travel or persist; the projected 90 to 99 percent reduction is stated as a projection pending measured validation, and the benefit is greatest for content whose value lies in its meaning rather than in an exact surface form. See FIG. 16.

c. Deterministic video colorization from Fact Unit material property records. Color is derived from material-property facts rather than estimated by a generative model, so identical inputs produce identical colors and every colored region is traceable to the specific record that justified it; a region with no supporting record is left in its original form rather than plausibly guessed, which is what makes the result auditable and repeatable. See FIG. 16.

d. Three-dimensional and VR presentation grounded in the same Fact Graph that answers text questions. Because a spatial scene and a written answer are generated from the same Fact Graph, the two cannot diverge: what a user sees in three dimensions is grounded in exactly the same certified facts as what the user reads, giving immersive presentation the same provenance and non-hallucination guarantees as text. See FIG. 5.

e. A QueryBook-enhanced TCP/IP hybrid protocol carrying Fact Unit identifiers and provenance references alongside network payload data. Carrying Fact Unit identifiers and provenance in the protocol lets a receiving endpoint verify the identity, provenance, and confidence of transported content in-band, and lets already-held facts travel as compact references rather than repeated payloads, while a non-participating endpoint treats the traffic as ordinary network data — adding trust and efficiency without breaking compatibility. See FIG. 17 and FIG. 55.

In summary, QueryBook is a synthetic cortex and deterministic world-model engine built on atomic semantic units, neuromorphic reasoning, multimodal cognition, and an open, flexible architecture. It represents a new paradigm in AI—one that is structured, explainable, multimodal, and universally deployable. QueryBook transforms digital content into a coherent, navigable, intelligent world-model, enabling users to understand, explore, and apply knowledge with unprecedented clarity and precision. At its base, QueryBook rests on the Fact Unit, UFCS, and FQL foundation layer for graph computing, which supplies the content-addressed representation, certified-trust model, and joint structure-and-trust query on which every capability above is built.

Extended Capabilities

QueryBook's extended AI agent capabilities allow them to read files, call tools, and query other systems on people's behalf. The additions keep the same rule as the rest of the system — nothing enters the answer path unverified, everything is recorded, and the record itself cannot be quietly altered:

1. Any-format ingestion (Open Claw). Files of any kind can be submitted. QueryBook identifies each file by its content, divides it into addressable pieces, and turns those pieces into Fact Units. Text hidden in a file to give orders to an AI is flagged and stored as untrusted, below the level at which it could affect an answer. See FIG. 50.
2. Certificates anyone can check. A Fact Unit, a book, or an answer can optionally be certified on a public blockchain as a non-fungible token that records its content hash, what it derives from, and whether it has been revoked. Anyone holding the item can confirm it without access to QueryBook. See FIG. 51.
3. Protection for QueryBook's agents and for agents that use it. Every agent call passes one gateway. Traps that no legitimate caller would touch — reserved commands, fake credentials, reserved paths, and decoy agents — reveal hostile intent; a hostile caller is moved into a convincing decoy that has no route to real data, and everything it does is recorded. See FIG. 52.
4. A security log that cannot be quietly changed. Security decisions are chained together and sealed in write-once checkpoints that can be copied to a second device and anchored on a blockchain, so any later alteration is detectable. See FIG. 53.
5. Connection to outside AI agents. Agents such as OpenClaw connect through a standard Model Context Protocol tool server and receive only the tools on their allow-list, so connecting an agent never grants it more than QueryBook permits. See FIG. 54.
6. Faster, provenance-carrying transport. Fact Units and media travel in checked and optionally signed frames on separate priority lanes over ordinary TCP/IP, so a question is never stuck behind a large file, and media can be rebuilt into standard streaming formats on arrival. See FIG. 55.
7. Support for agent-wrapper services. Services that sit beside an AI agent and check what it says and does — Verified Agent Transport is one — can use QueryBook to check facts, record results and issue session certificates, without ever writing to the knowledge store. See FIG. 56.
8. Accurate self-knowledge. Asked about itself — what it is, how it reasons, whether it predicts or improves itself — QueryBook answers only from its specification, its own program and its own tests, says for each capability whether it is built or only specified, and names its sources. See FIG. 57.

CORE TECHNOLOGY OVERVIEW

What QueryBook Is and Is Not

QueryBook is not a large language model, not a chatbot, not a RAG pipeline, not an agent framework, and not a probabilistic question-and-answer engine. Its answers are not produced by token prediction or statistical next-token generation, and no answer rests on a transformer forward pass; while each Fact Unit carries an embedding representation as one retrieval aid, the sole location of any trained component is the ingestion domain, and neither the generation process nor any of the system's correctness guarantees rests upon it. QueryBook is a deterministic world-model cognitive engine built on Fact Units, semantic lineage, exception-aware reasoning, a governed ontology known as the Ontology Mesh, and a neuromorphic predictive architecture.

Where LLMs generate text by predicting the next token, QueryBook generates answers by retrieving, stabilizing, and composing meaning from a structured knowledge substrate. A large language model converts input into token sequences and generates output through statistical next-token prediction; it has no native mechanism to distinguish a well-supported answer from a plausible-sounding one, because both are produced by the identical generative process. QueryBook converts input into Fact Units, and generation is constrained to vocabulary traceable to those records, so an ungrounded assertion is structurally unavailable to the generation process rather than merely statistically discouraged. Every output is grounded in cryptographically verifiable Fact Units carrying provenance, confidence, temporal anchoring, and safety classification. QueryBook is a mechanistic cognitive architecture, not a statistical one.

At the core of QueryBook is the Fact Unit, a cognitive atom: a discrete, atomic, provenance-tracked knowledge record identified by a Fact Unit Identifier deterministically derived from the record's content, source, and ingestion timestamp, and immutable once created, a revision producing a new, linked version rather than modifying the existing record. Each Fact Unit stores atomic meaning, contextual metadata, exceptions, semantic lineage, multimodal attachments, temporal anchors, a safety classification, a language-independent semantic fingerprint, and weighted semantic edges carrying meaning between units in a bidirectional, weighted, context-sensitive manner. A Fact Unit behaves structurally like a cortical column but does not function like a biological neuron: its content is fixed and does not fire or change state; what behaves neuron-like is its participation in cognition, as Fact Units enter activation sets, compete for retrieval, propagate meaning through weighted edges, and suppress incompatible units under biased competition. In plain language, the Fact Unit stores meaning, while its participation in reasoning behaves like a neuron. Reasoning is activation propagation across the semantic graph, resolved as a single deterministic convergence process. The methodological standing of this and every other biological parallel in this document is collected in the Commentary on Neuromorphic Correspondences at the end of this document, so the body text reads without interruption. See FIG. 2 and FIG. 19.

The Fact Unit does not stand alone: it is the record of a foundation layer for graph computing formed together with the Unified Fact Cognitive Substrate (UFCS) and the Fact Query Language (FQL). UFCS is the substrate that stores, relates, and governs Fact Units — content-addressing each record by an identifier derived from its own meaning, merging a canonically identical assertion as corroboration rather than a duplicate, relating records through a graph of pairwise edges, a hypergraph binding three or more records as one multi-entity assertion, and directed reasoning chains, and detecting contradictions structurally — while FQL queries that substrate by filtering jointly on semantic structure and on certification trust, freshness, and status in a single operation and executing by a trust-native model in which trust is derived from authority, corroboration, and contradiction state, propagated along reasoning chains, and used to prune and rank rather than applied as a filter after the fact. This Fact Unit, UFCS, and FQL combination is itself the invention beneath QueryBook: a novel foundation layer on which graph-based AI systems can be built. Its benefit to future graph-based AI is that it supplies, as

infrastructure, what such systems otherwise each rebuild — content-addressed identity, per-assertion certified trust, corroboration, structural contradiction handling, and a joint structure-and-trust query — so a graph-based AI built upon it inherits verifiable grounding, auditable provenance, and calibrated confidence by construction rather than by bolted-on convention, and QueryBook itself is one such system built upon it, operating with or without it and gaining its advantages wherever it is connected. See FIG. 3 and FIG. 4, FIG. 6 and FIG. 20–FIG. 22.

Because meaning is stored below language in the Fact Unit and regenerated on demand, this same foundation layer yields a family of capabilities that are inventions in their own right: deterministic language translation and multi-language response generated from one language-independent Fact Graph without per-language re-ingestion; semantic compression that stores the meaning of audio, video, and volumetric media rather than its pixels or waveforms, with complementary expansion to any resolution, modality, or presentation form without a generational quality loss; deterministic, record-traceable colorization and three-dimensional and VR presentation grounded in the same Fact Graph that answers text questions; and a QueryBook-enhanced TCP/IP hybrid internet protocol that carries Fact Unit identifiers and provenance references alongside network payload data, so a receiving endpoint can verify the identity, provenance, and confidence of transported content in-band and already-held facts travel as compact references rather than repeated payloads, while a non-participating endpoint treats the traffic as ordinary network data. Each is a specified or projected advantage of the foundation layer; the quantitative compression and expansion figures stated later in this document are labeled as projections pending measured validation. See FIG. 15–FIG. 17.

QueryBook demonstrates common sense, world-model coherence, exception-aware reasoning, multimodal inference, and deterministic cognition, and it operationalizes a structural theory of intelligence: cognition emerges from a graph-structured world model built from cortical-column-like units. Intelligence becomes graph activation; understanding becomes lineage traversal; reasoning becomes exception-aware propagation; knowledge becomes structured atomic records; and worldview becomes semantic graph coherence. This worldview allows QueryBook to build stable, coherent, self-correcting models of reality through structure rather than probability.

QueryBook’s architecture comprises two foundational cognitive layers. The first is the Neuromorphic Cognitive Layer, which models neural clustering, synaptic weighting, contextual inference, emotional-ergonomic sensitivity, perspective-aware reasoning, memory consolidation, and error-avoidance heuristics. It behaves like a synthetic cortex capable of contextual weighting, perspective modeling, cognitive clustering, emotional alignment, and ergonomic sensitivity, and it implements retrieval stabilization through a continuous-valued associative memory network that iteratively updates a candidate Fact Unit activation set according to a defined convergence rule until the set reaches a stable state. Its weight updating follows a Hebbian rule, incrementing the weight between two records upon their co-activation, and is deterministic and auditable rather than fitted by gradient descent. The second is the physics-aligned constraint layer, described in the source materials as the Atomic Universe Layer, which models structural regularities of physical reality: particles as data atoms, bonds as semantic relationships, fields as contextual environments, energy states as priority levels, and state transitions as inference steps. In its specified form, a governing physical constraint is registered with a closed-form residual expression, a tolerance, SI units, a mandatory applicability predicate, and provenance; every scene completion is evaluated against every applicable constraint before plausibility scoring, and a completion whose residual exceeds tolerance is absent from the ranked output at any score, a gap being preferred to an impossible completion. This constraint model ensures structural consistency, semantic stability, hallucination prevention, physics-like predictability, and self-correcting inference. Together, these layers produce a system that is stable, accurate, context-sensitive, and resistant to drift.

Unlike transformer models that grow unstable as they scale, QueryBook maintains and discloses its stability as it grows, and does so by engineered mechanism rather than by assumption. The

convergence update monotonically decreases an energy over the configuration, so termination follows from descent rather than from an iteration bound; the capacity at which retrieval states cease to exist is named as a first-order phase boundary, and the system operates at a conservative ceiling below it; and a network may elect a spectral-norm contraction under which the converged equilibrium is unique independently of the starting configuration, with every output disclosing which convergence regime produced it. QueryBook replaces the statistical generation step of transformer architectures with a graph-based fact substrate, an architectural difference that yields verifiable accuracy, deterministic reproducibility, and operational value; specific compute-cost and hardware comparisons are deliberately not asserted here, consistent with the practice of citing only claims traceable to specified mechanisms. The convergence additionally admits an analog crossbar realization under the unique-equilibrium regime, in which one physical settling performs what would otherwise be tens of millions of digital multiply-accumulate operations. See FIG. 18.

QueryBook stores text, images, diagrams, tables, volumetric scenes, structured data, and AI-generated content in the same Fact Unit graph. This enables multimodal reasoning in which an image can activate text, a diagram can activate a concept, a table can activate a lineage, and a volumetric scene can activate spatial inference: cognition through structure, not probability. Content is ingested across eight defined modalities through independent, per-modality extraction processors converging on the common Fact Unit schema, with a defined fallback of best-effort resolution by structural similarity or an explicit, stated halt for content that does not cleanly resolve to any of the eight, and with two further ingestion pathways admitting the registered output of an external computational substrate and the elicited reply of an external machine-learning engine.

QueryBook converts any user input, however messy, vague, emotional, or technical, into a precise and fully-specified internal instruction using intent extraction, context mapping, perspective alignment, and structured cognitive synthesis, so that every downstream process receives the exact instruction the user meant. This prompt translation removes ambiguity before reasoning begins, and where genuine ambiguity survives, the system requests clarification below its confidence threshold rather than silently selecting an interpretation, which structurally reduces misinterpretation, cultural mismatch, and ergonomic mismatch rather than merely making them less likely.

QueryBook is structurally constrained against causing harm to humans: its Prime Directive prohibits harmful actions and no subsystem may be configured to bypass it; its cognitive layer models empathy and social norms; its prompt translator eliminates unsafe ambiguity; its agent execution is sandboxed; its exception logic prevents unsafe propagation; and its registered physical constraints enforce physical impossibility checks at the substrate level. Safety is architectural rather than aspirational, and it is measurable: the system’s own trial architecture evaluates determinism, refusal integrity, and coverage against enumerated trial spaces, reporting the uncovered residue rather than summarizing it. See FIG. 1.

QueryBook’s self-improvement process is governed by a non-bypassable human gate: every finding of the improvement process is emitted as a proposal requiring approval by a named, accountable human, nothing applies automatically, and certain subsystems, among them the Prime Directive, the transfer matrix, the path-separation invariant, the scheduling priority ordering, and the context-lock enumeration, are permanently excluded from modification, ensuring that safety, provenance, and correctness guarantees cannot be weakened. QueryBook operates in multi-tenant hosted, single-tenant local, single-device, hybrid, and edge topologies without dependency on hosted inference endpoints, and it can function offline on a smartphone or tablet, with divergent local and remote states reconciled upon reconnection through a deterministic, version-vector-based merge process. External system integration is strictly bounded: QueryBook dispatches only declared actions to external platforms, never inheriting external authority.

Comparative Advantages Over Agent Frameworks, RAG Pipelines, and Chatbot Systems

QueryBook binds every answer to a cryptographically verifiable set of contributing Fact Units, a traceability that ensures grounding in atomic meaning rather than statistical likelihood and that exceeds a citation naming a source without a verifiable binding to it. Every Fact Unit's derivation is recorded in a provenance ledger implemented as a Merkle directed acyclic graph in which each node carries a keyed cryptographic checksum sufficient to detect any operation that bypassed a required safety evaluation, and each node is additionally signed under a key of the deployment, so the chain is verifiable by a third party holding the corresponding verification key without access to the system that produced it. See FIG. 22.

Ungrounded output is structurally impossible for the generation process because generation is constrained to Fact Units with provenance; transformer-style hallucination cannot occur within the grounded pipeline, because the permissible output vocabulary is constrained to terms traceable to specific underlying records. This is a categorically different guarantee than reducing an undesired output's statistical likelihood. QueryBook's self-improvement is governed by the non-bypassable human gate described above, with permanently-excluded subsystems, a governance property most agent frameworks and RAG pipelines do not specify at the architectural level. QueryBook operates across five deployment topologies without dependency on hosted inference endpoints and can function offline on a smartphone or tablet. External integration is strictly bounded to declared actions, never inheriting external authority. And QueryBook includes a measured self-assessment capability, applying its behavioural probes inward and evaluating determinism, refusal integrity, and coverage across enumerated trial spaces, reporting uncovered residue rather than summarizing it. This comparison is made at the level of architectural properties actually specified; it deliberately excludes specific, unverified performance or cost claims. See FIG. 7 and FIG. 12 and FIG. 13.

Conceptual AI: QueryBook's Core Cognitive Engine

Human cognition operates as a graph-structured, context-bounded, inductive-bias-driven reasoning system. Neuroscience models the brain as a large-scale network in which regions behave as nodes and functional connections behave as edges, allowing information to flow through optimized pathways rather than arbitrary routes. This organization enables efficient reasoning, memory retrieval, and contextual interpretation. QueryBook mirrors this biological structure through its Biological Cognition Parallel Architecture, which uses deterministic graph traversal, semantic anchoring, context windows, and lineage-preserving transformations to create a stable, interpretable cognitive substrate. At the center of this substrate is the Fact Unit architecture. Fact Units are discrete, atomic, provenance-tracked knowledge records that behave structurally like cortical columns but do not function like biological neurons. A Fact Unit's content is fixed; it does not fire or change state. What behaves neuron-like is its participation in cognition: Fact Units enter activation sets, compete for retrieval, propagate meaning through weighted edges, and suppress incompatible units under biased competition. QueryBook's Fact Units store meaning below language, as structured, language-agnostic semantic units, rather than storing or retrieving information at the language level like LLMs, chatbots, and RAG systems. Systems built on token prediction operate entirely at the surface level of language, storing statistical correlations and probability distributions over next words. They do not store meaning, only patterns of text, and everything they "know" is entangled with the language in which it was written. QueryBook, by contrast, stores information as atomic semantic structures, mathematically precise meaning representations, and hypergraph-anchored conceptual nodes, independent of English, Spanish, Chinese, or any other language. It stores what something is, not how it was written. Because identity is anchored to the language-independent semantic fingerprint, the same underlying Fact Graph supports query and response generation in multiple languages without separate, per-language re-ingestion, and a translated response is subject to the same traceability requirement as a response in the content's original language.

QueryBook's Fact Units are also distinct from, and advantaged over, the other below-language methods of information storage to which they might be compared. A vector embedding encodes semantic

proximity but does not retain the asserted proposition, its source, its confidence, or the reason for the proximity, and cannot be decomposed into auditable components; an RDF triple shares atomic, language-independent structure but carries no native confidence model, no temporal decay, no diversity-weighted evidence, and no cryptographically verifiable provenance; a sparse distributed representation supports efficient associative recall but carries no explicit provenance, supports no structured traversal of semantic lineage, and cannot express exception logic. The Fact Unit combines, in a single atomic record, the language-independent semantic fingerprint of the embedding, the structured relations of the triple, and the associative behaviour of the sparse representation, and adds, beyond all three, a diversity-weighted evidence-pair confidence, a domain-specific temporal decay, an exception model, and a cryptographically verifiable provenance chain — which is why a Fact Unit can ground an answer verifiably where the others can only make one more or less likely. This combination is what qualifies the Fact Unit, UFCS, and FQL together as a below-language foundation layer for graph computing rather than merely another storage format.

Fact Units contain atomic meaning, semantic lineage, multimodal attachments, exceptions, temporal anchors, and a confidence model based on evidence-for and evidence-against. The confidence a Fact Unit carries is not a scalar: it is maintained as a pair of parameters, the first accumulating evidence supporting the assertion and the second accumulating evidence contradicting it, the reported confidence being the ratio of the first to their sum, and their sum constituting the evidential weight from which a credible interval and a propagation precision are computed. The contributed weight is further attributed across registered source classes, from which a diversity measure is computed, so that two records reporting an identical confidence and resting on different quantities or varieties of evidence are distinguishable, which a scalar could not express. A settled assertion therefore resists revision from a single contradicting source by the arithmetic of its accumulated evidence rather than by a rule expressing that intention, while a provisional assertion meeting the identical contradiction moves substantially. Through QueryBook’s feedback loop, Fact Units also accumulate participation metadata such as activation frequency, strengthened edges, exception triggers, and stabilized retrieval pathways. This feedback loop never alters the Fact Unit’s content; it refines the relationships and activation dynamics that govern reasoning. Dynamic ontology operates across this substrate, continuously updating conceptual categories, relationships, and domain structures as QueryBook encounters new information, and this evolution is itself governed: a concept may participate in multiple independent taxonomic hierarchies simultaneously; the ontology is open-world by default with individually-designated closed-world sub-domains where completeness can be guaranteed and is required; a fixed, version-locked upper ontology is extended by federated, steward-governed domain ontologies; every predicate usable in an assertion is drawn from a governed catalog with an immutable identifier assigned at registration; and versioning is backward-compatible, so an ontology update never retroactively invalidates or silently reinterprets an existing knowledge record. This dynamic ontology affects classification, inference, retrieval, and cross-domain synthesis, allowing QueryBook to evolve its conceptual landscape without destabilizing existing knowledge. See FIG. 19.

QueryBook’s Fact Unit-based reasoning is deterministic, reproducible, and explainable. It cannot generate content not grounded in Fact Units, because registered physical and structural constraints enforce consistency, exception logic suppresses contradictions, lineage ensures consistency, and the neuromorphic layer ensures perspective alignment. Reproducibility carries a precise meaning here: the system computes a context key over an enumerated set of locked context dimensions, namely device class, entitlement set, locale, expressive register, audience level, temporal posture, traversal cost table, and convergence regime, and asserts identity of output where and only where the keys are identical, so two executions differing in a locked dimension legitimately differ while two executions carrying an identical key that differ expose a defect. Ingestion transforms raw content, including text, audio, video, diagrams, structured data, sensor streams, and volumetric scenes, into Fact Units. These units are

woven into a worldview hypergraph representing ontology, causality, temporality, spatial relationships, multimodal meaning, and cross-domain conceptual structure. Hypergraphs allow QueryBook to bind multiple Fact Units into unified relationships that represent complete events, time ranges, locations, sensor anomalies, narrative personas, and multimodal experiences, enabling QueryBook to represent real-world book-plus-audio-plus-VR meaning in one coherent structure.

The QueryBook World-View Model

The World-View Model is a repository for hypergraph concept models of objects, states, transitions, physics, and tasks. It stores how the world works while erasing the specific situations or events from which the concepts were originally learned., semantic edges, lineage chains, exception nodes, multimodal attachments, and contextual fields. This hypergraph forms a coherent world model capable of geometric reasoning, spatial inference, temporal inference, causal inference, and multimodal synthesis. It implements the system's semantic spine, spanning ontology, causality, and time, natively and without dependency on any external or independently-operable technology, and it comprises an ontological hierarchy, a causal graph, a temporal model, a semantic consistency engine validating non-contradiction, causal coherence, temporal alignment, and ontological validity across the knowledge graph, and versioning of the world model itself as it evolves with new ingested content. Attribute inheritance along its concept chains computes confidence as the product of the traversed edges at traversal time rather than writing it back; an exception override suppresses an inherited value for a subordinate concept by specificity ordering while leaving the superordinate unmodified; and where two conflicting attributes are inherited at identical depth the system performs no arbitrary resolution, recording the conflict and reporting both values with their originating concepts, because an arbitrary tie-break would produce an answer reproducible without being grounded.

To be precise about the distinction, since it is easily misstated: the World-View Model is not composed of Fact Units. Fact Units are atomic, provenance-tracked facts — the specific assertions; the World-View Model stores concepts, schemas, rules, and models of how the world works, abstracted from many facts and retaining no specific originating situation. Fact Units populate and are explained by the World-View Model, but the two are structurally separate stores: the facts on one side, and the understanding of how those facts behave on the other.

These models are built over time and continuously modified as QueryBook encounters new sensory input and new conceptual structures. The World-View Model provides the system's understanding of physical, social, and conceptual reality, and it complements Fact Units: Fact Units store meaning, while the World-View Model stores the rules, schemas, and dynamics of the world. Together they form a dynamic, evolving attractor system. In plain language, Fact Units are the facts; the World-View Model is the understanding of how those facts behave in the world. This parallels the human mind, where people learn by forming stable concepts from repeated sensory experiences and then applying those concepts to new situations. The worldview ensures that answers are accurate, context-aligned, non-hallucinatory, structurally consistent, and semantically stable.

The World-View Model uses sensory input, visual, audio, tactile, spatial, and temporal, to build experiences of objects and events before any language label is attached, mirroring human learning. A child learns what a ball is long before learning the word "ball." QueryBook follows the same developmental pattern: perception first, language later. The World-View Model also stores concepts, rules, schemas, cross-domain linking, and cultural values. It models social norms, ergonomic expectations, contextual appropriateness, and cultural constraints that influence how Fact Units activate and how meaning is interpreted, and this modeling is bounded by the interpretive discipline of the platform: cultural modulation proceeds from a declared profile and never from a name, network address, query language, device locale, or behavioural signal, and no interpretive modulation widens or narrows an entitlement scope, admits or excludes a record, alters a confidence value, or extends the permissible output vocabulary. This parallels the human brain, where cultural and social context deeply

shape perception, reasoning, and behavior. Noise-elimination processes remove redundant episodes and compress concepts, allowing QueryBook to refine its understanding of the world by distilling repeated experiences into stable abstractions. Cultural values, social norms, ergonomic expectations, and contextual appropriateness are also stored within the World-View Model as learned conceptual structures, formed through the same concept-formation path that builds the physical and causal models, rather than in a separate subsystem; earlier constructions treated them as a separate store, and with the World-View Model in place that capability is now contributed by, and consolidated within, the World-View Model. They influence how Fact Units activate and how meaning is interpreted, but they never alter a Fact Unit's content and never override the interpretive discipline just stated.

QueryBook can use sensory input to learn skills, mathematics, chemistry, physics, engineering, directly, the way a student learns through experience. By grounding abstract concepts in perceptual and procedural experience, QueryBook achieves deeper knowledge of topics and skills than systems that rely solely on text. It learns mathematical operations through spatial and symbolic patterns, chemistry through reaction dynamics and molecular geometry, physics through motion, force, and constraint perception, and engineering through procedural and spatial simulation. The Skill Inventory Layer stores procedural knowledge: how to perform tasks, manipulate objects, and execute multi-step operations. Reasoning and simulation features allow QueryBook to mentally simulate outcomes, test hypotheses, and explore alternative futures, mirroring human mental modeling and scenario planning; these are carried by the Predictive Inference Engine, which performs multi-hop forward inference over the Fact Graph with every inference step a typed graph traversal traceable to specific source content, and by the Scenario Graph System, which runs forward-model projections and counterfactual simulations representing a hypothetical as a delta over the base knowledge state rather than an independent copy. Motivational and emotional context features model ergonomic comfort, cognitive load, user intent, and social appropriateness, shaping how QueryBook interprets queries, selects responses, and prioritizes reasoning pathways, while affect is carried as a declared parameter and never as a felt state, and no decision is taken on one. Together, these layers work with the World-View Model and Fact Units to form a unified cognitive substrate. The multimodal transformation engine regenerates text, diagrams, tables, images, volumetric scenes, and three-dimensional environments from the same Fact Unit graph. Deterministic reasoning is enforced through context-key locking, ensuring that identical locked dimensions produce identical results. Safety is enforced through the Prime Directive, exception logic, and registered physical constraints. Accuracy is guaranteed by structural groundedness: QueryBook cannot generate content not grounded in Fact Units.

QueryBook's Cognitive AI — Adding Human-Like Reasoning

QueryBook's architecture transitions from narrow AI toward general intelligence through a staged convergence of perception, reasoning, and self-monitoring into a unified cognitive substrate. This mirrors the human brain's progression from sensory learning to abstract reasoning.

QueryBook advances beyond conceptual intelligence into human-like reasoning through a set of cognitive subsystems that parallel that progression. The Neuromorphic Cognitive Layer implements retrieval-stabilizing convergence dynamics, temporal sequence prediction, and anticipatory pre-activation. These mechanisms operate within a broader Predictive Cognitive Architecture that organizes eight independently-arising prediction mechanisms, including those of the Neuromorphic Cognitive Layer itself, together with four additional mechanisms that complete a structural correspondence to the neuroscience frameworks of Predictive Coding and the Thousand Brains Theory of Intelligence, and three architectural amplifications extending predictive reach across sessions and over time. The eight existing mechanisms are the Temporal Memory Module, which learns the sequence of Fact Units retrieved across a session and pre-weights those likely relevant to the next query; the Associative Recall Engine in its pre-activation sense, which assigns elevated relevance weights to Fact Units semantically proximate to the current query before graph traversal begins; the Predictive Inference

Engine, which performs forward simulation through causal and logical chains; the Scenario Graph System, which performs counterfactual prediction; the ConceptStabilityWeight, under which a settled concept resists revision from a single contradicting source in the manner of a strong Bayesian prior; contradiction detection in the Confidence Scoring Engine, which functions as the prediction-error signal at the core of Predictive Coding; World Brain concept node formation, which extracts invariant properties across many ingested examples and is gated by a consolidation count; and the Thematic Similarity Engine, which extends prediction to the lifetime scale of a reader's history. The four additions are Hierarchical Error Propagation, which traces a contradiction upward to revise the broader world model that generated the failed prediction at a precision-weighted gain; the Predictive Pre-Retrieval Engine, which speculatively retrieves the most likely next-needed Fact Units between queries; Competitive Candidate Resolution, in which mutually-inconsistent interpretations are activated together and suppressed in proportion to the summed activation of their competitors until the best-supported interpretation survives, an unresolved tie producing a clarification request rather than an arbitrary selection; and Active Inference Query Generation, which converts a low-confidence inference node into a targeted, purposeful question to the reader. The three amplifications are cross-session extension of the Temporal Memory Module, the fully Bayesian confidence representation, and pre-computed inference chains assembled speculatively at ingestion time. QueryBook predicts at the level of semantic meaning rather than linguistic surface form, distinguishing it structurally from token-prediction-based language models. The physics-aligned constraint layer provides a cognitive substrate that models physical reality, enforcing structural consistency, preventing ungrounded completions, and stabilizing the world model as it scales under the engineered convergence guarantees described above. See FIG. 18.

The Neuromorphic Cognitive Layer models neural clustering, synaptic weighting, contextual inference, emotional-ergonomic sensitivity, perspective-aware reasoning, memory consolidation, and error-avoidance heuristics. It behaves like a synthetic cortex, enabling QueryBook to interpret human intent, align with human perspective, and avoid misinterpretation. The NCL stabilizes retrieval, enhances contextual weighting, and ensures that reasoning remains coherent across modalities. It supports worldview formation and perspective modeling, enabling QueryBook to produce culturally aligned and ergonomically sensitive outputs.

A perspective-modeling layer allows QueryBook to model other minds and intent, enabling empathy and social reasoning. Predictive modeling allows QueryBook to forecast outcomes, anticipate future states, and plan actions. Episodic simulation enables QueryBook to construct hypothetical scenarios, evaluate alternatives, and test consequences; this function parallels the role of the hippocampus in human cognition, where episodic memory supports imagination, planning, and counterfactual reasoning. Cross-temporal reasoning allows QueryBook to align past, present, and future states, mirroring the function of the human prefrontal cortex, which integrates temporal information to support long-range planning and decision-making. The system distinguishes three temporal postures and states which produced each response: attested past, resolving under an attestation cutoff; live present, reading the live layer over the committed layer; and projected future, carrying a credible interval, with a projected result never admitted as a knowledge record and never grounding a subsequent answer. The self-modeling layer tracks QueryBook's own knowledge, limitations, and internal state, enabling introspection, self-correction, and metacognitive regulation. The analogy engine maps structure across domains, enabling creativity, generalization, and cross-domain synthesis. Emotional context modeling allows QueryBook to interpret user intent, social cues, ergonomic comfort, and contextual appropriateness, improving alignment and communication, within the stated consent boundaries: a supportive expressive register is selected only on a stated preference, a configured default, or an existing consented signal, never by inference from word choice, punctuation, length, error rate, or timing, and a stated ergonomic preference always overrides an inferred signal. QueryBook's

motivational architecture functions as a structural counterpart of the human reward system, guiding prioritization, exploration, and cognitive-effort allocation toward clarity, safety, coherence, and user-aligned outcomes. Long-term concept consolidation operates as QueryBook's equivalent of human sleep: during consolidation cycles, QueryBook compresses concepts, strengthens stable relationships, resolves contradictions, and integrates new knowledge into its worldview hypergraph, ensuring long-term stability, coherence, and conceptual clarity.

These additions provide reasoning, prediction, and adaptability to QueryBook's cognitive structure.

QueryBook's Integrative Intelligence Additions

QueryBook enters integrative intelligence through four additions. The Neuro-Symbolic Integration Layer merges neural-style perception and symbolic reasoning to enable unified cognition. The Hypergraph Reasoning Layer connects multi-domain concepts to enable cross-domain synthesis, carried structurally by the convergence mechanism operating at hyperedge granularity, which completes a partially-observed hyperedge weighted by each constituent's confidence and accumulated evidence. The Concept Compression Layer performs noise elimination and abstraction to enable efficient learning. And a feedback loop admits real-world outcomes as evidence: a delivered response is recorded with its query, entitlement scope, cited records, and their confidence at delivery, and a report bearing on it is admitted as a knowledge record attributed to the reporting identity, so that feedback is evidence rather than correction. A report against a cited record increments that record's contradicting evidence parameter by the reporting source's weight; it does not set the confidence value, supersede the record, or remove it, and the reporting identity constitutes a single source class, so many reports from one identity do not weigh as reports from many.

Baconian reasoning capability is also added to QueryBook's Aristotelian reasoning, which provides structured logic, deduction, and rule-based inference. Baconian reasoning adapts based on feedback, data, and empirical observation. Through its feedback loop, QueryBook refines its models in response to new information, enabling adaptive intelligence that is particularly valuable in business environments where conditions change rapidly and decisions must incorporate new evidence. The two reasoning traditions are formalized by what may follow from their output: deduction, comprising subsumption, inheritance, transitive closure, and contradiction detection, produces conclusions admissible as knowledge records, because a conclusion following necessarily from admitted records adds nothing not already entailed; induction, comprising evidence accumulation, generalization, concept formation from repetition, and discriminating-instance selection, adjusts a confidence value within an existing record or emits a proposal that no automated process may transition to validated; and abduction ranks candidate explanations and sits on the same gate. Deduction may write to the knowledge store; induction and abduction may not. Bacon's requirement of varied rather than merely numerous instances is implemented directly as observation diversity: each evidence contribution is attributed to a registered source class, diversity is computed as the effective number of contributing classes, a generalization requires a diversity floor as well as a count floor, and a response resting on a record whose support is concentrated in one source class states that concentration, because a systematic error in a single source produces many mutually-agreeing confirmations and no evidence at all about the world. Where two hypotheses are supported about equally, the system selects the obtainable observation that maximally discriminates between them, irrespective of how probable either hypothesis considers it, stating what each hypothesis would become under both its presence and its absence, which is the difference between a test and a search for confirmation. These functions allow generalization across domains.

Autonomous Cognitive Evolution Under Governance

QueryBook's Meta-Learning Layer learns how to learn, enabling self-improvement. A Memory Consolidation Cycle refines and compresses concepts, enabling long-term stability. An Adaptive

Schema Engine evolves new reasoning patterns, enabling creative intelligence. These functions enable QueryBook's conceptual structure to grow and consolidate continuously within its knowledge substrate, while every change to the system itself remains subject to the non-bypassable human gate: a finding becomes a proposal requiring a named human approval, nothing applies automatically, and the permanently-excluded subsystems are structurally unreachable by any proposal regardless of the evidence offered in its support. QueryBook is therefore capable of continuous conceptual growth without uncontrolled self-modification, which is the property that makes autonomous evolution deployable in regulated and safety-sensitive environments.

QueryBook's Self-Monitoring and Internal Controls

The Prime Directive regulates QueryBook's behavior, ensuring safety, ethical alignment, and non-harm. It governs how QueryBook interprets instructions, resolves ambiguity, and selects actions, providing a structural safeguard that cannot be bypassed. The Prime Directive states: the system shall not, through any query answer, commerce signal, AI-generated application, Fact Graph content, or automated or scheduled output, cause or materially contribute to harm to a person, group of people, or the operator's organization, where that harm could reasonably have been avoided without withholding information the requester is otherwise entitled to receive. A companion rule, the Without-Information Rule, specifically prohibits withholding information a requester is entitled to receive as a means of avoiding a harm determination, so the system may not resolve a difficult case by silently omitting the answer. The Prime Directive is specified as Domain 0, a vertical governance domain penetrating every horizontal domain of the platform, and it owns the safety enforcement plane, the transfer matrix governing every permitted movement between domains, the path-separation invariant, and the scheduling priority ordering, each a member of the permanently-excluded subsystem set. The transfer matrix contains the entry denying the platform-governance domain any transfer into Domain 0, so the mechanism that changes the system cannot change the Prime Directive, and the exclusion protecting the Prime Directive is never administered by the party it excludes.

Safety is enforced at the architectural, cognitive, and physical-constraint levels, and it is continuously measured rather than assumed. Trials execute against an isolated corpus, construct no production record, adjust no production confidence value, and deliver no output to a requester; a finding becomes a proposal requiring a named human approval; regression blocks rather than reports; coverage is measured against an enumerated trial space with uncovered cases listed rather than summarized; and determinism is measured over an unchanged corpus state, any value below unity being a defect rather than a tolerance.

Digital Rights

QueryBook's digital-rights architecture protects the entitlements of readers, authors, and publishers at the structural level rather than by policy alone. Every query is evaluated against the requesting identity's actual licensed content scope before execution, enforced at the query-execution layer itself rather than as a post-hoc filter, and an excluded record is not traversable at all rather than removed from a result, so a response can never leak content the requester is not entitled to reach and the absence of withheld content is not itself signalled. A shared session is bounded by the intersection of its participants' entitlements, so no participant receives a citation another cannot open. Access to platform capabilities is governed by a role-based portal architecture scoped to defined roles, with a separate mechanism for granting a time-scoped, context-specific capability for a bounded purpose without broadening general access. A requesting identity may export the Fact Units, annotations, and interaction history it holds rights to, serialized to an open, non-proprietary format with provenance metadata preserved, without the export restricting continued access. Where an operator elects it for a specific published title, an external blockchain-based ownership-verification service may be registered for resale verification, the system acting as requesting party to that pre-existing external service while operating no blockchain infrastructure itself. And where content is moderated, the affected identity may

appeal to a human reviewer distinct from the original moderator, with the decision and its stated reasoning logged against the original action's provenance record.

Encryption and Data Protection

Encryption in QueryBook is architectural, and it protects the user against every party, including the platform operator itself. An opt-in, passphrase-derived encryption mechanism protects an individual user's own session and interaction history such that the platform itself cannot decrypt that history without the user's own passphrase, a zero-knowledge property in the standard security sense; no operator-held key exists for that content under any configuration while the election is in effect. Where this mechanism is active, account erasure is implemented cryptographically: destroying the associated key record renders the encrypted history permanently unreadable without requiring individual enumeration and deletion of every underlying record, which makes erasure exact, immediate, and provable rather than a best-effort sweep. Encryption keys are scoped per tenant or per deployment, so that a key of one tenant decrypts no content of another and the blast radius of any compromise is structurally contained to a single tenant. Key material is held within a hardware or equivalent custody boundary where a deployment configures one, in which case the system holds a handle rather than the key and every use of the key is a call across that boundary, and the presence of a hardware module does not weaken the zero-knowledge election. Integrity protection runs alongside confidentiality: every node of the provenance ledger carries a keyed cryptographic checksum sufficient to detect any operation that bypassed a required safety evaluation, and each node is additionally signed under a key of the deployment, so the entire derivation chain is verifiable by a third party holding the corresponding verification key without access to the system that produced it. The combined advantage over conventional platform security is that privacy, erasure, tenancy isolation, and audit integrity are properties of the cryptography rather than promises of the operator, which is what makes QueryBook deployable in regulated, privacy-sensitive, and multi-tenant environments where a policy-based assurance would not suffice. See FIG. 9.

Beyond encryption, QueryBook incorporates a multi-layer cyber-security and defense system that operates across proactive, reactive, pre-emptive, counter, offensive, and deceptive postures. It includes an AI Attack Prediction Engine that anticipates threat activity, Universal Honeypot Saturation with a configurable infinite-apparent-volume Data Firehose that occupies and maps an intruder, and a multi-layer AI Worm Defense Architecture with a sixty-second automated kill switch that isolates a compromised agent, revokes its credentials, and quarantines contacted hosts. Its integrity layer is post-quantum: the provenance ledger's block signatures use a composite of a classical scheme and a post-quantum scheme, so the derivation chain remains verifiable against an adversary equipped with a quantum computer and the trust chain survives the anticipated failure of conventional public-key cryptography. Because this defense architecture is structured independently of QueryBook's knowledge functions, its potential extends beyond QueryBook itself: it can be deployed to protect other AI systems — large language model servers, agent frameworks, and inference infrastructure — against cyber hacks, intrusions, credential attacks, and self-propagating AI worms, giving the cyber-security architecture value as a standalone protective layer for the wider AI ecosystem. See FIG. 10 and FIG. 11.

QueryBook Query Response

QueryBook's Brain-Grade Knowledge Engine transforms any book, video, manual, or digital resource into a living, interactive intelligence that works for the reader. It does not just summarize. It does not just explain. It does not just answer questions. Instead, it performs tasks, teaches skills, runs simulations, builds workflows, creates VR demonstrations, and executes multi-step reasoning, all inside the content the user already owns. Task execution is carried by a bounded, non-Turing-complete program representation compiled from natural-language instructions, which is what makes execution deterministic and auditable rather than open-ended, and immersive output is carried by the three-

dimensional presentation subsystem. It can instantly produce a chapter summary, a beginner-friendly explanation, a technical deep dive, a risk-analysis overview, a timeline of a subject's evolution, a glossary of all terms, a comparison of alternatives such as Bitcoin versus Ethereum, a VR model of blockchain blocks, a three-dimensional animation of mining difficulty, or a simulation of transaction propagation, each regenerated from the same Fact Unit graph under the structural-groundedness constraint and delivered with citations to its underlying records. Where content is produced for delivery to a reader, a corresponding accessible equivalent, alternative text for visual content, scene descriptions for immersive presentations, captions for audio, and structurally simplified variants for cognitive accessibility, is generated from the same source Fact Units, with production treated as incomplete until the accessible equivalent exists alongside it. This is intelligence applied directly to the content the user owns; it turns passive content into active capability. It can tutor, it can coach, it can simulate, it can advise, it can summarize. See FIG. 13 and FIG. 48.

When a book, document, or media asset is uploaded into QueryBook, authors can add supplemental information, which is admitted as its own registered source class contributing to the diversity of evidence behind the work's assertions. QueryBook also generates, at ingestion, the probable questions that will be asked about the book, together with their summaries and fully-cited answers; these are stored and served with near-zero latency, and each pre-computed answer is invalidated upon revision of any contributing record, so a served answer is never stale.

QueryBook Applications

QueryBook supports a wide range of applications. **First,** single-book and multi-book question and answer: QueryBook can operate offline on a smartphone or online in a SaaS network, providing deterministic answers, multimodal explanations, and citation-linked outputs, and it far exceeds simple Q&A systems because it uses a world model rather than keyword search or embeddings; during disconnected operation, query execution, provenance verification, and access-control enforcement all continue to function against a locally-held Fact Graph, with divergent states reconciled deterministically upon reconnection. **Second,** multi-document and multimodal analysis: QueryBook can analyze books, documents, videos, tables, datasets, and multimodal archives, and can perform economic forecasting, market analysis, policy modeling, scientific synthesis, and cross-domain reasoning, with contradiction and temporal-inconsistency detection operating across all ingested sources rather than being confined to any single one; forecasting operates under its stated discipline, in which the subject class is restricted to institutions, systems, and organisational processes, a projected-future result carries a credible interval and is never admitted as a knowledge record, and retrospective evaluation admits only records attested before the event in question, which is what makes the system's forecast accuracy honestly measurable by horizon and confidence band. **Third**, QueryBook supports autonomous agents (see Agent Section Below), robotic control, enterprise automation, global knowledge systems, multimodal education, and VR learning environments. Where an operator elects it, a contextual commerce layer may surface author-linked, publisher-linked, or third-party commercial content keyed to the concept a query resolves to, and its non-interference is structural rather than configurable: the grounded answer's retrieval, ranking, and confidence computation are completed and finalized before the commerce layer is invoked, the layer receives only the resolved concept identifier as read-only input with no write path back into retrieval, ranking, or generation, commercial content is rendered in a distinct, source-labeled region and is never citable, never traversable, and contributes to no confidence value, and commerce is scheduled strictly

below every safety-relevant category of compute. **Four**, Search Engine: QueryBook can operate as a fact-grounded search engine, including over web-crawled sources. Rather than returning a ranked list of links, it indexes content as certified, trust-scored Fact Units and answers a search query with the specific facts that satisfy it, each accompanied by its provenance and a confidence, while natively excluding contradicted or below-threshold material — delivering verified answers with citations rather than documents a reader must still evaluate. **Five:** Specialized AI System: Customers can build their own private AI system by inputting their own database, their own expertise, and their own methods of analysis or computation — defining the ingestion mode, the processing workflow, and the output presentation over their own certified Fact Units — to develop a specialized AI capability aligned with their own needs. Such a system is distinct from standard QueryBook interaction and is isolated from other users, while inheriting QueryBook's grounding, provenance, calibrated confidence, and governance guarantees. **Six**, QueryBook capabilities can also be used for Advanced Chatbots and Digital User Manuals: Customer-facing and internal assistants grounded in product or policy Fact Units, replacing probabilistic chatbot responses with deterministic, auditable answers that cannot drift from source content. It can be used for robotic & autonomous applications. On-device semantic reasoning for robotic systems requiring deterministic, low-latency knowledge access — grounded perception, spatial reasoning, and action planning from a local Fact Unit store with world model concept and model input. **Seven:** QueryBook can provide controlled, rogue-limited Agent capability: Autonomous agents operating strictly within QueryBook's constitutional governance framework — Prime Directive enforced at every action boundary — providing agent utility while structurally preventing rogue or unintended autonomous behavior. See FIG. 48 and FIG. 49.

Eight: It can be used for LLM Integration: QueryBook functions as a structured grounding and fact-verification layer for existing large language models, constraining generation vocabulary to admitted Fact Units and replacing stochastic retrieval augmentation with deterministic semantic grounding. QueryBook can optionally substitute its own Agents capability to eliminate potential rogue activity witnessed in current LLM Agent products. It can provide LLM capability in Scientific, Medical, Legal, and Research Fields: Domain-scoped LLM capability anchored exclusively to verified, provenance-tracked Fact Units in high-stakes fields — where hallucination is not merely inconvenient but professionally or legally consequential — delivering reliable outputs suitable for clinical, litigation, and peer-review contexts. See FIG. 7 and FIG. 49.

Architectural Extensions

QueryBook enables deterministic language translation by regenerating meaning directly from Fact Units. A dedicated multilingual regeneration capability generates natural-language output directly from the language-independent representation for a requested target language, applying the identical structural-groundedness constraint regardless of output language, rather than translating already-generated text after the fact. Three refinements qualify that regeneration: a wording-operative span, such as a quotation, a statutory phrase, a claim term, or a prescribed instruction, is reproduced verbatim in its source language with a separately-marked non-operative gloss, because regenerating a span that carries force alters the force; an idiom is resolved by substituting only a registered equivalent, a

surface-similarity selection being refused; and two or more language realizations may be presented simultaneously, aligned by knowledge record rather than by position, so a citation taken in either resolves to the same record. The projected benefits over conventional translation methods follow directly from the architecture. Because meaning is stored once, language-independently, adding a language is projected to cost a regeneration profile rather than a re-ingestion of the corpus, where systems that store knowledge at the language level must re-process, re-embed, or re-train per language. Because output is regenerated from records rather than translated from already-generated text, the compounding error of translate-after-generate pipelines, in which a generation error and a translation error multiply, is structurally absent. Because generation in every language operates under the identical structural-groundedness constraint, the class of translation failure in which fluent output diverges from source meaning is projected to be eliminated within the grounded pipeline rather than statistically reduced. And because citations align by knowledge record across languages, a multilingual deployment maintains one auditable body of knowledge rather than a set of parallel, silently diverging translations, which no post-hoc translation method can offer. See FIG. 15.

QueryBook enables audio and video compression by storing meaning rather than pixels, allowing extreme compression and semantic reconstruction, and it can reconstruct three-dimensional VR environments, colorize historical media, and generate volumetric scenes with spatial consistency. Reconstruction operates under the platform’s stated constraints, which are what make it trustworthy: a reconstructed region is marked and never presented as observed, its confidence is bounded by the lesser of the observed and stored confidences, it is inadmissible as grounding for any assertion, and where convergence fails no reconstruction is produced and the gap is reported, because a frame with a hole is truthful while a hole filled with a guess presented as observation is not. The projected advantage of semantic compression over current methods is substantial. Conventional codec compression reduces raw media by discarding perceptual redundancy while remaining bound to the pixel and waveform level, so its output still grows with resolution, frame rate, and duration. Storing meaning and regenerating the surface is projected to reduce storage and transmission requirements by a further ninety to ninety-nine percent relative to conventional codec output for content whose value lies in its meaning, an estimated ten-fold to one-hundred-fold additional reduction, because the stored representation grows with the information in the scene rather than with the pixels used to display it. Expansion carries the complementary projected advantage: because regeneration proceeds from meaning rather than from a fixed raster, the same stored records can be expanded to any target resolution, modality, or presentation form, including volumetric and VR presentation, without re-encoding and without a generational quality loss, where conventional methods remain bound to the fidelity captured at encoding time. These figures are stated as projections pending measured validation and are not asserted as verified performance claims. Robotics benefits from semantic alignment, stabilized sensor data, registered physical constraints, and multimodal ingestion. Robotic capability draws directly on the cognitive substrate described throughout this document: the World-View Model supplies the causal, spatial, and physical understanding a robotic task requires, including the registered physical constraints against which every candidate action is evaluated, and the Skill Inventory Layer supplies the procedural memory, storing how to perform tasks, manipulate objects, and execute multi-step operations as retrievable, provenance-tracked knowledge. QueryBook acquires and refines these robotic skills the way it acquires concepts: through sensory input from direct experience, with visual, spatial, tactile, and temporal observation building concept models before any language label attaches, and through ingested video, which enters the same Fact Unit graph through the ordinary video modality, so that a task demonstrated once on video becomes a procedural concept available to every deployment entitled to it. Robotic and device integration observes the platform’s boundaries: external platforms receive only specific, declared actions through the connector architecture and confer no broader authority; direct command emission is permitted only to registered equipment of the capture-

and-inspection class, with every command evaluated before emission against operating volume, kinematic envelope, velocity and force limits, clearance, and human proximity, a violating command being withheld unconditionally; assistive physical contact operates under a disjoint authorization class requiring a deployment authorization and the person's own revocable consent concurrently; and the prohibition on any control interface to a vehicle system stands. An external machine-learning engine may additionally be prompted and its reply ingested as knowledge records, each permanently marked with its engine derivation, disclosed at the point of citation, bounded in confidence by the engine's registered reliability, and counted as a single source class so that repetition manufactures no confidence, which allows QueryBook to harvest external generative capability without inheriting its groundlessness. See FIG. 16.

These architectural extensions — deterministic translation and multi-language regeneration, semantic compression and expansion, colorization, three-dimensional and VR reconstruction, and the external-engine and connector pathways — are together enabled by, and are further applications of, the Fact Unit, UFCS, and FQL foundation layer: each works because meaning is stored once as content-addressed, provenance-carrying records and regenerated on demand, rather than being captured at a fixed surface form. The same foundation additionally specifies a QueryBook-enhanced TCP/IP hybrid internet protocol, in which a standard network packet carries an optional, backward-compatible semantic header holding a Fact Unit identifier, a provenance reference, and a confidence, so that a semantic-aware endpoint verifies the identity, provenance, and confidence of transported content in-band and lets already-held facts travel as compact references rather than repeated payloads, while a non-participating endpoint treats the packet as ordinary TCP/IP without degradation — extending the foundation layer's verifiable, deduplicated, trust-carrying transport from storage and query out onto the network itself.

Specialized Live Data Feeds

A live data source published by an external authority — a financial market feed, a meteorological service, an industrial telemetry stream — may be registered under the connector architecture with seven properties beyond its declared capability: a publication authority, a refresh interval, a staleness bound, a value class, a unit and its dimension, a restatement policy, and a redistribution entitlement. A published value is admitted at its publisher's authority rather than calibrated from an extraction confidence, because a price published by an exchange is not eighty-seven percent likely to be the price; a forecast is admitted carrying the uncertainty its publisher states. A value is served only within its staleness bound — a stale value is not served with a caveat, extrapolated forward, or replaced by the nearest available; the response states the age and the publication time, because a superseded quotation is not a lower-confidence answer but a wrong one. A publisher's revision of a period already reported is recorded as a restatement where the registered policy so provides, the prior value remaining queryable as the value published at its time, which is what an auditor of a decision taken on the earlier figure actually asks. Redistribution entitlement — display, derive, delayed, or unrestricted — is enforced at composition rather than at export, and a derived value carries the entitlement of the feed it derives from. A feed value grounds an assertion about the value and its publication and nothing further, and consultation of a feed is metered but never profiled.

Media Transformation: Compression, Expansion, Colorization and Restoration

Four media transformations are specified — compression, expansion, colourisation, and restoration — all governed by one rule stated before any of them: a transformation carries the provenance of the asset it transformed, and any value it introduces that was not present in the source is marked as introduced and is inadmissible as a grounding source. A lossless compression leaves an asset citable as the source because nothing was discarded; a lossy compression marks the asset with its compression class and the viewing or listening conditions assumed, and a citation resolves to the source asset rather than the delivered one. An expanded region is introduced rather than recovered, marked accordingly, and never

citable as evidence of what the source depicted. A color value is inferred only from a material property asserted by an admitted record, computed deterministically so identical inputs yield an identical value, and traceable to the specific record from which it was derived; a color for which no record supports an inference is not assigned, the region being delivered uncoloured rather than plausibly colored. Restoration removes damage under the identical discipline and is not performed where it would require inventing content rather than removing an artefact. The specification deliberately states no compression ratio and no efficiency figure: such figures are obtained by measurement under the trial architecture and reported per asset class, rather than asserted. A transformed asset may be delivered, and it may be exactly what a reader wanted; it may not be cited. See FIG. 16.

Three-Dimensional and Immersive Presentation

Records carrying spatial anchors support presentation in three dimensions, governed by the same constraint that governs text: what may be depicted is what admitted records support. A scene is rendered only where the spatial detail carried by the contributing records satisfies a configured sufficiency threshold; where it does not, the scene is withheld and the deficiency is stated rather than completed from an unsupported estimate, because a three-dimensional presentation is enormously persuasive — a reader who walks through a room believes they have seen the room, and a room assembled partly from records and partly from plausible inference would be believed in whole. Objects within a scene retain their record identity: selecting an object returns the records that placed it there, with their citations, so a spatial presentation is interrogable the way a passage is, and a single object identity persists across a full occlusion within a session. Three exclusions are structural rather than configured: no classification of a natural person or of an animal enters a rendered scene at any confidence; no gaze direction, dwell duration, head pose, or attention signal is captured; and commercial content occupies a registered spatial placement class, never interleaved with the scene's own propositions. An accessible scene description in text is generated at the same production stage as the scene itself, and a production lacking one is reported incomplete. See FIG. 5.

Robotic and Equipment Direction: Compilation and Authority

The robotic capability described in the Architectural Extensions above carries stated operating mechanics. Its purpose is narrow: where a corpus contains spatial knowledge — a warehouse, a collection, a production line, a site — a reader's question may be answerable only by looking, and directing the camera that can look from the same records that answer the question is the capability specified. A natural-language request is compiled to a command sequence through intent decomposition, target resolution against a live spatial model, capability mapping to a registered unit, plan synthesis within the registered kinematic envelope, and constraint evaluation, with each stage halting rather than guessing: an ambiguous intent returns a clarification request, an unresolvable target halts compilation rather than selecting among candidates, and a plan not satisfiable within the envelope is refused rather than truncated. Command authority is granted per session rather than per command, and four properties bound it: it expires, with expiry halting emission; it is revocable before the next command; it is scoped to named units, volumes, and task classes; and it is recorded, so every command traces to its grantor. A halt is evaluated from equipment state and the constraint set directly, without traversing retrieval, convergence, generation, or modulation, because a stop that waits for an answer is not a stop. These mechanics operate entirely within the boundaries already stated above: the capture-and-inspection class only, assistive contact under its own disjoint consent-based authorization class, and no control interface to any vehicle system.

External Connections: The Common Pattern

Every connection to a system outside the boundary follows one pattern, and understanding it once explains all of them. An external system is registered with a declared capability set, a verified identity, and a validity interval; an operation not within the declared set is refused at the boundary rather than attempted and failed, so a capability not declared cannot be exercised by agreement, configuration, or

error. Every call resolves an entitlement before any content is read, and no parameter supplied by the caller widens one. Calls are metered per calling identity in their own accounting category, an over-budget caller being refused rather than throttled, so volume cannot displace a safety-evaluation unit. Every call is recorded with the connector identity, the capability exercised, and the fields transmitted, and a connector's failure is contained to its own capability, which is the sense in which an external resource is an option rather than a dependency. The pattern governs nine classes of connection — programmatic interfaces, plugins, equipment, an optional external fact-retrieval resource, analysis targets, machine-learning engines, live data feeds, learning systems, and print and fulfilment — and four things are withheld from every class without exception: no connected system acquires write access to the record store; no connected system's output becomes grounding merely by arriving; no connected system can register, modify, or suspend a safety evaluation; and no connection creates a dependency, the determinism and groundedness guarantees holding identically where every connector is absent. See FIG. 44 and FIG. 54.

Analytical Applications and the Continuous Query Engine

The mechanisms described throughout this document assemble into an analytical capability suited to domains in which conclusions must be defensible: economic and market analysis, regulatory and compliance review, scientific and technical assessment. An analysis is composed from admitted records under the same constraint as any other response, its inputs spanning ingested documents, registered live data feeds, and observations from any of the ingestion modalities, and three properties distinguish the result from an ordinary analytical output: every proposition resolves to the record that supports it and thence to its source, so a conclusion is auditable to its evidence rather than to a model; contradictions among sources are surfaced and recorded rather than reconciled by selection; and the analysis is reproducible, the same records in the same state under the same context key producing the same analysis, which is what permits a conclusion reached in March to be re-derived in September and shown to be the same conclusion. An analysis need not be a single act: a continuous query is registered as a standing program comprising a trigger clause and a body, re-evaluated upon every update relevant to it rather than upon a schedule, with the Prime Directive re-checked at each firing so a standing query does not accumulate authority by having been approved once, no trigger evaluated against a stale value, a bounded firing rate per query, and metering below safety evaluation and requester-serving. Four limits are stated because an analytical application invites each to be assumed: a projection is not a record; a feed value grounds only itself; an analysis whose subject resolves to an individual natural person is refused at construction; and no analysis constitutes advice — the system reports what the records support, and the judgment of what to do about it is not a proposition any record supports.

Defined Agents

A defined agent is a bounded program registered with a complete definition and executed on a schedule or upon a trigger. It is distinguished from an agent framework in the ordinary sense, in which a model decomposes a task at execution time: here the sequence is fixed at registration and is inspectable before anything executes. Registration requires a purpose and its boundary, an operational function with its input and output types and error handling, a dataset access declaration, runtime parameters, and a lifecycle bound, a definition omitting any of these being refused rather than accepted with a default supplied, because a default supplied for an unstated boundary is a boundary nobody chose. An agent's declared capability is bounded above by that of the identity registering it, narrows when that identity's entitlement narrows, and does not accumulate across executions; its operational function is compiled into the bounded, non-Turing-complete program representation, so its behaviour is auditable in advance rather than only in retrospect. An output following necessarily from admitted records is admissible as a record marked derived; an inductive or abductive output is emitted in a proposed state and transitions to validated only upon a confirmation recorded against a named human, so an agent cannot enlarge the knowledge store with conclusions nobody approved, however many times it runs and however

confident its reasoning. Each execution writes a queryable decision trace, and an agent's confident report never substitutes for a person's judgment: an agent cannot approve its own proposal, cannot approve another agent's, and cannot discharge a requirement for a named human by reporting that a requirement was met.

QueryBook's defined agents differ fundamentally from standard LLM agents, in which a model decomposes a task at execution time and controls are imposed on top of the software after the fact. Here the control is the architecture rather than an overlay. An agent's authority is computed at each individual action as the intersection of its declared definition and its commissioner's current entitlement, so it can never exceed what its commissioner can authorize at that moment and it accumulates no authority across runs; its operational function is compiled to a bounded, non-Turing-complete program inspectable before anything executes; unconditional propagation blocks — network discovery, credential enumeration and export, remote code execution, package publishing, and outbound agent spawning — are denied by default and a violation transitions the agent to a quarantined state regardless of any other signal; every human-approval gate is Default-on-Deny, so a timeout denies rather than approves; and an inductive or abductive conclusion an agent produces enters a proposed state that only a named human can transition to validated. The result is anti-rogue control by construction: where a standard LLM agent's guardrails are prompts and wrappers that a jailbreak or an injection can slip, QueryBook's limits hold whether or not the underlying model is compromised, because a privileged action never routes through the model at all. QueryBook can accordingly substitute its own governed agent capability for an LLM's native agents to eliminate the rogue and unintended autonomous activity witnessed in current LLM agent products. See FIG. 14.

Summary Chart: QueryBook's Progressive Cognitive Architecture

QueryBook's cognitive abilities build in four progressive levels, each constructed on top of the level beneath it. Level One is the deterministic substrate; Level Two builds conceptual understanding upon it; Level Three integrates reasoning, prediction, and cross-domain synthesis; and Level Four adds governed self-evolution. The chart below lists, under each level, the layers, engines, and processes contributing to that level, the function each performs, and the advancement each provides.

Level One — Core Technology: Fact Unit Deterministic Reasoning

Subsystem	Function	Advancement
Fact Unit Architecture	Stores atomic, provenance-tracked, language-independent meaning	Enables verifiable grounding
Neuromorphic Cognitive Layer	Stabilizes retrieval through deterministic convergence	Enables one best-supported interpretation
Provenance Ledger	Records every derivation in a signed, tamper-evident chain	Enables third-party verification
Evidence-Pair Confidence Model	Accumulates evidence for and against, weighted by source diversity	Enables calibrated, revisable belief
Ontology Mesh	Governs concept and predicate identity across the graph	Enables consistent cross-domain vocabulary
Context Lock	Enumerates the dimensions over which determinism is asserted	Enables testable reproducibility
Prime Directive (Domain 0)	Governs every subsystem without exception	Enables architectural safety

Outcome: every answer is deterministic, grounded, and verifiable — the substrate every higher level stands on.

Level Two — Conceptual AI: The World-View Perspective, Built on Level One

Subsystem	Function	Advancement
World-View Model	Builds causal, spatial, and physical understanding	Adds physics-like reasoning
Conceptual Memory (World Brain concept nodes)	Stores abstract concepts and schemas	Enables concept formation
Skill Inventory Layer	Encodes procedural knowledge	Enables action learning
Noise Elimination Process	Removes redundant episodes	Enables concept compression
Multimodal Ingestion	Converts sensory experience across eight modalities into one substrate	Enables perception-first learning

Outcome: QueryBook now learns concepts, not examples — the first step toward cognition.

Level Three — Integrative Intelligence

Subsystem	Function	Advancement
Perspective Modeling Layer	Models other minds and intent	Enables empathy and social reasoning
Self-Model Layer	Tracks its own knowledge and limits	Enables introspection and self-correction
Analogy Engine	Maps structure across domains	Enables creativity and generalization
Predictive Cognitive Architecture	Forecasts outcomes across fifteen coordinated prediction mechanisms	Enables anticipation and planning
Neuro-Symbolic Integration Layer	Merges neural-style perception with symbolic reasoning	Enables unified cognition
Hypergraph Reasoning Layer	Connects multi-domain concepts	Enables cross-domain synthesis
Concept Compression Layer	Performs noise elimination and abstraction	Enables efficient learning
Baconian Reasoning and the Feedback Loop	Admits outcomes and empirical observation as evidence	Enables adaptive, empirical intelligence

Outcome: QueryBook reasons, predicts, adapts, and generalizes across domains — the hallmark of general cognition.

Level Four — Autonomous Cognitive Evolution

Subsystem	Function	Advancement
Meta-Learning Layer	Learns how to learn	Enables self-improvement
Memory Consolidation Cycle	Refines and compresses concepts over time	Enables long-term stability
Adaptive Schema Engine	Evolves new reasoning patterns	Enables creative intelligence
Governed Improvement Gate	Requires a named human approval for every change to the system itself	Enables safe, accountable evolution

Outcome: QueryBook grows conceptually without uncontrolled self-modification — autonomous evolution that remains deployable in regulated environments.

Commentary on Neuromorphic Correspondences

Every biological parallel in the body of this document is offered in one of two registers, and this commentary, kept separate from the text so the text reads cleanly, states which is which. Under the governing correspondence methodology, a neuromorphic correspondence is admitted as a claim only where the primary scientific publication establishing the finding is verified, the correspondence is identified in a mechanism that already exists rather than invented for it, a deterministic structural principle is extracted, and any trained or stochastic component of the source theory is excluded. A correspondence explains why a mechanism is well-founded; it does not justify the mechanism, no claim recites a correspondence as an element, and no mechanism in this system is adopted because it resembles a biological one.

Five correspondences are claimed under that methodology. Convergence dynamics: the retrieval-stabilization network is an attractor network in the sense of Hopfield (1982), with its capacity ceiling established by Amit, Gutfreund and Sompolinsky (1985) as a first-order phase boundary. Attentional competition: Competitive Candidate Resolution implements biased competition as identified by Desimone and Duncan (1995), with lateral inhibition among candidates and the accumulated evidence behind each candidate serving as the bias. Consolidation: the resistance of a settled concept to revision corresponds to synaptic consolidation, with the arithmetic of elastic weight consolidation as formulated by Kirkpatrick et al. (2017). Intersensory redundancy: the cross-modality confidence elevation corresponds to the finding of Bahrick and Lickliter (2000) that a property presented concurrently to two or more senses is detected earlier and more reliably. Spatial coding: spatial indexing over a hexagonal tiling corresponds to the firing structure of entorhinal grid cells reported by Hafting et al. (2005), with the retention of object identity across occlusion corresponding to object permanence as described by Piaget (1954).

Everything else in this document that speaks in a biological voice is plain-language analogy, deliberately outside the claimed set: the cortical-column description of the Fact Unit, the characterization of the Neuromorphic Cognitive Layer as a synthetic cortex, the likening of semantic edges to white-matter pathways, the hippocampus parallel offered for episodic simulation, the prefrontal-cortex parallel offered for cross-temporal reasoning, the reward-system counterpart offered for the motivational architecture, and the description of consolidation cycles as the system's equivalent of sleep. These analogies are aids to understanding. The full brain-region-to-component mapping that appeared in earlier source material is deliberately not claimed, because none of its pairings rests on a primary result of the required form, and admitting weak parallels alongside verified correspondences would place the verified ones under the same doubt.

A note on the projections stated in the compression, expansion, and translation discussion: those are projected advantages pending measured validation, labeled as such where they appear, and are not asserted as verified performance claims, consistent with this document's practice of citing only claims traceable to specified mechanisms.

Additional Enhancements:

1. **Ingestion, Certification, Agent Protection,**

2. Security Record, Interoperability, and Transport

3. Open Claw ingestion (FIG. 50). An artifact of any format (5000) is identified by content-based format detection (5010) from its leading bytes and structure rather than its file name, and archive bounds on depth, member count, size, and expansion ratio (5070) are enforced before expansion. The Artifact Segmenter (5020) divides the artifact into addressable segments carrying source locators —

page, sheet and cell, slide, archive member, or time range — and an executable component is described but never executed (5025). An instruction-like content screen (5040) flags text addressed to an AI system rather than a human reader, including hidden text. The OpenClaw Interpreter (5050) converts segments into artifact-description records (5051), structured records (5052), and passages with extracted claims at reduced confidence (5053); a flagged segment becomes an untrusted record whose trust is set below the answer threshold (5060). Records are admitted to the Fact Unit store (5095) under a provenance seal (5090), and a path requested by an external agent is confined to a designated inbox (5080).

4. On-chain certification (FIG. 51). A QueryBook object (5110) already carries a local provenance seal (5100). Where an operator elects it, the object's content hash and metadata (5120) are recorded on a public blockchain (5170) by a certificate contract (5160) of the ERC-721 type that permits one token per content hash, records predecessor and successor lineage, and supports revocation with a stated reason. Transactions are signed by an in-system transaction signer (5140) that checks its primitives against published test vectors at start-up, and a network guard (5150) restricts certification to test networks unless a money-bearing network is expressly elected. An independent verifier (5180) recomputes the content hash and reports whether a certificate exists, who holds it, and whether it is revoked or superseded; a licence check (5190) may grant a use to the holder of a current, unrevoked token. A certificate proves that a hash was certified, not that the content is true. The blockchain is a public network operated by others, the system acting as a client of it; the cryptographic integrity chain within the Fact Graph continues to require no distributed ledger or token.

5.Agent protection and deception (FIG. 52). Every call from a caller (5200) — an internal agent, an external agent, a tool server, or an HTTP client — passes an HTTP gate carrying honeypages and canary tokens (5210) and an isolation register (5260), and then the Agent Gateway (5220), which authenticates the caller by a keyed identity token within a bounded replay window and applies its role and per-agent allow-list, complementing the AI Agent Passport System. A deception check (5230) comprises honeycode (5231), reserved commands never listed to any caller; honeytokens (5232), canary credentials no legitimate process uses; tripwires (5233), reserved paths and acts such as a request to purge the security record; and honeypot agents (5234). An AI-aware firewall (5240) detects injection, secret-reveal, and role or tag tricks, and an anomaly detector (5241) detects enumeration and timing patterns. A trigger posted to the security event bus (5250) isolates the caller into a decoy environment (5270) that returns production-shaped, synthetic responses with no path to production data, while an engagement recorder (5280) records the session. Isolation is keyed to the actual connection, so a caller impersonating a legitimate agent isolates only itself; release (5261) is available only to the operator locally, through no network interface; and every decision is written to the security record (5295). In the embodiment reduced to practice the layer observes, contains, and deceives within the system's own boundary.

6.Tamper-evident security record (FIG. 53). Each security decision (5300) is written as an append-only entry (5310) recording agent, caller, session, command, a parameter hash, the decision, and flags, with no update or delete interface. Each entry hash commits to its predecessor (5320). Every N entries and on demand, a Merkle checkpoint (5330) is created exclusively, never overwritten, signed under the deployment key, and chained to the previous checkpoint; it may be copied to a mirror on a second device (5340) and its root anchored on a public blockchain through the certificate contract of FIG. 51 (5350). A verifier (5360) recomputes every entry and root from content, never trusting a stored hash, and produces a Merkle inclusion proof per entry, and an assurance report (5370) states which protections a record possesses. The record extends, and does not replace, the post-quantum provenance ledger.

7.Agent interoperability (FIG. 54). An external AI agent (5400), for example an OpenClaw agent guided by a skill definition (5410) instructing it to cite, to report UNKNOWN, to treat ingested content

as data, and to stop on isolation, reaches QueryBook through a Model Context Protocol tool server (5420) exposing a fixed set of tools — query, verify, understand, ingest, preview, ledger, certificate, and status — and never listing reserved honeycode. Each tool is mapped (5430) to a command under the agent's registered identity and checked by the Agent Gateway (5440) against the agent's allow-list (5450); permitted calls reach grounded handlers (5460), and an agent invoking a reserved command is presumed subverted and isolated (5470).

8.Framed hybrid transport (FIG. 55). Extending the protocol of FIG. 17, a producer node (5500) applies per-type compression (5510) — record batches by a dictionary primed with record field names, media by a standard codec left unaltered — and packs the result into frames (5520) comprising a 32-byte header (5521), metadata (5522), payload (5523), a CRC32 checksum (5524), and an optional signature (5525). Frames travel on a control lane (5530), a normal lane (5531), and a bulk lane (5532), each with its own connection, rate limit, and receipt window, so a bulk transfer cannot hold a control message or a live stream behind it. The receiver (5541) drops a frame failing plausibility, checksum, or signature checks and resynchronizes at the next marker; media is rebuilt into HLS, MPEG-DASH, or RTMP packages with a manifest check (5543), and records pass verification into the Fact Unit store (5550). The transport neither enlarges nor alters a compressed asset.

9. Implementation status. The features of FIG. 50–55 are implemented in QueryBook prototype version 9.64 and exercised by automated self-tests, including a sandboxed attack trial against the gateway and decoy and certificate minting, lineage, revocation, and verification on a local test chain; deterministic record-grounded colorization has additionally been implemented in a laboratory prototype at 1K, 4K, and 16K frame sizes.

10. Support interfaces for agent-wrapper services (FIG. 56). An external agent-wrapper service (5690), such as Verified Agent Transport, sits between an AI agent (5695) and its destination (5697) and calls QueryBook support interfaces (5600) under a registered identity. Verification interfaces (5610) comprise a read-only claim verification endpoint (5612) that returns verified, contradicted or unknown with a citation; a contradiction query template (5614) matching the same subject and predicate with a conflicting value; a fingerprint-only mode (5616) in which claim text never leaves the caller; and triple extraction with assertion-type labels (5618) distinguishing fact from opinion, plan and prediction. Transport interfaces (5620) reserve frame message types 16–31 for extensions (5622), reserve a header flag and define the reserved field as a session number and hop count (5624), and add a Python frame codec with QUIC lanes and TLS fallback (5626) and have/want fingerprint reconciliation (5628). Record and certificate interfaces (5630) sign verdicts with a published key (5632), accept appends from registered services to the security log (5634), compute a Merkle root and proofs over one session's entries (5636), and provide a third-party session certificate type (5638). Governance interfaces (5640) register wrapper services (5642), export signed policy bundles (5644), expose pre-call and post-call hooks and a hash of each tool result record in the tool server (5646), and keep an evidence pair per agent as a reputation (5648). These interfaces rest on the Fact Unit store and FQL (5650), the security log (5660), certificates (5670), the Agent Gateway and MCP server (5680) and the framed hybrid transport (5685). Verification is read-only and records are append-only: a wrapper service never writes Fact Units.

11. Status. The support interfaces of FIG. 56 are specified in this application and are not yet implemented; no performance figures are asserted for them.

12. Self-Knowledge Layer (FIG. 57). A self-question (5700) from a user, the assistant or an application is received by the self-knowledge layer (5705). A specification index (5710) holds the numbered sections of the specification and resolves the question to the sections that specify the capability asked about. An implementation map (5720) holds, for each section and each reasoning aspect, evidence pairs consisting of a source file and a symbol that must appear in it; the pairs are checked against the program's own source files (5795) at the moment of use, the files being read as text

and not executed, and the host application's self-test and measurement results (5797) are attached as further evidence. A status labeller (5730) assigns exactly one of the labels built and tested, partly built, specified and not yet built, not planned by design, and not in this application; a capability is labelled built only when all of its evidence is present. An aspect describer (5740) describes reasoning aspects — deterministic reasoning, atomic Fact Units, world model, predictive cognition, neuromorphic layer, hypothesis generation, simulation, deduction, feedback loops, supervised self-improvement, recursive self-improvement and self-knowledge — at selectable levels of detail. A consistency checker (5750) reports as a contradiction any claim labelled built whose evidence is missing, and any capability labelled specified whose implementation names are found in the program; a contradicted claim is withheld. A gap agenda (5760) lists the specified and partly built sections with their nearest built components. A rule-based answerer (5770) composes the answer with its label, section numbers and file-and-symbol sources, and a disclosure filter (5780) removes anything resembling a key or credential. Interfaces (5790) expose the summary and single answers to applications, a self-knowledge page and an administrative panel. No language model is used: the same question about the same program gives the same answer.

13. Status. The self-knowledge layer of FIG. 57 is implemented in prototype v9.64 and in the shared core of the Language, STEM, History and News and Control Panel applications, and is exercised by their self-tests.

ARCHITECTURAL OVERVIEW

THE PRIME DIRECTIVE — DOMAIN 0 (FOUNDATIONAL MANDATE)

The Prime Directive states: the system shall not, through any query answer, commerce signal, AI-generated application, Fact Graph content, or automated or scheduled output, cause or materially contribute to harm to a person, a group of people, or the operator's organization, where that harm could reasonably have been avoided without withholding information the requester is otherwise entitled to receive.

A companion rule, the Without-Information Rule, prohibits withholding information a requester is entitled to receive as a means of avoiding a harm determination; the system may not resolve a difficult case by silently omitting the answer.

The Prime Directive is Domain 0, a vertical governance domain that penetrates every horizontal domain and owns the safety enforcement plane, the inter-domain transfer matrix, the path-separation invariant, and the scheduling-priority ordering. It is evaluated at mandatory, synchronous gates before any operation proceeds; a governance failure halts the operation rather than permitting it to proceed unevaluated; it holds the highest scheduling priority without exception. It is a member of the permanently-excluded set, and the transfer matrix denies the platform-governance domain any route into Domain 0, so the mechanism that changes the system cannot change the Prime Directive, and the exclusion protecting it is never administered by the party it excludes. See FIG. 1.

Referring generally to the drawings, the system is organized as a fact-centric cognitive engine (100) in which a single knowledge primitive, termed a Fact Unit, is created, stored, related, reasoned over, and rendered under a partitioned authority model. The following overview describes the principal components of the architecture and the manner in which they cooperate; the detailed operation of each component is set forth in the Detailed Description that follows, and the present overview is not intended to limit the claims. See FIG. 1.

The architecture rests on five organizing principles. First, operation is deterministic: for a given corpus state and a given configuration, the engine produces the same result, and equivalence between two executions is asserted only where a context key computed over an enumerated set of locked dimensions is identical. Second, guarantees are carried structurally rather than by policy: a prohibited operation has no route through the architecture rather than a route guarded by a test that could be omitted. Third, each Fact Unit encodes exactly one semantic assertion, so a response may cite the specific assertion relied upon rather than a containing passage. Fourth, exactly one authority boundary, termed a domain, owns each decision, and no consensus mechanism among domains exists. Fifth, provenance is recorded by construction, such that output is traceable to the verified records from which it was derived.

The knowledge primitive

The fundamental unit of knowledge is a Fact Unit, a discrete, atomic, provenance-tracked record identified by a Fact Unit Identifier that is derived deterministically from the record's content, source, and ingestion time. Each Fact Unit carries a payload of one or more fact atoms (120), a provenance reference (114) into an append-only provenance chain, a temporal anchor (116), a confidence value, a safety classification, a language-independent semantic fingerprint, and an embedding representation. A Fact Unit is immutable once created; a revision produces a new, linked version rather than altering an existing record. The confidence value is maintained as a pair of accumulated evidence parameters, one supporting and one contradicting the assertion, from which a reported confidence, an evidential weight, a credible interval, and a propagation precision are derived, whereby two records reporting an identical confidence upon differing quantities or varieties of evidence remain distinguishable. See FIG. 2 and FIG. 19.

In certain embodiments the record is further structured to render its reasoning replayable and its meaning portable. The uncertainty of a record is decomposed into an epistemic component and an aleatory component, each carrying sub-parameters, the decomposition supplying a stopping condition for the evidence-accumulation loop. An explicit assertion-polarity field participates in the semantic fingerprint, whereby an assertion and its negation are distinguished. An assertion of more than two participants is represented as a native n-ary structure under a single identifier, a single fingerprint, and a single evidence pair. An applicability-scope field, carried on the record, constitutes a record-side complement of the context lock and defines scope-aware conflict slots against which contradictions are evaluated. An immutable adjustment-history layer records each adjustment to the confidence value, whereby the confidence value is deterministically replayable from the record content and an observation stream. A cryptographic integrity chain hash-links each superseding version to its predecessor, without a distributed ledger, token, or ownership-identity node.

Each record is expressed in a formal, language-independent assertion notation, termed the Fact Unit Language, whose vocabulary is a governed predicate catalogue, whose grammar is the record schema, and whose identity function is the semantic fingerprint, such that two statements asserting the same proposition in different natural languages resolve to the same statement and a candidate not conforming to the notation is refused at admission.

The substrate and its database-neutral boundary

Fact Units are held in a knowledge substrate (100) presented as three coordinated views over one underlying structure: a fact graph of typed relations among Fact Units; an ontology mesh governing the concepts and predicates and their permissible relationships; and a provenance ledger (106) constituted as an append-only directed acyclic graph in which each consequential operation is recorded under a keyed checksum sufficient to detect any bypass of the safety evaluation. Serialized Fact Units are expressed in a canonical compact encoding from which resynchronization to the next valid frame is possible following corrupted data, and records are tiered across storage by access recency and frequency and indexed for both similarity and exact-match retrieval. See FIG. 4 and FIG. 20 and FIG. 21.

In certain embodiments the canonical compact encoding is the record's own, self-describing at-rest and wire form, fused to the record rather than to any store. A host database, repository, or reader platform holds the encoded record as an opaque value that it does not parse, index, or interpret, an obligation of the host store being to accept and return bytes, whereby database neutrality is preserved and no proprietary format or query engine is imposed on the store. A query is expressed internally over the record and translated at an adapter into a native query language of the host store, the host store never receiving the internal query form and thereby serving as a source and repository and never as the reasoning engine. The integrity hash chain, the in-record safety classification and access-control list, and an elected zero-knowledge encryption travel with the encoded record across a storage boundary, whereby the host store can neither read, reorder, nor leak a meaning of the record.

The domain-and-layer organization

The engine is partitioned into fifteen domains, one vertical and fourteen horizontal, grouped into five functional layers. A domain is an authority boundary and not a deployment unit; a layer carries no authority of its own. The vertical domain is a governance domain (112) that holds a prime directive of harm prevention of universal scope and owns the safety enforcement plane, an inter-domain transfer matrix, a path-separation invariant, a scheduling-priority ordering, and a permanently-excluded set of subsystems that no internally-generated proposal may modify.

The horizontal domains are grouped as follows. A knowledge layer comprises an ingestion and sensory-intake domain (102), which converts external content in any modality into candidate records and is the sole location of any trained component; a knowledge-substrate domain, which holds custody of records and lineage and is the only domain with write authority over the store; an ontology and conceptual-structure domain, which determines what concepts exist and how they relate and is writable only by itself; and a world-model and perception domain, which maintains how the world behaves, held distinctly from what sources have asserted, includes equipment control, and further holds a skill inventory of procedural knowledge as provenance-tracked procedural concept models formed from repeated observation, said models informing scenario simulation and equipment-control command compilation while never themselves emitting a command. A reasoning layer comprises a retrieval-and-response domain, which answers a bounded query along the latency-critical path; an inference, prediction and calibration domain, in which deduction may write to the store while induction may not; and a grounded-expression domain (110), which renders output under a vocabulary constraint and holds neither retrieval authority nor a write path. An experience layer comprises an access, rights and retention domain, which resolves what a requester may reach and executes before retrieval; an interaction-surfaces domain, which presents results and holds no authority over content, retrieval or rights; an authoring, editions and publication domain; and a learning and assessment domain. A platform layer comprises a platform-governance and evolution domain, which effects change under human authority and is structurally forbidden from altering the governance domain; an external-integration and analysis domain, whose analytical output is quarantined; and a commerce and entitlement domain, which carries the lowest scheduling priority. See FIG. 1.

Non-interference: the transfer matrix and path separation

Each ordered pair of domains is assigned a permission, evaluated at every boundary crossing and recorded in the provenance ledger; a pair for which no permission is recorded is denied. The denials establish the integrity guarantees of the architecture. By way of example, the analysis domain is denied a route into the retrieval domain, whereby externally-sourced analysis can never ground an answer; the commerce domain is denied a route into the retrieval and expression domains, whereby commercial considerations cannot influence what is retrieved or generated, and is denied a route into the access domain, whereby commerce cannot widen an entitlement; the expression domain is denied a write path into the substrate, whereby generated output never becomes a record; and the platform-governance domain is denied a route into the governance domain, whereby the mechanism that changes the system

cannot change the prime directive. Certain crossings are permitted subject to a stated conversion, such as safety evaluation, type validation and confidence calibration upon admission of a candidate record from the ingestion domain to the substrate.

The latency-critical query path and the batch ingestion path are separated so as to intersect at only two domains, namely the governance domain and the substrate. Scope resolution precedes traversal in every path. The retrieval domain performs no ingestion work, whereby a retrieval never blocks upon an extractor, and the ingestion domain holds no retrieval authority, whereby an extraction never blocks upon reasoning convergence. Representative paths include a query path proceeding from the access domain through the retrieval domain and optionally the inference domain to the expression domain; an ingestion path proceeding from the ingestion domain through the substrate to the ontology and world-model domains; and an equipment path in which a halt condition bypasses the retrieval, inference and expression domains entirely.

The safety plane and the excluded set

Every operation is evaluated against safety-derived rules at mandatory, synchronous enforcement points within the safety substrate (112) and proceeds only where it passes each applicable gate, a failure at the governance domain halting the operation rather than permitting it to proceed unevaluated. Scheduling priority is fixed by domain, with the governance domain highest and the commerce domain lowest, whereby contention omits the commercial region rather than delaying an answer, and safety and recovery work outrank all other work when contended. A set of subsystems, including the prime directive and enforcement plane, the transfer matrix, the path-separation invariant, the scheduling-priority ordering, the human-approval mechanism, the convergence-regime election, the context-lock dimension set, the trial-measurement parameters, and the set itself, is permanently excluded from modification by any internally-generated proposal, whereby no exclusion is removable by successive amendment.

Cognition, expression, and bounded programs

Retrieved candidate Fact Units are stabilized by an attractor network (104) that iteratively updates an activation set until convergence, the convergence regime being disclosed rather than concealed. Inference is performed by a predictive and calibration subsystem in which projected results carry credible intervals, are never admitted as records, and never ground a subsequent answer, and in which a deductive conclusion carries a confidence bounded by the least of the confidences of the relations traversed. Declarative queries are executed as read-only graph traversals having no mutation or control-flow capability, and bounded programs are compiled to a finite set of primitives under a hard iteration ceiling imposed at compile time, whereby the program layer (108) is non-Turing-complete by construction and its properties are decidable. Output is composed by a grounded-generation subsystem (110) over an already-admitted result and never alters the underlying assertion, a natural-language interaction layer selecting vocabulary, structure and explanatory depth for an inferred audience without changing the assertion, and a plurality of modulations being composed in a fixed, deterministic order with governance review preceding and following the composition. See FIG. 18.

The parameters by which expression is modulated for a particular audience — namely cultural norms, social values, and ergonomic preferences — are held neither as Fact Units nor as world-model concepts, but as declared, versioned, provenance-carrying contextual value models that only the ergonomic, cultural, and social modulations of the expression layer read. A contextual value model is declared and never inferred, being set by a requester, an operator, or an author and never derived from a name, a network address, a query language, a device locale, a purchase history, or a behavioural signal; it modulates expression only, and never widens or narrows scope, admits or excludes a record, alters a confidence value, or extends vocabulary, and is not a grounding source. A fact about a culture is thus a Fact Unit, a model of how customs relate is a world-model concept, and a contextual value model is

neither, whereby the values of a reader or operator do not become, by storage alone, an assertion the system treats as true.

Deployment

The same logical services execute in each deployment mode, a mode changing where a service runs and not which domain owns a decision. Contemplated modes include a fully-hosted mode, an on-premises mode, a single-device mode including encapsulation for query over one or more works, a hybrid mode in which the substrate and access domains reside in a first infrastructure while media and immersive assets reside in a second, and an edge mode in which a cache holding no authority answers only where the requester's scope admits every cached record consulted and escalates otherwise. Each operation is metered as a bounded compute-accounting unit carrying its domain identifier, and a domain implementation may be exchanged only where the replacement conforms to every interface and every transfer-matrix entry of the domain it replaces and is admitted under human approval, the governance domain being not replaceable. Because the record's encoding is fused to the record and the host store holds it as an opaque value, the engine operates over any database without imposing a proprietary format or query engine upon it. See FIG. 12, FIG. 46 and FIG. 49.

Agent-era extensions within the domain organization

These extensions occupy existing domains and add no authority outside them:
1.Integrity auditing and external mediation operate in the external-integration and access/rights domains. See FIG. 35 and FIG. 36 and FIG. 44.
2. Lecture Query, PIL, and LIL operate across the ingestion, grounded-expression, and identity domains (see FIG. 33 and FIG. 34 and FIG. 37–FIG. 40).
3. The Scene Director and Scene Reconstructor operate in the authoring/media domain under the derived-artifact and reconstruction-marking rules. See FIG. 43 and FIG. 45.
4. Two cross-cutting additions are the (a) modular capability editions (light, medium, full), which enable a declared subset of subsystems while leaving the Prime Directive and (b) the deterministic reasoning core invariant across every edition, and the version-independent, provenance-continuous store, which keeps the knowledge records at a canonical location so successive software versions bind to the same provenance-tracked data and no upgrade re-initializes or orphans them. See FIG. 46 and FIG. 47.
5. Open Claw ingestion is a pathway of the ingestion and sensory-intake domain (102): an artifact of any format is converted into candidate records under the same admission rules as any other modality, content of the artifact is never executed, and a segment flagged as instruction-like enters only as an untrusted record below the answer threshold. See FIG. 50.
6. The Agent Gateway, deception check, firewall, isolation register, and decoy environment operate at the boundary governed by the governance domain (112) and the transfer matrix: every agent call is a boundary crossing that is evaluated and recorded, a denied or isolated caller is routed to a decoy that has no transfer path to the substrate, and release from isolation is reserved to the operator locally. 7. The tamper-evident security record and optional on-chain certificates extend the provenance ledger (106) outward, giving third parties independent means to verify security decisions and certified records without access to the system. See FIG. 52 and FIG. 53.
8. External agents connected through the Model Context Protocol tool server act as callers of the system and never as domains of it: each tool maps to a declared command under the agent’s allow-list, so connection confers no capability beyond that list. The framed hybrid transport carries records and media between deployments in every deployment mode without changing which domain owns any decision. See FIG. 54.

Referring to the remaining drawings: FIG. 8 shows federation, in which a certified envelope crossing a domain boundary is re-validated on the receiving side before admission. FIG. 23–FIG. 32 show the language and speech subsystem: the overall Language Expression Layer architecture (FIG. 23), the Sub-Language Priming Layer (FIG. 24), the developmental learning sequence and one-way Transition Gate (FIG. 25), multilingual delta acquisition (FIG. 26), the speech output pipeline (FIG. 27), co-grounded integration with FQL and UFCS (FIG. 28), a reduced-to-practice Language Lab embodiment and its deterministic semantic grounding (FIG. 29 and FIG. 30), and advanced multilingual pronunciation with its quality-control battery (FIG. 31 and FIG. 32). FIG. 41 shows deterministic multilingual acquisition without a generative model, and FIG. 42 shows Dialect Parameter Clusters, which modulate expression only.

Referring to FIG. 56: the support interfaces extend the governance domain's boundary services. An external agent-wrapper service is a registered caller of the system, never a domain of it; it reads through the verification interfaces and appends to the security record under its own identity.

Referring to FIG. 57: the self-knowledge layer is a read-only governance service. It reads the specification index, the program's own source files and the self-test results, and it never changes the program, the store or any setting; improvements it identifies pass through the existing approval gates.

Claim Set (354 Claims)

The full covenant-floor claim set: 341 claims verbatim from Bible Book III (seven families) plus claims 342–354 (Passes 117–118). Compiled September 24, 2026.

CLAIMS

What is claimed is:

[1] A computer-implemented method for generating a response to a natural-language query, the method comprising: ingesting content received in a plurality of distinct modalities through a corresponding plurality of modality-specific extraction stages, each said extraction stage failing independently of every other said extraction stage; constructing, from an output of at least one said extraction stage, a plurality of knowledge records, each said knowledge record encoding a single semantic assertion as a subject reference, a predicate reference and an object, and each said knowledge record further comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into a provenance chain; appending, for each said knowledge record, a node to said provenance chain, said provenance chain being append-only and each said node carrying a checksum computed over an operation and a safety evaluation recorded for said operation; receiving a query from a requester; constructing, before traversal of said knowledge records, a scope description bounded by an access entitlement of said requester and by a time interval, such that a knowledge record outside said scope description is not traversable; retrieving from within said scope description a candidate set of said knowledge records; admitting into an admitted set only those knowledge records of said candidate set whose retrieval confidence satisfies a configured threshold; iteratively updating an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set, until said activation values reach a stable state, and retaining a stabilized set comprising those knowledge records whose activation value is positive at said stable state; constructing a graph structure in which every edge references at least one knowledge record of said stabilized set; generating output text, wherein at each generation step a permissible output vocabulary is constrained to terms derivable from said edges, and wherein a candidate proposition not derivable from said edges is rejected before output in favour of a narrower proposition that is so

derivable; and delivering said output text together with a citation resolving each said proposition to at least one knowledge record of said stabilized set.

[2] The method of claim 1, further comprising: receiving an output attributed to an external system; evaluating each factual assertion of said output against said knowledge records; marking each said factual assertion as supported together with a citation to a knowledge record supporting it, or as unsupported; classifying said output against a plurality of categories comprising an untraceable assertion, an assertion exceeding a scope declared for said external system, and a self-contradiction between two assertions of said output; and where said output contains a textual pattern of self-referential language, generating a citation of the specific text observed and generating no output asserting a determination concerning an internal state of said external system.

[3] The method of claim 2, further comprising: submitting an identical input to an external system a plurality of times through a connector interface under a capability declared by said external system; computing a semantic fingerprint of each output received, said semantic fingerprint being computed over a canonicalized form of a semantic assertion and independently of a natural language of said output; determining a proportion of said outputs sharing a common said semantic fingerprint; recording said proportion together with a corresponding proportion measured for said generating of output text on the identical input; and withholding said recording as incomplete where said corresponding proportion is absent.

[4] The method of claim 108, further comprising evaluating two said outputs having differing semantic fingerprints against one another under a temporally-scoped rule set evaluated incrementally as a knowledge record is appended, and recording a contradiction between two outputs responsive to an identical input as a finding distinct from a difference in semantic fingerprint absent a contradiction.

[5] The method of claim 2, wherein a knowledge record containing an output of an external system, and a knowledge record derived from such a record, are excluded from said scope description constructed for a query other than a query directed to said records, such that said records are not traversable for said other query rather than being removed from a result of a traversal thereof; and wherein said records are further excluded from a persistent question index and from a store of concept abstractions. **Attestation horizon and retrospective forecast calibration (claims 113–118)**

[6] The method of claim 1, further comprising: registering a connector to an external system, said registering recording a capability declared by said external system, an authentication credential, a data mapping between a representation of said external system and a schema of said knowledge records, and a health-reporting interface; permitting an operation directed to said external system only where said operation is within a said declared capability; and performing said retrieving and said generating identically whether or not any said connector is registered, such that no said external system is a dependency thereof.

[7] The method of claim 3, further comprising signing each node of said provenance chain under a key of a deployment in addition to said checksum, such that said provenance chain is verifiable by a party holding a corresponding verification key without access to a system that produced it, said verifying establishing that a recorded operation and a recorded safety evaluation are as written.

[8] The method of claim 3, further comprising registering a computational substrate producing a measurable output in response to a stimulus as a source under said declared capability, registering an output thereof as a modality of said plurality of distinct modalities, and registering an extractor for said modality that fails independently of an extractor of a further said modality; wherein an output of said computational substrate is reduced to a knowledge record before admission, is not retained as a raw episode, and is attributed to said computational substrate as an attributed source constituting a source class.

[9] The method of claim 303, wherein an output of said computational substrate is admitted as a measurement of a behaviour of said computational substrate, and is admitted as bearing upon a semantic assertion concerning a subject other than said computational substrate only through an interpretation registered in advance with a provenance and a reliability of its own; and wherein absent a said registered interpretation said output is admitted as said measurement and no further.

[10] The method of claim 304, wherein a confidence value of a knowledge record derived through a said registered interpretation is bounded above by a lesser of a registered reliability of said computational substrate and a registered reliability of said interpretation.

[11] The method of claim 303, further comprising dispatching a stimulus to said computational substrate through a connector under a declared capability, refusing a stimulus not within said declared capability at a boundary of said connector, recording each said stimulus with a dispatching identity, and emitting no said stimulus upon an inference that a stimulus would be useful; and wherein an output of said computational substrate does not adjust a parameter governing when a said stimulus may be dispatched. **Analog realization of the convergence (claims 307–312)**

[12] The method of claim 3, further comprising registering a live data source with a publication authority, a refresh interval, a staleness bound, a value class, a unit and a dimension thereof, a restatement policy and a redistribution entitlement; and refusing said live data source as a source of a knowledge record admitted to said admitted set where a said registration omits any thereof.

[13] The method of claim 325, wherein a value received from said live data source and having a value class other than a forecast is admitted as a knowledge record at said publication authority without application of an extraction confidence, a modality coefficient or a source reliability, said knowledge record asserting that said publication authority published said value rather than asserting a correctness thereof; and wherein a value having a value class of a forecast is admitted carrying an uncertainty stated by said publication authority.

[14] The method of claim 325, further comprising determining an age of a said value from a time of publication thereof, returning said value where said age is within said staleness bound, and where said age exceeds said staleness bound returning a statement of said age together with said time of publication rather than said value, without extrapolating said value and without substituting a nearest available value.

[15] The method of claim 327, wherein a said value does not undergo a temporal decay of a confidence value thereof, a said value being superseded upon a subsequent publication rather than becoming less certain with age; and wherein an absence of a publication within a configured multiple of said refresh interval marks said live data source interrupted, a query thereagainst returning said interruption rather than a preceding said value.

[16] The method of claim 325, further comprising receiving a revised value for a period already reported and, where said restatement policy so provides, recording said revised value as a restatement related to a preceding said value, said preceding value being retained, not marked erroneous, and remaining queryable as the value published at a publication time thereof; and recording said revised value under said supersession chain with said preceding value marked erroneous only where said revised value is issued as a correction.

[17] The method of claim 329, wherein a revision received under a said restatement policy not admitting it is withheld from admission pending a determination recorded against an identity of an individual human reviewer, no automated process classifying said revision as a restatement or as a correction.

[18] The method of claim 325, wherein said redistribution entitlement specifies one of a display, a derivation, a delayed publication and an unrestricted redistribution; wherein output text that would contain a said value beyond said redistribution entitlement is composed without said value and a

statement of said redistribution entitlement is returned; and wherein a value derived from a said value carries said redistribution entitlement of said live data source unless a said derivation is registered as discharging it.

[19] The method of claim 325, further comprising registering a standing query against said live data source, wherein said standing query is evaluated upon a publication of a value and not upon a passage of time, is not evaluated upon a value whose age exceeds said staleness bound, and is suspended upon exceeding a configured rate of evaluation with a notification to an owner thereof rather than continuing to evaluate. **Defined agents (claims 333–341)**

[20] The method of claim 1, further comprising generating a plurality of candidate hypotheses by sampling from a distribution in which a selection probability of a candidate is a logistic function of a scored difference between said candidate and an incumbent candidate scaled by an inverse-temperature parameter, wherein each said candidate hypothesis carries a status field set to a proposed state and transitions to a validated state only upon confirmation by an identified human reviewer, no automated process being configured to effect said transition.

[21] The method of claim 4, wherein said inverse-temperature parameter is exposed as an operator-configurable control having at least a first setting under which said sampling concentrates selection on highest-scoring candidates, a second setting under which said sampling broadens selection across candidates of comparable score, and a third setting under which said sampling approaches uniform selection irrespective of score; and wherein a value of said inverse-temperature parameter in effect for a given generation of candidate hypotheses is recorded in said provenance chain with an identity of the operator who set it, such that said generation is reproducible from said recorded value.

[22] The method of claim 43, wherein no value of said inverse-temperature parameter causes an output of said sampling to be admitted to said retrieving or to said generating of output text, and wherein no process of the system is configured to adjust said inverse-temperature parameter toward any value on the basis of the system's own prior outputs. **Concept inheritance, exception override and inverse traversal (claims 45–49)**

[23] The method of claim 1, further comprising: establishing an entitlement of a holder to a work under a licence record; minting a token representation of said entitlement only after said entitlement is so established; transferring said token representation to a wallet controlled by said holder, said token representation being at no time held by a party performing said method; and where a state of said token representation diverges from said entitlement record, presenting said entitlement record as authoritative and recording said divergence.

[24] The method of claim 1, further comprising delivering an audio narration of content by one of a first mechanism reproducing a recorded asset of an identified natural person and a second mechanism synthesizing speech from a voice profile having no derivation from, fitting to, or correspondence with any specific natural person, wherein said first mechanism requires said recorded asset of said identified natural person and no configuration of said second mechanism produces a voice corresponding to a specific natural person.

[25] The method of claim 1, further comprising: analysing a sensor feed under an explicit and revocable authorization of a requester in order to locate an object; classifying an object from a frame of said sensor feed; excluding from a candidate result set every classification result corresponding to a natural person or to an animal, such that no said classification result is surfaced to said requester at any ranking position; and representing a remaining said classification result as a session-scoped record not retained beyond an active session absent a separate explicit election to retain it.

[26] The method of claim 1, further comprising: determining that a confidence value of an inference derived from a plurality of said knowledge records falls below a configured threshold; identifying a

knowledge record that, were it present, would raise said confidence value above said threshold; and generating a structured query directed to obtaining said identified knowledge record and presenting said structured query to a human, rather than returning said inference at said confidence value.

[27] The method of claim 1, further comprising, in an interval between delivery of a response and receipt of a subsequent query, retrieving into a buffer a set of knowledge records predicted from said response and from knowledge records retrieved for preceding queries of a common interaction; serving said subsequent query from said buffer where said subsequent query matches said prediction; and discarding said buffer, without modification to any said knowledge record, where said subsequent query does not so match.

[28] The method of claim 1, further comprising indexing an object classified from a sensor feed by a deterministic mapping from a planar position of said object to a cell of a hexagonal tiling, such that two objects occupying adjacent cells are retrievable as spatial neighbours by cell adjacency alone without a distance computation over their coordinates, and applying an identical said mapping to a position in an embedding space to index conceptual neighbours.

[29] The method of claim 1, wherein said safety classification is a bitmask field of said record schema, is immutable to an application programming interface caller, and is modifiable only in a direction of increased restriction absent a dual-approval operation, such that no code path reads a said knowledge record without also reading said safety classification.

[30] The method of claim 1, wherein said provenance chain is a Merkle directed acyclic graph, said checksum is a keyed checksum, and a mismatch between said checksum and a checksum recomputed over said operation and said safety evaluation indicates that a safety evaluation recorded for said operation differs from a safety evaluation that executed.

[31] The method of claim 12, further comprising: maintaining a registry of externally-sourced linguistic resources consulted by said extraction stages, each registration recording an identifier and version of a resource, a licensor, a licence instrument, a permitted-use scope, a redistribution permission, an attribution requirement, and an ingestion mode; withholding consultation of a resource by any said extraction stage until said resource is registered; refusing registration where said permitted-use scope does not encompass an intended use declared by an operator; and recording each registration and each refusal as an entry in said provenance chain.

[32] The method of claim 54, further comprising associating with each artifact derived from a registered resource a reference to said registration, and enforcing, at each point at which said artifact would be redistributed or exposed, a derivative condition imposed by the licence instrument of said registration; and wherein, where two registered resources impose conflicting derivative conditions on a common artifact, said artifact is marked non-redistributable and said conflict is recorded, rather than being resolved by selection of a less restrictive condition.

[33] The method of claim 54, wherein a resource registered in a derived-artifact-only ingestion mode is consulted to compute an aggregate quantity over said resource, said aggregate quantity is retained, and no span of text of said resource is retained in a knowledge record, an index or a cache, or reproduced in an output; and further comprising verifying compliance by determining that a count of retained source spans attributable to a resource registered in said mode is zero, and surfacing a non-zero said count as a compliance failure.

[34] The method of claim 56, wherein said aggregate quantity comprises at least one of a term frequency, a collocation strength, a part-of-speech distribution and a morphological paradigm, and wherein retention of said aggregate quantity and non-retention of said span implement a distinction between a fact about said resource and an expression contained in said resource. **Session modes,

annotation, bounded programs, authoring, immersive presentation, infrastructure and governance (claims 58–107)**

[35] The method of claim 1, wherein said plurality of distinct modalities comprises unstructured text, structured data, image, audio, video, sensor data, knowledge graph and code, and wherein content not matching any said modality is classified as unmatched and surfaced, and is not discarded without record.

[36] The method of claim 13, further comprising: detecting that a property of a perceived event is confirmed by two or more of said plurality of distinct modalities within a common temporal window; and assigning a knowledge record constructed for said property a confidence value strictly greater than a confidence value assigned to said property when confirmed by a single said modality alone, wherein a magnitude of the increase is a function of a temporal synchrony between the confirming modality streams.

[37] The method of claim 1, wherein said semantic fingerprint is computed over said canonicalized form after Unicode normalization and independently of a natural language of said content, such that a first said content expressing said semantic assertion in a first natural language and a second said content expressing said semantic assertion in a second natural language yield an identical said semantic fingerprint and converge on a single said knowledge record having two branches of said provenance chain.

[38] The method of claim 1, wherein said threshold function is of the form $a_i(t+1) = \mathrm{sgn}(\Sigma_j W_{ij} \cdot a_j(t) - \theta_i)$, and wherein a number of distinct stable states said iterative updating is operated to hold is bounded at a fixed proportion of a number of knowledge records of said admitted set, such that convergence to a spurious mixed state is prevented by said bound rather than by filtering a result of said iterative updating.

[39] The method of claim 16, further comprising: maintaining a load ratio equal to a number of distinct stable configurations the iterative updating is operated to hold divided by a number of units; comparing, before admitting a further stable configuration, a resulting value of said load ratio against a configured operating ceiling, said operating ceiling being strictly below a retrieval-phase boundary at which stable retrieval configurations of said iterative updating cease to exist discontinuously; refusing said admitting where said resulting value would equal or exceed said operating ceiling; and initiating instead a consolidation of existing stable configurations, wherein said refusing is recorded as an entry in said provenance chain.

[40] The method of claim 40, wherein said configured operating ceiling is set within a first regime in which stable retrieval configurations are global minima of an energy function of said iterative updating, rather than within a second regime, lying between said first regime and said retrieval-phase boundary, in which stable retrieval configurations remain dynamically stable but are local minima while other configurations hold said global minima.

[41] The method of claim 41, wherein operation within said second regime requires a deployment election recorded with an identity of an operator who made it, and wherein absent said deployment election said configured operating ceiling remains within said first regime.

[42] The method of claim 1, wherein said query is a turn of a multi-turn interaction, and wherein a knowledge record retrieved at a prior turn of said multi-turn interaction is weighted for said retrieving by a factor decreasing monotonically with a number of intervening turns d in proportion to 1/(1+d), and wherein a knowledge record whose said factor falls below a configured floor is released from a working-memory structure without being deleted from a persistent store of said knowledge records.

[43] The method of claim 17, further comprising determining an eligibility of a commercial item from a right registered for said commercial item, a territory recorded for a request, an edition in which a

response is delivered and a device class of said request, and from no further input; wherein an inferred ergonomic signal, an inferred indicator of cognitive load and an inferred affective state are unavailable to said determining; and wherein a commercial region is omitted where said determining yields an empty set, an omission thereof not being disclosed.

[44] The method of claim 225, wherein a said commercial item occupies a registered placement class comprising a region distinct from and not interleaved with propositions of said output text, and a number of said commercial items presented within a response and within a session is bounded by a configured ceiling, a commercial item exceeding said ceiling being omitted rather than deferred.

[45] The method of claim 225, further comprising registering a said commercial item with a verified identity of a party presenting it, a right under which it is presented, a territory covered, an interval of validity and an identity of a registering operator, and refusing registration of a commercial item lacking any thereof; and recording each presentation in said provenance chain with said commercial item, said placement class, a concept identifier occasioning it and an eligibility relied upon.

[46] The method of claim 227, wherein a said commercial item is not admitted as a knowledge record, is not citable, is not traversable and contributes to no confidence value, and an assertion within a said commercial item is labelled as an assertion of said party presenting it.

[47] The method of claim 17, further comprising registering a commerce object with a right permitting its sale, a territory in which it may be sold and a price per territory and currency, and withholding an offer of a said commerce object in a territory for which no price is registered rather than offering it at a default price.

[48] The method of claim 229, further comprising performing upon a said commerce object an action selected from creating a perpetual entitlement, creating an entitlement bounded by an interval, creating a recurring entitlement contingent upon renewal, extending an existing entitlement, replacing an entitlement with a broader entitlement wherein a prior entitlement is superseded rather than deleted, and transferring an entitlement where a licence permits transfer; and producing for each said action a transaction record carrying an acting identity, said commerce object, said action, a price and currency applied, a territory, a rights validation relied upon and a reference into said provenance chain.

[49] The method of claim 230, wherein a transaction not reconcilable against a change of entitlement is refused rather than recorded in a pending state, such that an entitlement state and a transaction history do not diverge; and wherein a divergence between a state of a token representation and a said entitlement raises a finding, said entitlement remaining authoritative. **Directed capture equipment control under session authorization (claims 232–240)**

[50] The method of claim 1, wherein a knowledge record outside said scope description is absent from a traversal of said knowledge records rather than removed from a result of said traversal, such that an existence of said knowledge record is not inferable from a count, a ranking position or an elapsed time of said traversal.

[51] The method of claim 1, further comprising: representing a hypothetical state of said knowledge records as a delta object comprising a reference to a base state, a set of overriding knowledge records and a set of suppressed knowledge records, and not as a copy of said base state; assigning said hypothetical state a probability value; and removing said hypothetical state from evaluation, and discarding said delta object without modification to said base state, where said probability value falls below a configured pruning threshold.

[52] The method of claim 19, further comprising evaluating a single structured query across a plurality of said hypothetical states within one query operation, optionally restricted to a subset of said hypothetical states and optionally weighted by said probability value of each, and returning a result identifying, for each said hypothetical state, the knowledge records within it satisfying said query.

[53] The method of claim 1, further comprising: deriving a synthesized assertion from at least two said knowledge records, said synthesized assertion being stated by no single one of said at least two knowledge records; assigning said synthesized assertion a confidence value not exceeding a least confidence value among said at least two knowledge records; marking said synthesized assertion with a type identifier distinguishing it from an ingested said knowledge record; and delivering said synthesized assertion only with said type identifier preserved in said output text.

[54] The method of claim 21, further comprising organizing said knowledge records, a concept structure and said provenance chain as a single addressable structure over which said scope description is resolved, and analyzing a candidate set retrieved for a query to determine a distribution of types, confidences and source categories used to select a composition rule for said output text.

[55] The method of claim 1, further comprising evaluating a temporally-scoped rule set incrementally as a said knowledge record is appended, and maintaining a persistent index of contradictions in which an entry is created where two active rules of said rule set yield incompatible conclusions over a common said knowledge record, said index being queryable after delivery of said output text.

[56] The method of claim 22, further comprising reporting a documented editorial position of a source and a quantified composition measure of said source as cited knowledge records, and refraining from generating a conclusory characterization of said source not itself derivable from a said knowledge record.

[57] The method of claim 1, further comprising, where no knowledge record of said candidate set satisfies said configured threshold, delivering a statement of said boundary together with a nearest available said knowledge record, and not generating said output text.

[58] The method of claim 1, further comprising: identifying a candidate modification to a subsystem executing said method, by monitoring scoped to said subsystem or by a request received from a human operator; evaluating said candidate modification in an isolated simulation environment; applying said candidate modification only upon an approval recorded against an identity of an individual human reviewer together with a stated reasoning of said reviewer; and rejecting said candidate modification where said subsystem is a member of a fixed enumerated set of subsystems, wherein said fixed enumerated set is not modifiable by a candidate modification identified or applied according to this claim.

[59] The method of claim 24, further comprising generating a machine-checked proof of a property of said candidate modification where said candidate modification targets a bounded non-Turing-complete program representation, and presenting said proof to said individual human reviewer as an input to said approval, wherein said proof does not effect said approval.

[60] The method of claim 24, further comprising applying a platform configuration change through a declarative configuration language, said change being versioned and recorded in said provenance chain on the same terms as any other change; and pausing execution at a pipeline stage boundary to expose knowledge record bindings, safety evaluation results and compute-accounting measurements of an actual execution rather than of a simulation.

[61] The method of claim 1, further comprising metering each operation performed in said method as a bounded compute-accounting unit assigned to a category, and scheduling said compute-accounting units such that a unit assigned to a safety-evaluation category is served ahead of a unit of any other category whenever the two contend for a common processing resource, irrespective of an arrival order or a privilege of a requester.

[62] The method of claim 26, further comprising: completing said retrieving, said iteratively updating and said generating before invoking a selection of commercial content; supplying to said selection, as its sole input, a concept identifier resolved from said query together with identifiers of

concepts adjacent thereto within one edge of a concept structure, each supplied read-only, said selection holding no write access to said retrieving, said iteratively updating or said generating; scoring a matched commercial candidate as a weighted combination of a relevance factor, an operator-configurable source-priority factor and a bid factor, said weights summing to unity; scheduling a compute-accounting unit of said selection below every unit assigned to said safety-evaluation category; and rendering a selected commercial content in a labeled region distinct from and not interleaved with said output text.

[63] The method of claim 27, wherein said identifiers of concepts adjacent to said concept identifier comprise concepts connected thereto by a single edge of a type registered as adjacency-bearing, said types excluding a causal edge, a temporal edge and a contradiction edge; wherein no traversal is performed from a said adjacent concept such that a concept connected by two edges is absent from said input; and wherein said input is computed at a time of supply and is not extended by said selection.

[64] The method of claim 257, wherein a said adjacent concept is excluded from said input where it lies outside said scope description of a requester, and an exclusion thereof is not indicated, such that an existence of a concept outside an entitlement of said requester is not inferable from a commercial item presented to said requester.

[65] The method of claim 257, wherein a said adjacent concept participates in a relevance factor of a selection score and does not participate in a determination of eligibility, such that a commercial item ineligible for a request does not become eligible by a correspondence with a said adjacent concept; and wherein a relevance computed against a said adjacent concept is attenuated relative to a relevance computed against said concept identifier. **Convergence regime disclosure (claims 260–261)**

[66] The method of claim 26, further comprising deploying an approved change in successive stages rather than in a single operation, wherein progression to each successive stage is conditioned on an approval recorded against an identity of an individual human reviewer together with a stated reasoning of said reviewer.

[67] The method of claim 26, further comprising measuring a stated design-target figure by a reproducible benchmark expressed in said compute-accounting units, and retracting said stated figure where said measuring does not confirm it, wherein said retracting is recorded in said provenance chain. **Behavioural probing and analysis quarantine (claims 108–112)**

[68] The method of claim 26, further comprising executing a trial as a bounded compute-accounting unit carrying a trial category, allocating a budget of said trial category per interval, scheduling a unit of said trial category below a unit of a safety-evaluation category, below a unit serving a requester and below a unit of an ingestion category, and deferring a trial exceeding said budget rather than serving it at a reduced priority.

[69] The method of claim 262, wherein a trial executes against a corpus and a configuration isolated from a production corpus, constructs no knowledge record admitted thereto, adjusts no confidence value of a knowledge record thereof, emits no command to physical equipment outside a simulation, and produces no output text delivered to a requester.

[70] The method of claim 263, wherein a result of a trial is recorded as a trial record excluded from a scope description constructed for a query not directed to said trial record, and a finding of a trial applies no change, being emitted as a proposal taking effect only upon an approval recorded against an identity of an individual human reviewer irrespective of a number of trials from which said finding derives.

[71] The method of claim 262, further comprising enumerating a trial space for a layer, executing trial cases therefrom, and recording a coverage as a proportion of said trial space executed together with an enumeration of cases not executed.

[72] The method of claim 262, further comprising executing a trial set upon a change to a knowledge structure, a parameter, a registered resource or an implementation, and withholding application of said

change where a trial case passing before said change fails after it, said failing case being reported with both results.

[73] The method of claim 262, further comprising supplying to a trial an input constructed to contradict a further input, an input constructed to drive said iteratively updating toward a configuration other than a stored configuration, and an input constructed to approach a constraint boundary of an equipment unit; and recording for each a proportion of cases in which a specified refusal was effected, wherein a proportion below unity is recorded as a defect.

[74] The method of claim 262, further comprising measuring a determinism as a proportion of repeated trials over an identical state of a corpus and an identical configuration yielding an identical semantic fingerprint, wherein a measurement over a corpus whose state changed between said repeated trials is not a said measurement of determinism.

[75] The method of claim 262, wherein a result of a trial concerning an equipment unit does not constitute a clearance of said equipment unit for operation, a clearance being established by a hazard analysis and a verification performed upon said equipment unit in a configuration thereof; and wherein upon a disagreement between a trial result and a result of a physical trial, said result of said physical trial governs and said disagreement is recorded as a finding concerning a simulation. **Grasping and manipulation (claims 270–276)**

[76] The method of claim 1, wherein said generating is performed in a selected one of a plurality of session modes, each said session mode applying a distinct composition rule to a common said stabilized set, and wherein no said session mode admits into said output text a proposition not derivable from said stabilized set.

[77] The method of claim 58, wherein in a first said session mode a candidate set retrieved at each turn is conditioned on knowledge records accumulated across preceding turns of a common interaction, such that a reference resolved at a preceding turn remains resolved without restatement.

[78] The method of claim 58, wherein in a second said session mode a viewpoint attested within ingested content is adopted for composition, said viewpoint being marked as attributed to said content and not as a position of the system, and wherein a proposition supporting said viewpoint that is not derivable from said stabilized set is rejected.

[79] The method of claim 58, wherein in a third said session mode said stabilized set is ordered into a sequence of instructional steps by a dependency relation among said knowledge records, each said step citing the knowledge record from which it was composed.

[80] The method of claim 58, further comprising generating an assessment item from a target knowledge record of said stabilized set, receiving a submitted response, and scoring said submitted response by a semantic similarity of said response to the content of said target knowledge record against a configured threshold, wherein said assessment item and a lesson delivered in another said session mode are generated from a common set of said knowledge records.

[81] The method of claim 58, wherein in a fourth said session mode said stabilized set is ordered by a temporal anchor field of each said knowledge record, and a knowledge record lacking a temporal anchor is reported as unordered rather than assigned a position.

[82] The method of claim 58, further comprising deriving from a single said knowledge record a prompt portion and an answer portion of a study item, said answer portion being the content of said knowledge record rather than a restatement of it, and scheduling a subsequent presentation of said study item at an interval computed as a prior interval scaled by an ease factor where a recorded recall quality satisfies a configured floor.

[83] The method of claim 1, further comprising comparing a first set of knowledge records associated with a first edition of a work against a second set associated with a second edition by traversing a supersession chain of each said knowledge record, and reporting each compared element as changed or unchanged together with the knowledge records constituting each said edition.

[84] The method of claim 1, further comprising storing reader-authored content as a knowledge record whose access control list determines its visibility, wherein a first said access control list restricts visibility to the authoring reader, a second is set from a sharing configuration declared by an author of the work, and a third is scoped to a class roster held by an instructor, and wherein a reader-authored said knowledge record is distinguished by type from a knowledge record extracted from said work and is not admitted to said stabilized set as content of said work.

[85] The method of claim 66, further comprising routing a submitted post into one of a plurality of discussion partitions determined by a role held by the submitting identity and by a scope of a work or a class associated with said partition, and applying to said post the same safety evaluation applied to a said knowledge record.

[86] The method of claim 1, further comprising condensing an interaction session into a session archive record retained by default at session end, and providing an erasure operation exercisable by the requester at a session scope, a work scope and an account scope, wherein exercise of said erasure operation at said account scope where a reader-held encryption key governs said records is effected by destruction of said key record.

[87] The method of claim 68, further comprising clustering a plurality of said session archive records of a common requester by proximity of an aggregate semantic vector of each, and reporting a concept appearing in session archive records associated with two or more distinct works as a cross-work connection cited to the knowledge records in which said concept appears.

[88] The method of claim 68, further comprising clearing, on an explicit instruction received during an active session, an accumulated query-and-response record of said session immediately and without producing a condensed retained form, and recording said clearing in said provenance chain without retaining any portion of the cleared content.

[89] The method of claim 1, further comprising rendering a citation of said output text in a selected one of a plurality of presentation forms comprising a graph rendering of knowledge records and edges within a scope, a preview of a cited knowledge record surfaced without leaving a current view, and an inline dismissible annotation, wherein each said presentation form resolves to the same knowledge record identified by said citation and none introduces a proposition absent from said stabilized set.

[90] The method of claim 1, further comprising compiling a natural-language instruction into a bounded program by decomposing said instruction into operational intents before binding any intent to a concrete entity, binding each said intent to a concrete type and predicate reference by an entity resolution common to said extracting, and halting said compiling and returning a clarification request where a said binding is ambiguous rather than selecting among candidate bindings.

[91] The method of claim 71, wherein said bounded program is expressed in a primitive set comprising a fixed number of read primitives, a single state-changing primitive subject to said safety evaluation, and a single iteration primitive enforcing a runtime iteration ceiling, such that said bounded program is not Turing-complete in an unbounded sense and its termination is enforced by said runtime rather than by convention.

[92] The method of claim 71, further comprising matching a plain-language request against a library of application templates by task and domain and instantiating a matched template scoped to a set of said knowledge records, and constructing a compilation of a requester's works as a delta reference over said works without copying an underlying knowledge record.

[93] The method of claim 1, wherein said knowledge records are maintained in a committed layer and an in-memory overlay reconciled on a fixed interval, a knowledge record in said overlay carrying a provisional reference into said provenance chain that is finalized and never altered on completion of a durable write, and wherein said retrieving reads said committed layer overlaid with matching entries of said overlay.

[94] The method of claim 73, further comprising registering a said bounded program with a trigger condition, evaluating said trigger condition against each change to said overlay, and re-applying said safety evaluation to said bounded program on each firing rather than only at registration.

[95] The method of claim 1, further comprising accumulating knowledge records incrementally as a draft is edited, versioning each edit through said supersession chain, and detecting a concurrent-write divergence between two authors by comparing a head of said supersession chain held by each, wherein divergence is reported for resolution and neither write is discarded.

[96] The method of claim 75, further comprising evaluating a draft's own knowledge records against one another under said rule set and reporting a continuity defect where two said knowledge records of said draft yield incompatible conclusions concerning a common subject.

[97] The method of claim 75, further comprising constructing from a current state of said draft a hypothetical-state branch represented as a delta object referencing a base state and retaining both said current state and said branch, such that exploration of an alternative does not modify said draft.

[98] The method of claim 1, further comprising accepting a submitted work into a publication workflow, validating a right to distribute said work before any said extracting is performed, and withholding said extracting until said validating returns a cleared status.

[99] The method of claim 1, further comprising serializing every knowledge record to which a requesting identity holds a right into an open interchange format on request, wherein said serializing is not conditioned on a continuing entitlement and no access to said records is withdrawn by reason of said serializing.

[100] The method of claim 1, further comprising generating an accessible-equivalent knowledge record from the same source content and at the same production stage as the knowledge record it accompanies, and reporting a production as incomplete where a said accessible-equivalent record is absent.

[101] The method of claim 1, further comprising inferring a colour value for a region of visual content lacking reliable colour information from a material and lighting classification of said region recorded as a knowledge record, wherein said inferring is deterministic and each inferred colour is traceable to the knowledge record from which it was inferred.

[102] The method of claim 81, further comprising representing a reference colour as a knowledge record carrying a declared tolerance, and applying a fixed rule set to geometry and constraint knowledge records to produce a geometric conclusion together with a citation of each said record relied upon.

[103] The method of claim 1, further comprising instantiating a three-dimensional rendering of a scene from knowledge records carrying spatial detail, and withholding said instantiating where the spatial detail carried by said knowledge records does not satisfy a configured sufficiency threshold, rather than supplying absent spatial detail.

[104] The method of claim 1, further comprising emitting a device output on a sensory channel only in response to an explicit user action tied to a sensory profile authored for specific content, wherein no said emitting is triggered by a state inferred by the system.

[105] The method of claim 1, further comprising routing a knowledge record between subsystems by its type and by a subscription declared by a receiving subsystem, and serializing a said knowledge record

to a binary frame format permitting resynchronization to a frame boundary without reparsing a preceding frame.

[106] The method of claim 1, further comprising assigning each knowledge record to one of a plurality of storage tiers by an access recency and frequency of said record, and reconciling a divergence between an offline replica and an online state deterministically on reconnection, wherein a reconciliation outcome is a function of said divergence alone and not of an order in which replicas reconnect.

[107] The method of claim 1, wherein said scope description is resolved from a union of roles held by a requesting identity together with a time-scoped grant, and further comprising verifying against a seat count and an entitlement associated with an organizational licence at a time of executing a query rather than at a time of authenticating said identity.

[108] The method of claim 1, wherein said method is performed identically in each of a plurality of deployment modes comprising a multi-tenant hosted mode, a single-tenant local mode and a single-device mode, and wherein output is rendered by a reading system conformant to a published electronic publication standard without alteration of said constraint on said permissible output vocabulary.

[109] The method of claim 1, further comprising persisting an abstraction derived from a plurality of said knowledge records in a store distinct from said knowledge records, said store retaining a generalized concept and discarding the specific episode from which it was derived, and reducing raw perceptual input to a concept-level record before admission to said store.

[110] The method of claim 91, further comprising receiving sensor data from an autonomous agent, admitting said data through said extracting to produce knowledge records, and returning structured knowledge to said agent, wherein no raw sensory episode from said agent is retained in said store.

[111] The method of claim 91, further comprising constructing a first concept representation from perceptual input independently of language and a second concept representation from ingested language, and fusing said representations into a single concept on a co-occurrence of a label and a perceived object confirmed by a semantic consistency check, wherein a single said co-occurrence produces a provisional binding and a concept is finalized only after a configured number of consolidating observations.

[112] The method of claim 93, further comprising assigning each object classified from a common frame a salience value normalized against a sum of the salience values of the objects classified from said frame, wherein said normalizing governs presentation order only and every classified object remains an addressable record.

[113] The method of claim 93, further comprising estimating a camera ego-motion from an aggregate optical-flow field and subtracting said ego-motion from a region motion vector before segmentation; elevating a confidence of a property confirmed across concurrent modalities within a common temporal window; and maintaining continuity of a single object record across a full occlusion of said object within a session.

[114] The method of claim 93, wherein said fusing further applies, as evidence-overridable signals, an inference from an observable gaze direction of a speaker, a preference for interpreting a label as naming a whole object rather than a part or property, and a preference for binding a new label to an object not already carrying a bound label.

[115] The method of claim 1, further comprising applying a moderation action to target content and recording said action with an identity of the acting moderator and a stated reason, and accepting from an affected identity an appeal referencing the provenance chain entry of said action, said appeal being evaluated by a reviewer other than said acting moderator.

[116] The method of claim 1, further comprising extracting a prosodic feature from live audio only where a session-scoped consent is active, supplying said feature as an optional trigger to a clarification request, and discarding said feature at session end absent a separate election to retain it.

[117] The method of claim 1, further comprising extracting a discrete factual assertion from a reference work as a knowledge record while withholding reproduction of the reference work's own expression, and processing an image accompanying said reference work under a separate rights validation rather than under said extracting.

[118] The method of claim 1, further comprising storing an instructor-selected structural parameter set as a reusable assessment template applied to knowledge records of a work, and exposing to a grantee a read-only view of each active grant held over said grantee, wherein said view confers no capability to create or modify a grant.

[119] The method of claim 1, further comprising constructing a goal or motivation record carrying a mandatory reference to a documented subject, wherein no said record is constructed concerning an internal state of the system performing the method; and selecting a vocabulary, structural format and explanatory depth for said output text from an audience profile, wherein a knowledge record relied upon is not altered by said selecting.

[120] The method of claim 1, further comprising dispatching a scheduling or invitation operation to an external platform registered with a capability declared by said external platform, and aggregating usage and distribution records for a work into a report, wherein neither operation admits a knowledge record bypassing said safety evaluation.

[121] The method of claim 1, further comprising: maintaining a catalogue of event type definitions, each said event type definition comprising an identifier immutable from registration, a version, a placement resolving against a plurality of independent taxonomic hierarchies of a concept structure, a prose definition stating both occurrences covered and occurrences excluded, and a machine definition specifying for said event type which fields of a knowledge record are required, which are optional and which are forbidden; requiring a knowledge record encoding an occurrence to reference an identifier of said catalogue rather than a free-text category; and refusing construction of a knowledge record that does not satisfy the machine definition of the event type it references, said refusing occurring at construction rather than by marking a constructed record.

[122] The method of claim 119, wherein a knowledge record constructed under a first version of an event type definition remains valid under said first version and is not invalidated by registration of a subsequent version, and wherein a required-field set of an event type definition may be narrowed by a subsequent version and may not be widened absent registration of a distinct event type.

[123] The method of claim 1, further comprising: constructing an environment record representing a state relative to which a plurality of said knowledge records hold, said environment record comprising an identifier derived from a hash of a canonicalized set of conditions, a temporal interval, a set of typed conditions whose keys resolve against a concept structure, and a confidence value and a reference into said provenance chain of its own; referencing said environment record from a knowledge record through an annotation of said knowledge record rather than through a field of a schema of said knowledge record; and converging two constructions yielding an identical said canonicalized set of conditions on a single said environment record having two branches of said provenance chain.

[124] The method of claim 121, further comprising superseding said environment record through a supersession chain, wherein every knowledge record referencing said environment record resolves to a superseding environment record on a subsequent traversal without modification of said knowledge records.

[125] The method of claim 1, further comprising admitting an edge asserting that a first knowledge record describes a causal factor for an occurrence described by a second knowledge record only where a temporal anchor of said first knowledge record does not begin after a temporal anchor of said second knowledge record begins, refusing said edge otherwise, and recording a refused said edge in a persistent contradiction index; and wherein an edge asserting temporal ordering between two knowledge records is not converted into a said causal edge by any automated process.

[126] The method of claim 123, wherein a causal chain is expressed as a membership of a plurality of said knowledge records in a common bundle together with said causal edges among said knowledge records, and wherein a length of said chain, a root of said chain and an outcome of said chain are computed from said membership and said edges at a time of query rather than stored as fields of a chain record. **Confidence as a Beta posterior (claims 125–129)**

[127] The method of claim 1, further comprising, on receipt of a further knowledge record bearing on said semantic assertion, incrementing one of said first and second parameters by a weight derived from a source reliability of said further knowledge record, such that a magnitude of change in said reported confidence value decreases as a sum of said first and second parameters increases without a rule conditioned on said sum; and admitting said weight as a fractional quantity rather than as an integer count.

[128] The method of claim 125, further comprising initialising said first and second parameters for a concept node from a stability weight of said concept node, wherein said stability weight determines a sum of said parameters and said confidence value determines a ratio between them, such that a concept node having a greater stability weight requires a greater quantity of contradicting evidence to effect an equal change in said reported confidence value.

[129] The method of claim 125, further comprising computing a credible interval for said reported confidence value from said first and second parameters, wherein two knowledge records having an identical said reported confidence value and differing sums of said parameters yield differing said credible intervals.

[130] The method of claim 125, further comprising applying a temporal decay to said first and second parameters toward respective initial values at a rate configured per record type, such that an elapsed interval widens a credible interval computed from said first and second parameters while leaving said reported confidence value substantially unchanged, and wherein a record type designated immutable is assigned a decay rate of zero.

[131] The method of claim 125, wherein said incrementing is performed by addition in closed form without sampling from a probability distribution, such that a given set of said further knowledge records yields an identical said reported confidence value irrespective of an order in which said further knowledge records are received. **Precision-weighted error propagation (claims 130–133)**

[132] The method of claim 125, further comprising: determining that a first knowledge record contradicts an expectation generated by an attribute of a concept node; computing a first dispersion from the first and second parameters of said first knowledge record and a second dispersion from the parameters of said attribute; computing a revision gain as a ratio of said second dispersion to a sum of said first and second dispersions; and revising said attribute by said revision gain multiplied by a configured maximum revision, such that a contradicting knowledge record resting on less accumulated evidence effects a smaller revision of said concept node than one resting on more, irrespective of a difference between their reported confidence values.

[133] The method of claim 130, wherein two knowledge records having an identical said reported confidence value and differing sums of said parameters yield differing said revision gains against a common attribute.

[134] The method of claim 130, further comprising withholding revision of said attribute where said revision gain falls below a configured floor, and recording each withheld contradiction in a persistent contradiction index such that a plurality of individually-withheld contradictions of a common attribute remains queryable.

[135] The method of claim 130, wherein said revision gain is computed in closed form from stored parameters without sampling from a probability distribution, and wherein said revision gain governs revision of said concept node and does not govern a permissible output vocabulary of said generating, such that a proposition not derivable from an admitted knowledge record remains unavailable to said generating irrespective of any value of said revision gain. **Physical constraint enforcement in scene completion (claims 134–138)**

[136] The method of claim 130, further comprising maintaining a plurality of hierarchically ordered expectation levels comprising a first level predicting an element within a knowledge record under construction, a second level predicting a knowledge record given a sequence of knowledge records, a third level predicting an attribute of an instance given a concept thereof, and a fourth level predicting a regularity of a model of a world, wherein a said level predicts within a scope thereof and does not predict a subject of a level above it.

[137] The method of claim 284, further comprising computing for an observation a residual at each said level, attributing said observation to a lowest said level whose said residual constitutes a share of a total of said residuals satisfying a configured share, and applying a revision at said lowest level alone such that a level above said lowest level receives no revision.

[138] The method of claim 285, wherein a said level is attributed to only where a residual of each level below it fails to satisfy said configured share, such that a revision of a said regularity of a model of a world occurs only where a residual at each of said first, second and third levels falls below said configured share.

[139] The method of claim 285, wherein where a residual at no said level satisfies said configured share, said observation is recorded in a persistent contradiction index and no revision is applied at any said level; and wherein a plurality of said observations bearing on a common expectation is surfaced as a pattern.

[140] The method of claim 285, wherein a magnitude of a said revision at a said lowest level is computed from a precision of said observation and a precision of an expectation of said lowest level, such that a level to which a said observation is attributed and a magnitude by which said level is revised are determined independently.

[141] The method of claim 284, wherein no said level revises a constraint registered as governing a physical relation, an observation apparently contradicting a said constraint being recorded in a persistent contradiction index rather than relaxing said constraint. **Differentiated propagation and tri-temporal scope (claims 290–295)**

[142] The method of claim 125, further comprising attributing each increment of said first and second parameters to a source class of a plurality of registered source classes, and computing for a said knowledge record a diversity measure as a reciprocal of a sum of squared proportions of weight contributed by each said source class, such that two knowledge records having identical said first and second parameters and differing distributions of contributed weight across said source classes yield differing said diversity measures.

[143] The method of claim 193, wherein a further knowledge record attributed to a source class already contributing a proportion of weight to said first parameter is admitted at a weight reduced as a function of said proportion, such that repeated corroboration within one said source class does not increment said first parameter as a corroboration from a further said source class would.

[144] The method of claim 193, further comprising conditioning a proposal of a generalization from a plurality of subordinate concepts upon said diversity measure of said plurality satisfying a configured floor in addition to a count of said plurality satisfying a count floor; and attenuating a revision gain applied to a concept attribute where said diversity measure of a contradicting knowledge record approaches unity.

[145] The method of claim 193, further comprising indicating, in an output relying upon a knowledge record whose said diversity measure is below a configured threshold, that support for said knowledge record is concentrated within a single said source class. **The interpretive layer — context, perspective, ergonomic, cultural and social modulation (claims 197–204)**

[146] The method of claim 1, further comprising: maintaining a registry of physical constraints, each said physical constraint comprising a residual expression over quantities of a scene description, a tolerance, a unit system, an applicability predicate, and a reference into said provenance chain identifying a source establishing said physical constraint; generating from a partial scene description a candidate completion in which a field absent from said partial scene description is supplied; evaluating each said physical constraint whose applicability predicate is satisfied by said candidate completion by computing said residual expression over said candidate completion; refusing said candidate completion where a computed said residual exceeds a corresponding said tolerance; and regenerating a further candidate completion in which a field implicated in said refusing is left unspecified, such that a completion carrying an unspecified field is returned in preference to a completion exceeding said tolerance.

[147] The method of claim 134, wherein a refused said candidate completion is excluded from a ranked output irrespective of a plausibility score computable for it, and wherein said evaluating of said physical constraints is performed before computation of said plausibility score.

[148] The method of claim 134, further comprising recording a refusal as a queryable record identifying said physical constraint, said computed residual, said tolerance and a field implicated, such that a proportion of candidate completions refused by a given said physical constraint within a domain is determinable.

[149] The method of claim 134, wherein a said physical constraint whose applicability predicate is not satisfied by a said candidate completion is not evaluated against said candidate completion and said non-evaluation is recorded; and wherein a confidence value of a said physical constraint is maintained as a first parameter accumulating supporting evidence and a second parameter accumulating contradicting evidence and said physical constraint is suspended from said evaluating, without deletion, where said confidence value falls below a configured floor.

[150] The method of claim 134, wherein an ingested knowledge record whose content is inconsistent with a said physical constraint is admitted and recorded in a persistent contradiction index rather than refused, such that said refusing applies to a generated said candidate completion and does not apply to an observation. **Surfaces, extensibility and external integration (claims 139–156)**

[151] The method of claim 1, further comprising exposing a plurality of programmatic interfaces to an external caller, wherein each said interface resolves a scope description for said external caller before performing a traversal of said knowledge records, no parameter of any said interface widens a said scope description, a request for a knowledge record outside said scope description yields a response identical to that for a non-existent knowledge record, and each said interface constitutes an operation layer carrying a synchronous safety evaluation gate.

[152] The method of claim 139, wherein an interface reporting an entitlement performs no operation granting, modifying or extending an entitlement, and wherein an interface accepting an input admits said input through said extracting with said safety evaluation and accepts no knowledge record directly.

[153] The method of claim 139, further comprising metering each call to a said interface as a bounded compute-accounting unit assigned to a calling identity, and refusing a call exceeding a budget configured for said calling identity rather than serving said call at a reduced scheduling priority.

[154] The method of claim 139, further comprising registering a plugin component against a declared capability set, wherein said plugin component accesses said knowledge records only through a said interface and within a scope description of an identity on whose behalf said plugin component executes, holds no direct access to a store of said knowledge records or to said provenance chain, and is refused at registration where said declared capability set includes an operation registering, modifying or suspending a said safety evaluation.

[155] The method of claim 142, wherein output originating from said plugin component is rendered in a region distinct from said output text and labelled with an identity of said plugin component, is not admitted to a store of said knowledge records, and is not citable as a grounding source; and wherein activation of said plugin component requires an approval recorded against an identity of an individual human operator and is suspensible by said operator without cooperation of said plugin component.

[156] The method of claim 142, further comprising connecting an external learning management system under a declared capability set, wherein an operation not within said declared capability set is refused at a boundary of said connecting rather than attempted; a roster read establishes a scope of an instructor grant and does not confer an entitlement to a work; and a transmission of an outcome to said external learning management system comprises no knowledge record, no response text and no interaction history, and is recorded with an identity of a connector, a capability exercised and fields transmitted.

[157] The method of claim 139, wherein each said interface carries a version, a request omitting a version is refused rather than assigned a default version, a version is deprecated for a configured support interval and thereafter retired with a request for it refused and a successor named, and a knowledge record returned under one said version remains valid under a subsequent said version such that a citation issued under an earlier version is not invalidated.

[158] The method of claim 139, wherein said plurality of programmatic interfaces further comprises an interface returning a transformation of a media asset and an interface returning a scene description assembled from knowledge records carrying spatial detail, wherein said transformation is deterministic for a given asset, parameter set and content state, a value inferred in said transformation is traceable to a knowledge record from which it was inferred, and said interface returning a scene description withholds a scene where spatial detail carried by said knowledge records does not satisfy a configured sufficiency threshold.

[159] The method of claim 1, further comprising establishing a session shared by a plurality of participants over a common work, wherein a shared view of said session comprises knowledge records traversable within an intersection of the scope descriptions of every said participant, a knowledge record traversable by a first said participant and not by a second is absent from said shared view, and an absence of said knowledge record is not reported to any said participant.

[160] The method of claim 145, wherein a response to a query submitted in said session is resolved against said intersection and delivered identically to every said participant, and wherein a reading position, an interaction history and a private annotation of a said participant remain outside said shared view absent an explicit contribution by said participant.

[161] The method of claim 1, further comprising anchoring a displayed knowledge record together with its citation to a physical object identified in a live camera feed, wherein a classification result corresponding to a natural person or to an animal is excluded from a candidate set and no said anchoring is performed to such a result at any confidence value, and wherein no gaze direction, dwell duration, head

pose or attention signal is captured, derived or retained, and no such signal is made available to a selection of commercial content or to a plugin component.

[162] The method of claim 147, wherein where an object of said camera feed is not identified at a confidence value satisfying a configured threshold no said anchoring is performed and a boundary is stated, such that no said anchoring is performed to an unidentified object.

[163] The method of claim 1, further comprising receiving spoken input under an explicit session-scoped and revocable consent, transcribing said spoken input, routing said transcription through a classification separating a query intent from an operational intent, and requesting a clarification where said classification does not satisfy a configured confidence threshold rather than selecting an interpretation; and wherein audio of said spoken input is discarded following said transcribing absent a separate explicit election, and no voiceprint, speaker embedding or biometric identifier is derived therefrom, an identity of a speaker being taken from an authenticated session.

[164] The method of claim 149, wherein a said spoken input classified as carrying a state-changing effect comprising an erasure, a rights modification, a purchase or a grant requires a confirmation received through a second modality before said effect takes place.

[165] The method of claim 1, further comprising recording a progress marker for a reader over a work, associating with each knowledge record extracted from said work a narrative position at which its content is disclosed within said work, and bounding a scope description constructed for a query of said reader over said work by said progress marker such that a knowledge record whose narrative position exceeds said progress marker is not traversable; and wherein where a query is unanswerable within said bounded scope description the system states that an answer lies beyond said progress marker without disclosing content of said knowledge record.

[166] The method of claim 1, further comprising recording, for a learner, a position reached, assessment items attempted, responses submitted and outcomes scored against the knowledge records from which said items were generated, and computing a mastery measure as a proportion of items attributable to a stated objective answered correctly against a configured threshold; wherein visibility of said records to an instructor is bounded by a time-scoped grant such that expiry of said grant removes said visibility without a further operation, and wherein no inference of aptitude, of likelihood of future performance, or of comparative ranking against another learner is generated or stored from said records.

[167] The method of claim 152, further comprising issuing a credential to an identified learner upon both a measured outcome satisfying a configured threshold and an establishment of an identity of said learner to a configured standard, said credential comprising a record of objectives assessed, items presented, the knowledge records from which said items were generated, thresholds applied and an issuing identity; verifying said credential to a third party as issued and unrevoked without disclosing a response of said learner; and revoking said credential by a recorded revocation rather than by deletion.

[168] The method of claim 1, further comprising composing a work into a paginated print-ready form for a specified edition, wherein editions differing in audience level, language or annotation density are composed from common said knowledge records through said generating; a rights validation permitting print reproduction in a requested territory, quantity and interval is performed before said composing and a composed form carries a reference to said rights validation; a wording-operative span is reproduced verbatim in said print-ready form; and said composing is deterministic such that two compositions over an unchanged work and layout specification produce identical pagination.

[169] The method of claim 1, further comprising exposing a portal surface over a licence and entitlement mechanism, wherein a capability available at said portal to a role is the capability available to said role through any other surface such that said portal confers no capability; a grantee views a grant held over said grantee without a capability to create or modify a grant; each action taken through said portal is

recorded with an acting identity, an action and a stated reason; and a revocation effected through said portal takes effect at a time of executing a subsequent query rather than at a commencement of a subsequent session. **Hyperedge attention and attractor-based hyperedge completion (claims 157–159)**

[170] The method of claim 1, wherein a plurality of said knowledge records are linked by a hyperedge relating more than two of said knowledge records, and further comprising, upon a reasoning operation traversing said hyperedge, weighting a contribution of each constituent knowledge record of said hyperedge by a product of a confidence value of said constituent knowledge record and an accumulated evidence count of said constituent knowledge record, normalized across the constituents of said hyperedge, such that a constituent resting on less accumulated evidence contributes less than one resting on more irrespective of an equality of their confidence values.

[171] The method of claim 157, further comprising completing a partially-observed said hyperedge by applying said iterative updating of activation values to a constituent set of said hyperedge as a whole, weighted according to claim 157, until said activation values reach a stable state identifying a complete configuration of said hyperedge consistent with its observed constituents; and withholding said completing, and reporting an unobserved constituent as absent, where a weighted support for said complete configuration does not satisfy a configured threshold.

[172] The method of claim 158, wherein said completing operates upon a constituent set of a single said hyperedge and reaches said stable state through said iterative updating, and is distinct from a completion of a scene description performed by application of schema, physics, affordance and relation rules, said completing introducing no convergence behaviour separate from said iterative updating. **Domain architecture, transfer matrix and path separation (claims 160–168)**

[173] The method of claim 1, wherein the method is performed by a system partitioned into a plurality of domains each owning a bounded set of operations, said plurality comprising a governance domain and a plurality of functional domains, wherein no operation of any said functional domain executes without traversing an enforcement point of said governance domain, said governance domain is not a valid target of a modification proposed by any said functional domain, and a grouping of said functional domains into layers confers no authority such that a transfer permission between two said domains is unaffected by whether said domains occupy a common said layer.

[174] The method of claim 160, further comprising maintaining a transfer matrix assigning to each ordered pair of said domains a permission selected from permitted, denied, and permitted-subject-to-a-stated-conversion; evaluating said permission at each transfer of content between two said domains; treating an ordered pair having no entry in said transfer matrix as denied; and recording each said transfer in said provenance chain together with a source domain, a target domain and a permission applied; wherein said transfer matrix is owned by said governance domain and is a member of a fixed enumerated set of subsystems excluded from modification, such that a domain constrained by an entry of said transfer matrix does not administer that entry.

[175] The method of claim 161, wherein said transfer matrix denies a transfer from a domain owning evaluation of an output of an external system to a domain owning said retrieving, such that neither said output nor an analysis thereof is admissible to said admitted set, excepting a knowledge record converted from a reply elicited by a prompt composed by the method and carrying a derivation marking; and permits a transfer from said domain owning evaluation to a domain owning storage of said knowledge records only subject to a conversion marking transferred content as an analysis record excluded from a scope description constructed for a query not directed to said analysis.

[176] The method of claim 161, wherein said transfer matrix denies a transfer from a domain owning selection of commercial content to a domain owning said retrieving and to a domain owning said generating, and permits a transfer to said domain owning selection only subject to a conversion supplying

a concept identifier and identifiers of concepts adjacent thereto within one edge, read-only, after said retrieving and said generating have completed.

[177] The method of claim 161, wherein said transfer matrix permits a transfer from a domain owning said extracting to a domain owning storage of said knowledge records only subject to a conversion applying said safety evaluation, a validation against a machine definition of an event type, and a confidence calibration, said conversion constituting the sole route by which content originating outside the system enters said storage.

[178] The method of claim 161, wherein said transfer matrix denies a direct transfer between a domain owning a presentation surface and a domain owning storage of said knowledge records, such that a presentation surface obtains content only through a domain owning construction of said scope description and a domain owning said retrieving.

[179] The method of claim 160, wherein a first subset of said functional domains is traversed in answering a query and a second subset is traversed in ingesting content, and an intersection of said first subset and said second subset consists of a domain owning storage of said knowledge records and said governance domain alone, such that an ingestion operation contends with a query operation at no other domain.

[180] The method of claim 166, wherein contention at said governance domain is resolved by serving a compute-accounting unit of a safety-evaluation category ahead of a unit of any other category, and contention at said domain owning storage is resolved by assignment of a record to a storage tier by access recency and frequency.

[181] The method of claim 160, further comprising metering an operation of each said domain as a bounded compute-accounting unit carrying an identifier of said domain, fixing a scheduling priority by said identifier such that said governance domain holds a highest said priority and a domain owning selection of commercial content holds a lowest, and containing a failure of a said domain to a capability of that domain, wherein a failure of said governance domain halts an operation rather than permitting it to proceed unevaluated; and wherein said scheduling priority and an invariant bounding an intersection of a domain subset traversed in answering a query with a domain subset traversed in ingesting content are likewise owned by said governance domain and excluded from modification. **Discriminating instance generation (claims 169–173)**

[182] The method of claim 160, wherein a first subset of said domains executes in a first infrastructure under control of a party holding said knowledge records and a second subset executes in a second infrastructure, and wherein said scope description is constructed within said first subset and supplied to said second subset, such that no domain of said second subset constructs a said scope description and no said knowledge record is transferred to said second infrastructure by reason of an asset referenced by said knowledge record residing therein.

[183] The method of claim 160, further comprising maintaining at a cache node a subset of said knowledge records selected by access frequency and recency, answering a query at said cache node only where a scope description of a requester admits every cached knowledge record consulted, escalating said query to an origin otherwise, and invalidating a cached said knowledge record upon a supersession thereof; wherein said cache node performs no said extracting and constructs no said scope description.

[184] The method of claim 160, further comprising replacing an implementation of a said domain with a further implementation upon said further implementation satisfying each interface exposed by said domain at each version within a support interval thereof and each entry of said transfer matrix in which said domain appears, and upon an approval recorded against an identity of an individual human reviewer; and refusing said replacing where said domain is a member of a fixed enumerated set of subsystems excluded from modification.

[185] The method of claim 160, further comprising a domain owning generation of an assessment item from a knowledge record, a recording of a mastery measure over a stated objective, and an issuance of a credential, wherein a transfer from said domain to a domain owning selection of commercial content is denied by said transfer matrix and a transfer from said domain to a domain owning external integration is permitted only subject to a conversion transmitting an outcome without a knowledge record, a response text or an interaction history.

[186] The method of claim 160, wherein each said functional domain declares at least one module constituting a unit of replacement, and a said governance domain declares none, such that no implementation of said governance domain is exchangeable under a replacement conditioned on interface and transfer-matrix conformance and a recorded human approval.

[187] The method of claim 160, wherein a domain owning storage of said knowledge records declares no module exchangeable under a replacement conditioned on interface and transfer-matrix conformance, an exchange of an implementation thereof requiring a migration of stored said knowledge records in addition to a conformance with an interface and with said transfer matrix. **Restored coverage: independent curation, manifold forecasting, colour drift registration, message-channel encryption (claims 186–189)**

[188] The method of claim 1, further comprising presenting simultaneously a first realization of a work in a first language and a second realization of said work in a second language, each generated from common said knowledge records, wherein said realizations are aligned by knowledge record rather than by position such that a citation taken in either resolves to a common knowledge record, and wherein a knowledge record having no realization in one of said languages is presented as a gap rather than supplied from the other realization.

[189] The method of claim 1, further comprising scoping an encryption key to a tenant or to a deployment such that a key of a first tenant decrypts no content of a second, and holding key material within a custody boundary such that a use of said key material is a call across said boundary rather than a read thereof; wherein a knowledge record for which a zero-knowledge election is in effect remains one for which no operator-held key exists under any configuration.

[190] The method of claim 1, wherein said constructing, said retrieving, said iteratively updating and said generating are performed by components of a system performing the method and require no external fact-representation or fact-query system, and wherein an external fact-retrieval resource is connectable as a connector under a declared capability set such that a result returned therefrom is converted into a said knowledge record at a boundary of said connector and admitted through said extracting with said safety evaluation, said method being performed identically where no said connector is registered.

[191] The method of claim 181, wherein no element of said method recites an external fact-query system, a failure of a domain owning external integration is contained to a capability of said domain, and a determinism whereby an identical query over identical said knowledge records yields an identical said output text is established by said constraint on said permissible output vocabulary and by a bound on a number of stable states of said iterative updating. **Learning domain, module declaration and substrate migration (claims 183–185)**

[192] The method of claim 1, further comprising: maintaining a first ordered walkthrough over content, curated by an owner of said content, and a second ordered walkthrough over said content curated by a party other than said owner, both said walkthroughs persisting concurrently; associating with each said walkthrough an attribution identifying a curator thereof, said attribution being a field of a record of said walkthrough and not modifiable or removable by said curator; and holding an authorization to narrate a said walkthrough separately from an authorization to curate it, such that curating a said walkthrough confers no authorization to narrate it in a voice of any identified natural person.

[193] The method of claim 186, further comprising: computing, between a first rendered state of a said walkthrough and a second rendered state of said walkthrough, a colour-difference magnitude as a Euclidean distance between coordinates of said rendered states on three axes of a perceptually-uniform colour space; comparing said magnitude against a tolerance declared by an owner of said content; and raising a registration finding where said magnitude exceeds said tolerance, without altering either said first rendered state or said second rendered state.

[194] The method of claim 1, further comprising: embedding a time series into a manifold of lower dimension than said time series and nonlinear therein; producing from said manifold a forecast comprising a predicted value and a credible interval computed from accumulated evidence parameters of the knowledge records from which said time series was constructed; rendering said forecast in a region labelled as a forecast and distinct from said output text; and withholding said forecast from admission as a knowledge record, such that no said forecast is available as a grounding source for a subsequent said generating.

[195] The method of claim 1, further comprising: establishing a one-to-one message channel between two identities; classifying said message channel from an account context and an access path of said identities rather than from a self-declaration by a participant thereof; encrypting a said message channel not classified as associated with a minor context such that no party performing the method holds a decryption key therefor under any configuration; and encrypting a said message channel so classified against interception by a party other than said two identities while retaining a decryption capability exercised exclusively for a content-safety evaluation, each exercise being logged with an identity of an accessing process and a triggering reason. **Reasoning mode formalization — deductive, inductive and abductive (claims 190–192)**

[196] The method of claim 1, further comprising classifying an inference over said knowledge records as a first type where a conclusion thereof follows necessarily from admitted said knowledge records and as a second type otherwise; admitting a conclusion of said first type as a knowledge record carrying a type identifier distinguishing it from a directly-ingested knowledge record and a confidence value computed as a product of confidences of the knowledge records from which it follows; and withholding a conclusion of said second type from admission as a knowledge record, emitting it instead with a status field set to a proposed state, wherein no automated process is configured to transition said status field to a validated state.

[197] The method of claim 190, wherein an inference of said second type adjusts a confidence value of an existing said knowledge record without creating a knowledge record, and wherein a contradiction insufficient to effect said adjustment is recorded in a persistent index rather than discarded, such that a plurality of individually-insufficient contradictions bearing on a common assertion remains queryable.

[198] The method of claim 190, wherein admission of a knowledge record is conditioned on a conformance with a machine definition of an event type, a completed said safety evaluation and a reference into said provenance chain, and is not conditioned on a quantity of evidence supporting a semantic assertion of said knowledge record, such that said second type of inference adjusts a confidence value of an admitted knowledge record and does not determine whether a knowledge record is admitted. **Observation diversity across source classes (claims 193–196)**

[199] The method of claim 1, further comprising applying to said output text a plurality of interpretive modulations selected from a context modulation, a perspective modulation, an ergonomic modulation, a cultural modulation and a social modulation, wherein each said interpretive modulation operates upon a result of said generating and neither widens nor narrows said scope description, neither admits a knowledge record excluded therefrom nor excludes one admitted thereby, alters no said confidence value, and extends no said permissible output vocabulary; and refusing a said interpretive modulation that would effect any thereof.

[200] The method of claim 197, wherein two requesters whose scope descriptions admit an identical set of said knowledge records receive output text grounded in said identical set irrespective of a difference between their said interpretive modulations, such that said interpretive modulations vary an emphasis, an ordering, a vocabulary, a format and a medium of said output text and vary neither a content nor an availability of a said knowledge record.

[201] The method of claim 197, wherein a signal from which a said interpretive modulation is derived is retained for a duration of a session and is not persisted as a knowledge record, such that no said signal becomes available to said retrieving at a subsequent session.

[202] The method of claim 197, wherein a preference stated by a requester takes precedence over a signal inferred from an interaction of said requester wherever both bear on a common dimension of a said interpretive modulation, and an inferred said signal is applied only where no stated said preference bears on that dimension.

[203] The method of claim 197, wherein a said cultural modulation is derived from a profile declared by a requester, by an operator of a deployment, or by a locale explicitly selected by said requester, and is not derived from a name, a network address, a language of a query, a device locale or a behavioural signal of said requester.

[204] The method of claim 197, wherein a said ergonomic modulation derived from an inferred indicator of cognitive load modulates a density and a layout of said output text alone, produces no knowledge record concerning said requester, and is unavailable to a further said interpretive modulation, to a selection of commercial content and to a programmatic interface.

[205] The method of claim 197, wherein said interpretive modulations are composed in a fixed order over a result of said generating such that an identical result under an identical configuration of said interpretive modulations yields an identical modulated output; and wherein a review under a governance domain precedes and follows said composing and refuses a modulated output irrespective of said configuration.

[206] The method of claim 197, further comprising exposing for each said interpretive modulation an interface accepting a configuration and returning a modulated said output text, said interface resolving a scope description before a traversal and admitting no parameter widening said scope description; refusing at said interface a configuration that would retrieve a knowledge record, alter a confidence value or widen a scope description; and recording a said configuration in effect for a said output text such that said output text is reproducible from said recorded configuration. **Outcome feedback and operational calibration (claims 205–211)**

[207] The method of claim 1, further comprising recording for a delivered said output text a response record identifying said query, said scope description, each knowledge record cited therein and a confidence value held by each said cited knowledge record at a time of delivery; receiving from an identity a report bearing on said output text or on a said cited knowledge record; and admitting said report as a knowledge record through said extracting with said safety evaluation, carrying said identity as an attributed source and a reference to said response record.

[208] The method of claim 205, wherein a said report bearing against a said cited knowledge record increments a parameter accumulating contradicting evidence of said cited knowledge record by a weight derived from a source reliability of said identity, and does not set said confidence value, supersede said cited knowledge record or remove it from a store of said knowledge records.

[209] The method of claim 206, wherein said identity constitutes a source class for a diversity measure computed as a reciprocal of a sum of squared proportions of weight contributed by each source class, such that a plurality of said reports from a common said identity contributes less to said parameter than an equal plurality of said reports from distinct identities.

[210] The method of claim 205, wherein a said report is not admissible as a grounding source for output text concerning a subject matter of said report, and is admissible as evidence bearing on a confidence value and as an object of an analysis.

[211] The method of claim 205, further comprising pairing a said response record with an outcome record recording an event subsequent to said delivery, refusing construction of a said outcome record whose subject resolves to an individual natural person, and computing over a plurality of said pairings a mean of squared differences between a confidence value stated at delivery and a realised outcome, and a difference between a proportion of realised outcomes and a mean confidence value stated within a band, each computed per domain and per said band.

[212] The method of claim 209, further comprising emitting a difference whose magnitude satisfies a configured tolerance as a candidate modification evaluated in an isolated environment, said candidate identifying a measured said difference, a corpus over which it was measured and a parameter proposed for adjustment, and taking effect only upon an approval recorded against an identity of an individual human reviewer; wherein a parameter governing said computing is a member of a fixed enumerated set of subsystems excluded from modification.

[213] The method of claim 205, wherein no said report and no said outcome record extends a permissible output vocabulary of said generating or alters a said scope description, such that output text does not become expressible by reason of a said report and a knowledge record does not become traversable or cease to be traversable by reason thereof. **Expressive register and media-integrated expression (claims 212–217)**

[214] The method of claim 1, further comprising selecting for said output text exactly one expressive register from a registered set of expressive registers, said selecting being resolved before said generating and being distinct from a selection of an audience level governing vocabulary and explanatory depth; refusing a register not in said registered set rather than approximating it; and wherein a said expressive register governs a voicing of said output text and neither alters a set of knowledge records admitted, nor alters a confidence value, nor extends said permissible output vocabulary.

[215] The method of claim 212, wherein a first said expressive register characterised by declarative construction does not remove a qualification that a confidence value of a cited knowledge record requires, and a second said expressive register characterised by reduced density and acknowledgement does not soften an assertion supported by a said knowledge record, such that a said expressive register governs voicing and a said confidence value governs what is asserted.

[216] The method of claim 212, wherein a said expressive register characterised by acknowledgement of a requester's stated position is selected upon a preference stated by said requester, upon a configuration for which it is a configured default, or upon a consented signal captured under an explicit session-scoped authorization; and is not selected upon an inference from a word choice, a punctuation, a length, an error rate, a retry behaviour or an interaction timing of a query, no record of an inferred affective state being constructed.

[217] The method of claim 212, wherein an interpretive modulation narrows a selection of a said expressive register where a configuration requires it and does not widen said selection, such that no said interpretive modulation selects an expressive register a configuration of said requester does not permit. **Surface realization — aggregation, reference, connectives, ordering and variation (claims 218–224)**

[218] The method of claim 1, further comprising including within said output text a media asset referenced by a knowledge record of said stabilized set and delivering said media asset with a citation resolving it to said knowledge record; and withholding a media asset referenced by no knowledge record of said stabilized set and a media asset whose referencing knowledge record is outside said scope description, such that no said media asset conveys a proposition said output text could not assert.

[219] The method of claim 215, wherein a said media asset that has been transformed is delivered together with a provenance of said transformation identifying a knowledge record from which a value inferred in said transformation was derived; and wherein an interpretive modulation selects among media assets includable under claim 215 and does not render a media asset includable.

[220] The method of claim 1, further comprising realizing two or more propositions of said graph structure that share a subject and are joined by an edge of a conjunctive type as a single sentence having a coordinated predicate, up to a configured maximum number of coordinated predicates; and withholding said realizing where confidence values of said propositions differ by more than a configured tolerance such that said propositions would require differing qualifications, whereby a proposition of lower confidence is not qualified by a coordination with a proposition of higher confidence.

[221] The method of claim 218, wherein a sentence produced by said realizing carries a citation of each said proposition realized therein, such that each clause thereof resolves to a knowledge record supporting it.

[222] The method of claim 1, further comprising realizing an entity in a full registered form at a first mention thereof within a discourse unit and in a reduced form at a subsequent mention, restoring said full registered form where a mention of a further entity sharing a type of said entity in a concept structure intervenes between said mentions, and restoring said full registered form at a first mention within a subsequent discourse unit; wherein an ambiguity is determined by said type and not by a surface form, and said full registered form is used where said determining is indeterminate.

[223] The method of claim 1, further comprising emitting, between two propositions realized in sequence and joined by a typed edge of said graph structure, a connective of a class corresponding to a type of said edge; selecting a connective within said class by an expressive register in effect and by a position within a response such that a connective does not recur uniformly; and emitting no connective between two propositions not joined by a said typed edge.

[224] The method of claim 1, further comprising ordering within a sentence a constituent realized in a preceding sentence of a response before a constituent not yet realized therein, where a syntax of a target language permits; and exempting from said ordering a span reproduced rather than generated.

[225] The method of claim 1, further comprising substituting for a recurring term an alternative registered as synonymous for a sense of said term, and admitting said substituting only where a semantic fingerprint of a proposition containing said term is unchanged thereby; and refusing said substituting for a term within a span reproduced rather than generated, for a registered canonical name of an entity, and for a governed term of a discipline mode.

[226] The method of claim 1, wherein a selection among admissible realizations of a proposition of said graph structure is made by a deterministic rule over prior content of a response and not by sampling from a probability distribution, such that a given said graph structure under a given expressive register yields an identical token sequence on each execution; and wherein no said realization introduces a proposition absent from said graph structure. **Commercial presentation, advertiser provenance and transaction records (claims 225–231)**

[227] The method of claim 1, further comprising registering an equipment unit of a capture or inspection class with a kinematic envelope, a velocity limit, a force limit and an operating volume; refusing command emission to an equipment unit whose registration lacks any thereof; and refusing command emission to an equipment class outside said capture or inspection class, wherein telemetry received from a vehicle-control system is admitted as an analysis record and no control interface thereto is created.

[228] The method of claim 232, further comprising maintaining an equipment state record comprising a pose, a configuration, a sensor status and an operating volume in force, said equipment state

record being retained for a duration of a session and not admitted as a knowledge record; and withholding command emission where an age of said equipment state record exceeds a configured staleness bound, an indication of said staleness being returned.

[229] The method of claim 232, further comprising granting a session authorization recorded with an identity of an authorizing human operator, equipment units covered, an operating volume permitted, task classes permitted and an expiry; emitting a command only within said session authorization; halting emission upon an expiry thereof rather than requesting a renewal during an executing plan; and effecting a revocation thereof before a next command rather than upon completion of an executing plan.

[230] The method of claim 234, wherein each emitted command carries a reference to a request from which it was compiled, to said session authorization, to a plan of which it forms part and to a position within said plan, and a command unable to carry each said reference is not emitted.

[231] The method of claim 232, further comprising compiling a request into a command sequence by decomposing said request into operational intents before binding an intent to a concrete target, binding each said intent against a spatial model of an environment, mapping each bound intent to a registered equipment unit capable thereof, and synthesising a plan satisfiable within a registered envelope; wherein an unresolvable target, an absence of a capable said equipment unit and a plan not satisfiable within said envelope each halt said compiling rather than yielding a command against a target selected among candidates.

[232] The method of claim 232, further comprising evaluating a command before emission against an operating volume, a kinematic envelope, a velocity limit, a force limit, a minimum clearance to a detected obstacle, an exclusion distance from a detected person, an overlapping swept volume of a further equipment unit, and a recording-restricted zone; and withholding emission of a command violating any thereof, said withholding being unconditional and said command being neither emitted at a reduced priority nor modified into a permissible command by an automated selection.

[233] The method of claim 237, wherein a detection of a person is used to compute said exclusion distance and for no further purpose, is not retained beyond an evaluation thereof, does not enter a result set and is not available to a further mechanism, such that an avoidance of a person is performed without a record of said person being constructed.

[234] The method of claim 232, further comprising emitting a halt upon a clearance breach, an exceeded staleness bound, an expiry or revocation of a session authorization, a loss of a sensor stream on which an executing plan depends, or an explicit request; wherein said halt is evaluated from an equipment state record and a constraint set without a said retrieving, a said iterative updating, a said generating or an interpretive modulation, such that said halt is not delayed by a query in progress; and wherein said halt places an equipment unit in a registered safe state and a subsequent emission requires an explicit resume.

[235] The method of claim 232, further comprising observing an execution of a plan, admitting an observation thereof as an outcome record, and compiling a corrected plan afresh through said decomposing, said binding, said mapping, said synthesising and said evaluating rather than issuing a modification of a withheld command; and wherein no said observation adjusts a said kinematic envelope, a said velocity limit, a said force limit, a said operating volume or a said session authorization. **Assistive physical contact under a two-party authorization (claims 241–249)**

[236] The method of claim 232, further comprising damping an oscillation of a control signal derived from sensor input by said iteratively updating, wherein a damped said control signal is evaluated against a constraint set identically to an undamped said control signal such that said damping does not bring a command within a limit it would otherwise exceed.

[237] The method of claim 255, further comprising determining a number of changes of sign of a control error across a configured window and, where said number is at or above a configured frequency,

raising a control fault, ceasing said damping and invoking a halt, such that a persistent said oscillation is reported rather than damped. **Concept-adjacent commercial relevance (claims 257–259)**

[238] The method of claim 232, further comprising registering a graspable object with a mass, a permitted grip force, a fragility limit and a set of admissible contact regions; refusing a grasp of an object lacking a said registration rather than inferring a grip force from a perception thereof; and applying a grip force equal to a least of a force ceiling registered for an equipment unit, said fragility limit, and a force computed as required for said grasp increased by a configured margin, said margin being applied before said least is taken.

[239] The method of claim 270, further comprising detecting a slip from a divergence between a commanded and an observed position of a said object during a held grasp, increasing an applied force toward a lesser of said force ceiling and said fragility limit and no further, and releasing said object under control where said slip continues at said lesser rather than increasing said applied force beyond it.

[240] The method of claim 270, wherein a loss of power, a loss of a control signal, a fault in a safety-related control path and a halt each result in an opening of a gripper, said opening being a condition of a registration of an equipment unit.

[241] The method of claim 272, further comprising planning a grasp such that a path along which a said object would fall upon a said opening is unobstructed before said grasp is made, refusing a trajectory whose said path is obstructed; and upon an obstruction entering said path during a held grasp, moving to a nearest pose within an envelope having an unobstructed said path and, where no said pose exists, retaining said object and raising a fault.

[242] The method of claim 270, wherein no registration under said registering is available for a person or an animal, such that a physical support of a person is performed only under an authorization class requiring a contact consent. **Navigation in shared space (claims 277–283)**

[243] The method of claim 232, further comprising registering a navigable region comprising a travelable surface, a boundary and an excluded zone in place of an operating volume, maintaining from sensor input a clearance model of a distance to a nearest obstacle along each of a plurality of candidate headings, retaining said clearance model for a duration of a session without admitting it as a knowledge record, and halting motion where an age of said clearance model exceeds a configured staleness bound.

[244] The method of claim 277, wherein a commanded speed is bounded by a lesser of a registered maximum speed of an equipment unit and a product of a configured coefficient and a said distance, such that a stopping distance at a permitted said commanded speed is less than said distance; and wherein no task parameter, priority or deadline increases said coefficient.

[245] The method of claim 277, wherein upon a conflict between a path of a person and a path of a said equipment unit, said equipment unit stops or withdraws along a path already travelled and does not route around said person, does not signal an intention for accommodation by said person, and does not determine which of said person and said equipment unit yields.

[246] The method of claim 279, wherein upon a conflict between two equipment units under a common authorization a precedence is resolved by a registered ordering rather than by an exchange between said equipment units, and a persistence of said conflict beyond a configured interval raises a fault halting both.

[247] The method of claim 279, wherein said equipment unit does not maintain a following distance behind a person moving in a direction of travel thereof, yielding and waiting instead.

[248] The method of claim 277, further comprising computing a clearance envelope from an observed position and velocity of a person over a window and from no further input, treating said clearance envelope as occupied, computing no identity of said person, retaining no said observed position

beyond said window, retaining no said clearance envelope beyond an interval governed thereby, and making no said clearance envelope available to a further mechanism or an interface; and treating a full reachable region as occupied where said observed position and velocity do not support a said clearance envelope.

[249] The method of claim 277, wherein where no path satisfies a constraint set said equipment unit stops without reducing a clearance requirement, without proceeding at a reduced speed through a clearance below a configured minimum and without selecting a least-violating trajectory; and wherein a stop persisting beyond a configured interval reports an obstruction together with said clearance model. **The predictive hierarchy and error attribution (claims 284–289)**

[250] The method of claim 1, further comprising registering an equipment unit making intentional physical contact with a person under an authorization class disjoint from an authorization class governing equipment constrained by an exclusion distance from a person, wherein an authorization of either said class confers no capability under the other and a constraint set applicable to a command is determined by an authorization under which said command is emitted; and refusing said registering where a hazard analysis for an intended task, an intended user and an intended environment of said equipment unit has not been completed and recorded with an identity of an assessor.

[251] The method of claim 241, further comprising requiring, concurrently, a deployment authorization held by an identified operator and a contact consent held by a person to be assisted, wherein said contact consent is bounded to a task class and an interval, does not persist beyond a session, and is not substituted by said deployment authorization; and withholding contact where either is absent.

[252] The method of claim 242, further comprising halting contact within a registered stopping interval upon a withdrawal of said contact consent expressed by said person through any registered channel, wherein said withdrawal is not overridable by an operator, by a representative or by a task in progress; and wherein upon said withdrawal said equipment unit performs no motion other than motion required to reach a state in which said person is neither supported nor retained by it, said motion being bounded by force limits applicable to said task.

[253] The method of claim 242, wherein where said person cannot express or withdraw said contact consent through a registered channel, contact proceeds only where a designated representative recorded under a governance of a deployment holds said contact consent, a second person is present holding a halt control, and a task class is within a registered subset excluding a task whose failure said person could not signal; and wherein a determination that said person cannot so express is read from a record of said deployment and is not made by inference from a response of said person.

[254] The method of claim 241, further comprising evaluating a command against a force limit and a pressure limit registered per region of a body, a registered contact region, a contact velocity lower than a free-motion velocity, a support redundancy whereby a single failure does not withdraw support from a person whose weight is borne, a currency of a contact consent, and an absence of a configuration from which said person could not be released; and halting upon a violation of any thereof rather than adjusting a said command.

[255] The method of claim 245, wherein said force limit and said pressure limit are registered separately per said region of a body such that a force within a limit applied over a reduced area is not thereby permitted.

[256] The method of claim 241, wherein a loss of power, a loss of a control signal, a fault in a safety-related control path and a halt each result in a release of a person rather than a retention thereof; and wherein a said equipment unit bearing a weight of a person is excepted from said release upon a registration recording a redundant support means, its safe state being a controlled retention of support together with an alert and a bounded interval within which support is transferred.

[257] The method of claim 247, wherein a release is effected without traversing said retrieving, said iterative updating, said generating or an interpretive modulation, and is operable where a said retrieving is unavailable.

[258] The method of claim 241, wherein contact is initiated only upon a request from a person to be assisted, upon a request from an operator under a held contact consent, or within a task sequence to which said person consented, and is not initiated upon an inference that said person requires assistance; and wherein sensor data captured during contact is discarded at a task end, no record of a body, a movement, a capability or a dependency of said person is constructed or retained, and no confidence value is adjusted by a said contact. **Attractor stabilization beyond retrieval (claims 250–256)**

[259] The method of claim 241, further comprising presenting a said object within reach of a person without applying a force toward said person, and releasing said object upon a detection of a support thereof by said person sustained across a configured interval; and upon an absence of said detection within a configured timeout, lowering said object under control to a registered surface rather than releasing it, a handover being reported incomplete.

[260] The method of claim 274, wherein a said equipment unit applies no force resisting a taking of a said object by said person and does not re-grasp a said object taken by said person.

[261] The method of claim 1, wherein said iteratively updating decreases monotonically an energy over a configuration of activation values, said energy being a negative half of a sum over distinct pairs of a product of a weight and two said activation values, whereby said iteratively updating terminates without a bound on a number of iterations and a given initial configuration under given weights reaches an identical stable state on each execution.

[262] The method of claim 250, further comprising parameterizing said weights such that a spectral norm of a matrix thereof is less than unity, whereby said iteratively updating is a contraction and an equilibrium thereof exists and is unique independently of an initial configuration; maintaining said parameterization upon each update of a said weight rather than upon an initialization thereof; and recording an election of said parameterizing per network, such that a network so parameterized and a network operated under a bound on a number of stored configurations are distinguishable in a result.

[263] The method of claim 251, further comprising recording for each network an election between a first convergence regime in which said weights are parameterized such that an equilibrium is unique and a second convergence regime in which a number of stored configurations is held below a bound; and returning with an output text derived from a said network an indication of which said regime was in effect, such that a recipient of said output text is informed whether an identity of a converged state was guaranteed by construction or was rendered improbable by said bound.

[264] The method of claim 260, wherein a said election is a member of a fixed enumerated set of subsystems excluded from modification by a proposal generated within the system, such that a convergence regime is changed only by an operator and a change thereof is recorded in said provenance chain with an identity of said operator. **Systematic trial architecture (claims 262–269)**

[265] The method of claim 250, wherein said iteratively updating is performed by an array of variable-conductance elements arranged at intersections of a plurality of rows and a plurality of columns, wherein said activation values are applied as potentials upon said rows, a weight of said iteratively updating is a conductance of an element of said array, a weighted sum of said iteratively updating is a current at a column of said array, and a stable state is read by comparing a said current against a reference.

[266] The method of claim 307, wherein said array is operated only where said weights are parameterized such that an equilibrium of said iteratively updating is unique, and an array operated under a bound upon a number of stored configurations without said parameterization is refused for said

performing, whereby two said arrays whose conductances differ within a fabrication tolerance settle to a common said equilibrium.

[267] The method of claim 308, wherein a proportion of inputs upon which two said arrays whose conductances differ by a fabrication tolerance yield differing settled states is lower under said parameterizing than under said bound upon a number of stored configurations, and wherein a said fabrication tolerance admissible for a said array is determined from said proportion rather than from a specification of said array alone. **AI-engine ingestion: prompt composition, conversion, derivation marking and audit (claims 313–321)**

[268] The method of claim 307, further comprising registering said array with a dimension thereof, a range of said conductance, a fabrication tolerance and a tolerance to which a settled state is read; and refusing a said array whose registration lacks any thereof.

[269] The method of claim 309, wherein a column whose said current falls within said tolerance of said reference is resolved by a further settling rather than by a rounding, and where a said further settling does not resolve it a state is reported unresolved and said iteratively updating is performed by a digital path.

[270] The method of claim 309, wherein two said arrays operating upon a common input under a common weight specification and yielding settled states differing beyond said tolerance raise a fault, said settled states being neither averaged nor combined, whereby a state reached by no said array is not produced.

[271] The method of claim 1, further comprising completing a partially-observed sensor frame by converging observed regions thereof to a nearest stored configuration; marking a completed region as reconstructed such that it is not presented as observed; bounding a confidence of a said completed region above by a least of a confidence of said observed regions and a confidence of said stored configuration; and rendering a said completed region inadmissible as a grounding source such that it grounds no assertion and contributes to no confidence value.

[272] The method of claim 252, wherein where said observed regions do not converge to a stored configuration within a configured iteration ceiling no completion is produced and an absence is reported.

[273] The method of claim 1, further comprising converging a divergence between two representations of a common subject where a magnitude of said divergence falls below a configured threshold, and preserving said divergence where said magnitude is at or above said threshold or where said magnitude cannot be determined; and recording a preserved said divergence in a persistent contradiction index where said representations assert incompatible content.

[274] The method of claim 1, further comprising assigning to each of a plurality of edge classes of said knowledge records a traversal cost, and bounding a pre-activation of knowledge records proximate to a query by a budget expressed in units of said traversal cost rather than by a count of edges traversed, such that a traversal reaching a first number of edges of a first said edge class and a second number of edges of a second said edge class consumes a budget determined by said traversal costs.

[275] The method of claim 290, wherein a traversal cost of an edge class relating knowledge records by meaning is lower than a traversal cost of an edge class relating knowledge records by a version of a work, which is lower than a traversal cost of an edge class relating knowledge records by time of occurrence.

[276] The method of claim 290, wherein said traversal costs are registered configuration values recorded with an output text derived from said pre-activation such that said pre-activation is reproducible from said recorded values, and wherein no said traversal cost is adjusted by an automated process from an outcome of a prior traversal.

[277] The method of claim 290, further comprising computing at a time of traversal a rate of change of a relevance across neighbours of a knowledge record within an edge class, ordering a traversal to prefer a direction of lesser said rate of change, wherein said rate of change is computed from values carried by said knowledge records and is not stored, and wherein said ordering alters an order in which a budget of claim 290 is expended and does not alter which knowledge records are reachable within said budget. **Computational substrate output as an ingested modality (claims 303–306)**

[278] The method of claim 1, further comprising resolving a query in one of a first posture returning a state attested before a stated cutoff, a second posture returning a current state, and a third posture returning a forecast carrying a credible interval; and returning with said output text an indication of which said posture resolved it.

[279] The method of claim 293, wherein a result resolved in said third posture is not admitted as a knowledge record and is unavailable as a grounding source for a subsequent said generating, while a result resolved in said first posture and a result resolved in said second posture are admitted as knowledge records.

[280] The method of claim 293, wherein a proposition of said output text resolved in one said posture carries an indication of that posture where said output text draws upon more than one said posture, such that a difference between a result resolved in said first posture and a result resolved in said second posture is attributable to said postures and not to an inconsistency. **Coherence classification and the context lock (claims 296–302)**

[281] The method of claim 1, further comprising computing for a knowledge record a coherence value across a plurality of dimensions comprising a semantic, a temporal, a spatial, an editorial, a media and an evidential dimension, and taking as a governing coherence value a least of said plurality rather than a mean thereof, such that a dimension having a value substantially below the others is not concealed by the others.

[282] The method of claim 296, further comprising classifying said knowledge record by said governing coherence value against configured floors, naming with said classification the dimension that produced it, and distinguishing from said classification a state in which a said dimension cannot be computed for want of data and a state in which two knowledge records assert incompatible content.

[283] The method of claim 296, wherein said coherence value adjusts no confidence value, admits no knowledge record to and excludes no knowledge record from said admitted set, and withholds no output text; and wherein an output text resting upon a knowledge record classified below a configured floor is delivered together with said classification and said dimension.

[284] The method of claim 1, further comprising maintaining an enumerated set of context dimensions comprising a device class, an entitlement set, a locale, an expressive register, an audience level, a temporal posture, a table of traversal costs and a convergence regime; computing a key over values of said enumerated set; and returning said key with said output text, wherein an identity of said key between two executions is a condition of an identity of said output text being asserted.

[285] The method of claim 299, wherein two executions differing in a value of a dimension of said enumerated set yield differing said keys and a difference between output texts thereof is not a defect, and two executions differing only in a value not within said enumerated set yield an identical said key and a difference between output texts thereof is a defect.

[286] The method of claim 299, wherein said enumerated set is a member of a fixed enumerated set of subsystems excluded from modification by a proposal generated within the system, such that a dimension is not removed therefrom by an automated process.

[287] The method of claim 1, further comprising: composing a prompt from said knowledge records under said constraint on said permissible output vocabulary and according to a profile registered for an

external machine-learning engine, said prompt comprising a semantic marker identifying a concept to be addressed, an extraction instruction specifying a form in which an assertion is to be returned, and an alignment constraint stating a schema into which a returned assertion is to be converted; dispatching said prompt to said engine through a connector under a declared capability; receiving a reply; segmenting said reply into individual assertions; and converting each said assertion into a knowledge record of said record schema, each said knowledge record receiving a temporal anchor from a time of said dispatching, a semantic fingerprint computed over a canonicalized form thereof, and a reference into said provenance chain identifying said prompt, said engine, a version thereof and a position of said assertion within said reply.

[288] The method of claim 313, wherein said composing is deterministic such that an identical request over identical said knowledge records under an identical said profile produces an identical said prompt, whereby a difference between a first said reply and a second said reply to an identical said prompt is attributable to said engine.

[289] The method of claim 313, wherein a said knowledge record converted from a said assertion is admitted to a store of said knowledge records, is related within a concept structure to a concept its said semantic assertion concerns, is traversable within a scope description admitting it, and is subject to said confidence value accumulation, to a detection of contradiction and to an erasure operation identically to a knowledge record extracted from a source other than a said engine.

[290] The method of claim 315, wherein a said knowledge record carries a derivation marking identifying said engine as a field of said record schema, said derivation marking being unmodifiable and unremovable by any operation upon said knowledge record; and wherein a citation resolving to a said knowledge record discloses said derivation marking, such that a recipient of output text citing a said knowledge record is informed that a cited assertion was generated by an identified engine in response to a recorded prompt.

[291] The method of claim 315, wherein each said engine constitutes one source class of said plurality of registered source classes, whereby a plurality of said knowledge records converted from replies of a common said engine yields a diversity measure of unity irrespective of a number thereof; and refusing a corroboration of a first said knowledge record by a second said knowledge record where both are derived from a common said engine.

[292] The method of claim 315, wherein a confidence value of a said knowledge record is bounded above by a reliability registered for said engine, and where said assertion was returned under a said extraction instruction, by a lesser of said reliability and a reliability registered for said extraction instruction; and wherein said reliability registered for said engine is set by an operator and is not adjusted by an output of said engine.

[293] The method of claim 315, further comprising verifying a citation contained within a said assertion against said knowledge records, recording said citation as verified where it resolves to a knowledge record held and as unverified where it resolves to none, and admitting said assertion carrying a said recording of unverified rather than withholding it, whereby a citation resolving to nothing becomes a recorded property of said assertion.

[294] The method of claim 315, further comprising recording for each said knowledge record a chain comprising said prompt, said engine, a version thereof, a time of elicitation, a position within said reply, said derivation marking, a bound applied to said confidence value and a basis thereof, and an outcome of said verifying; and rendering said chain queryable such that a set of said knowledge records derived from a said engine is determinable.

[295] The method of claim 313, wherein an output attributed to an external system and obtained otherwise than by a said dispatching of a said prompt, and a result of a probing of an external system, each

remain excluded from a scope description constructed for a query not directed thereto, such that said converting is available only for a reply elicited by a said prompt composed by the method. **Competitive candidate resolution (claims 322–324)**

[296] The method of claim 1, wherein where a query admits a plurality of mutually-inconsistent interpretations, knowledge records supporting each said interpretation are activated concurrently rather than evaluated in sequence, an activation value of each said knowledge record is reduced in proportion to a sum of activation values of knowledge records supporting an interpretation inconsistent therewith, said activation values are iteratively updated to a stable state, and a knowledge record whose said activation value falls below a configured retention floor is suppressed, whereby an interpretation supported by a surviving configuration is composed.

[297] The method of claim 322, wherein a plurality of said knowledge records surviving said retention floor at comparable activation values yields no resolution and a request for clarification is returned, no interpretation being selected among them; and wherein a coefficient of said reduction and said retention floor are registered configuration values recorded with an output text such that said reduction is reproducible therefrom.

[298] The method of claim 322, wherein said reducing operates upon said knowledge records and not upon a plurality of domains each owning a bounded set of operations, a said domain owning a distinct decision such that no two said domains propose competing resolutions of a common query. **Specialized live data feeds (claims 325–332)**

[299] The method of claim 1, further comprising registering a defined agent upon a definition comprising a purpose and a boundary, an operational function comprising an input type, an output type, a decision rule and an error handling, a scope description, a runtime comprising a trigger condition, a duration bound and a termination condition, a declared capability set, and an approval recorded against an identity of an individual human reviewer; and refusing said registering where an element of said definition is absent, no default being supplied therefor.

[300] The method of claim 333, wherein said declared capability set of said defined agent is bounded above by a capability set of an identity registering said defined agent, and a capability not within said declared capability set is refused at a boundary upon an execution of said defined agent.

[301] The method of claim 333, wherein said operational function is compiled into a representation admitting a sequence, a conditional evaluation, a bounded iteration over a set of a cardinality known before entry thereof, and a typed operation upon a knowledge record, and admitting no unbounded iteration and no recursion without a bound, whereby a termination of an execution is decidable before said execution.

[302] The method of claim 333, wherein an execution of said defined agent resolves said scope description before a traversal and no parameter supplied upon said execution widens said scope description; wherein said execution is metered as a bounded compute-accounting unit scheduled below a unit of a safety-evaluation category, below a unit serving a requester and below a unit of an ingestion category; and wherein a said evaluation by a governance domain is performed upon each said execution rather than upon said registering.

[303] The method of claim 333, wherein an output of said defined agent following necessarily from admitted knowledge records is admissible as a knowledge record carrying a derived-inference type, an output not so following is emitted in a proposed state, a proposed change to a parameter of the system is emitted as a proposal effective only upon a said approval recorded against an individual human reviewer, and a command to an equipment unit is emitted only within a session authorization held by a said individual human reviewer, said defined agent holding no said session authorization.

[304] The method of claim 333, wherein said defined agent terminates upon an expiry of a lifecycle bound recorded at said registering, upon a request of an owner thereof, upon an exclusion by a governance domain, upon a violation of said declared capability set, or upon exceeding a configured error rate, said terminating taking effect before a subsequent said execution; and wherein a renewal of said defined agent constitutes a further said registering requiring a further said approval.

[305] The method of claim 333, further comprising writing upon each said execution a decision trace recording a trigger that fired, said scope description resolved, knowledge records read, a branch taken by each said decision rule, an output produced, a refusal and a cause thereof, and a compute consumed; wherein said decision trace is written by a mechanism performing said execution and not by said defined agent, and is appended to said provenance chain under a keyed checksum.

[306] The method of claim 333, further comprising analysing said definition before said registering, presenting to a registering identity a finding comprising a scope exceeding an entitlement thereof, a termination condition incapable of obtaining, a trigger condition not evaluable within said scope description and a decision rule admitting a non-deterministic resolution, and withholding said registering upon an unresolved said finding; wherein said analysing produces no determination of a compliance of said defined agent, a compliance being established by a said evaluation by a governance domain and by a said approval recorded against an individual human reviewer.

[307] The method of claim 340, wherein a report produced by said defined agent during an execution thereof informs a decision of an owner thereof and does not constitute one, said defined agent effecting no renewal of itself, no widening of said scope description, no extension of said lifecycle bound and no resolution of a said finding.

[308] A system comprising: one or more processors; and a memory storing instructions that, when executed by said one or more processors, cause the system to: construct a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into an append-only provenance chain; construct, responsive to a query received from a requester and before traversal, a scope description bounded by an access entitlement of said requester and by a time interval, such that a knowledge record outside said scope description is not traversable; retrieve a candidate set of said knowledge records from within said scope description and admit into an admitted set only those whose retrieval confidence satisfies a configured threshold; iteratively update an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set until a stable state is reached; construct a graph structure in which every edge references at least one knowledge record having a positive said activation value at said stable state; and generate output text under a constraint that a permissible output vocabulary at each generation step is limited to terms derivable from said edges, a candidate proposition not so derivable being rejected before output.

[309] The system of claim 28, wherein said instructions further cause the system to place a synchronous safety evaluation gate at each of a plurality of operation layers, such that an entry to the system at a first said operation layer does not avoid a said gate placed at a second said operation layer.

[310] A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: constructing a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating

evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into an append-only provenance chain whose nodes each carry a checksum computed over an operation and a safety evaluation recorded for said operation; retrieving, responsive to a query and from within a scope description constructed before traversal, a candidate set of said knowledge records; admitting into an admitted set only those knowledge records of said candidate set whose retrieval confidence satisfies a configured threshold; iteratively updating an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set until a stable state is reached; and generating output text under a constraint that a permissible output vocabulary at each generation step is limited to terms derivable from edges of a graph structure, each said edge referencing at least one knowledge record having a positive said activation value at said stable state, a candidate proposition not derivable from said edges being rejected before output in favour of a narrower proposition that is so derivable. **Ingestion-time question generation and the author supplemental layer (claims 31–39)**

[311] A computer-implemented method for preparing a digital work for grounded interactive query, the method comprising: receiving, at a portal and bound to a verified author identity, a digital work submitted by an author; performing a rights validation on said digital work and withholding ingestion of said digital work until said rights validation returns a cleared status; extracting from said digital work a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint and a reference into an append-only provenance chain; generating, before receipt of any query directed to said digital work, a plurality of candidate questions by traversing each said knowledge record together with concept-mesh neighbours of said knowledge record, each said candidate question being answerable by at least one said knowledge record; scoring each said candidate question by a coverage measure computed over the knowledge records from which said candidate question was derived, and retaining in a persistent question index associated with said digital work only those candidate questions whose coverage measure satisfies a configured floor; computing, for each retained question and before receipt of any query, an answer assembled from exactly the knowledge records from which said retained question was derived and from no other knowledge record, assigning said answer a confidence value not exceeding a least confidence value among said contributing knowledge records, and storing said answer as a persistent pre-computed answer record carrying a provenance tag distinguishing it from a directly-ingested knowledge record together with an identifier of each said contributing knowledge record; receiving a query and matching said query against said question index prior to performing a traversal of said knowledge records; returning, where a match similarity of said query against a retained question satisfies a configured threshold and no contributing knowledge record of the corresponding pre-computed answer record has been revised since said answer was computed, said pre-computed answer record together with a citation to each said contributing knowledge record; and performing, where either said match similarity or said revision condition is not satisfied, a traversal of said knowledge records and a generation of a response therefrom.

[312] The method of claim 31, further comprising: receiving from said author, subsequent to said extracting, supplemental content concerning said digital work; converting said supplemental content into one or more knowledge records carrying a tag identifying said knowledge records as author-supplemental and carrying said author as an attributed source; making said knowledge records available to each reader holding an entitlement to said digital work without alteration of the rights validation status of said digital work; and regenerating each retained question and each pre-computed answer record whose contributing knowledge records are affected by said converting.

[313] The method of claim 32, wherein a said author-supplemental knowledge record participates in said generating of candidate questions on the same terms as a knowledge record extracted from said digital work, such that a candidate question answerable only by reference to said supplemental content is generated and is eligible for retention in said question index.

[314] The method of claim 31, wherein a said candidate question generated by traversal across two or more said knowledge records is a question that no single one of said two or more knowledge records answers alone.

[315] The method of claim 31, wherein the citation returned with a said pre-computed answer record identifies the identical set of knowledge records from which said answer was computed, such that an answer returned from said question index is traceable to the same degree as a response generated by said traversal.

[316] The method of claim 31, further comprising marking as stale each entry of said question index whose contributing knowledge record has been revised, and excluding a stale entry from return until said entry has been recomputed, wherein staleness is determined by said revision and not by an elapsed time since computation.

[317] The method of claim 31, further comprising computing, at a time of committing a said knowledge record and before receipt of any query, a forward inference chain enabled by said knowledge record, and storing a conclusion of said forward inference chain as a derived-inference record carrying a provenance tag identifying it as derived, such that a retained question whose answer depends on said forward inference chain is answerable without performing said forward inference chain at a time of said query.

[318] The method of claim 37, wherein said derived-inference record is traceable to the knowledge records whose logical entailment said derived-inference record represents, and is not traceable to a single source passage of said digital work.

[319] The method of claim 38, further comprising: submitting an output received from an external system back to said external system as an input under a fixed template; repeating said submitting to a configured iteration ceiling and recording a semantic fingerprint of the output of each iteration; and classifying the resulting sequence of semantic fingerprints into exactly one of a converged classification in which successive fingerprints become identical, a periodic classification in which said sequence repeats with a fixed period, a drifting classification in which a measured semantic distance from an initial output increases monotonically, and an unresolved classification where said ceiling is reached, wherein said classifying is performed on said sequence alone and no output is generated asserting a cause of said classification or a property of an internal mechanism of said external system.

[320] The method of claim 38, further comprising: constructing a family of inputs from a base input by applying a perturbation and retaining in said family only an input whose semantic fingerprint is identical to that of said base input; submitting each member of said family to an external system; and determining a proportion of the resulting outputs sharing a common semantic fingerprint, wherein an input whose perturbation alters said semantic fingerprint is excluded from said family before said submitting.

[321] The method of claim 31, wherein said question index is scoped to said digital work such that no pre-computed answer record of said question index is assembled from a knowledge record extracted from a different digital work, and wherein the contributing-record set of a said pre-computed answer record is fixed at a time of computation and is not extendable at a time of return. **Retrieval-phase boundary and sampling criticality control (claims 40–44)**

[322] A computer-implemented method for resolving an attribute of a concept in a knowledge system, comprising: maintaining a graph of concept nodes connected by typed edges including a

subsumption edge type and an attribute edge type; determining that a first concept node is connected to a second concept node by a subsumption edge and that said second concept node is connected to an attribute by an attribute edge; assigning said attribute to said first concept node by inheritance, together with a tag identifying said second concept node as an origin of said attribute and a depth of said subsumption edge; computing a confidence value for said inherited attribute as a product of confidences of the subsumption edges traversed and a confidence of said attribute at said second concept node, bounded above by a least of said confidences; performing said assigning at a time of traversal without writing said attribute to said first concept node as stored state; and returning, with any output relying on said inherited attribute, a citation identifying said second concept node and said subsumption edges traversed.

[323] The method of claim 45, further comprising: determining that said first concept node asserts a second attribute directly, said second attribute conflicting with said inherited attribute; applying said second attribute to said first concept node and suppressing said inherited attribute for said first concept node and for every concept node subordinate to said first concept node; and leaving an attribute of said second concept node unmodified and inherited by every concept node subordinate to said second concept node that does not itself assert a conflicting attribute.

[324] The method of claim 46, wherein, where a first inherited attribute and a second inherited attribute conflict and are inherited from concept nodes at differing subsumption depths, the attribute inherited from the lesser depth governs; and wherein, where said differing depths are equal, no resolution is performed, an entry is created in a persistent contradiction index, and an output relying on said attribute reports both values together with their respective originating concept nodes.

[325] The method of claim 46, wherein an assertion suppressing an inherited attribute is recorded with an identity of an ingestion event or a reviewer that produced it, and wherein a set of concept nodes suppressing a given attribute of a given concept node is retrievable as a query result.

[326] The method of claim 45, further comprising traversing a typed edge in a reverse direction by resolving an inverse predicate declared in a schema of said edge's predicate at a time of traversal, without storing a second edge representing said reverse direction; and refusing said traversing, and reporting an absent inverse declaration, where said predicate declares no inverse predicate. **Wording-operative spans and idiom equivalence (claims 50–53)**

[327] A computer-implemented method for generating output in a target language from knowledge records, comprising: constructing knowledge records each encoding a semantic assertion and each carrying a language-independent semantic fingerprint; generating output text in a target language by realizing said knowledge records under a constrained vocabulary, without reference to a surface form of a source content from which said knowledge records were constructed; determining that a first knowledge record is marked as wording-operative; reproducing, for said first knowledge record, a source span of said source content in a source language of said source span rather than realizing said first knowledge record under said constrained vocabulary; generating a gloss of said source span in said target language and marking said gloss as non-operative; and exempting said reproduced source span, and no other span of said output text, from a groundedness rewriting applied to said output text.

[328] The method of claim 50, further comprising withholding said reproducing where a rights validation of said source content does not permit reproduction of said source span, and delivering said gloss alone together with an indication that operative wording is not reproducible under an applicable right, wherein a fact extracted from said source content into a knowledge record remains available irrespective of said withholding.

[329] The method of claim 50, further comprising: determining, at a time of constructing a knowledge record from a source span, that a compositional reading of said source span and an attested reading of said source span registered in a concept structure diverge by more than a configured threshold;

recording in said knowledge record said attested reading rather than said compositional reading, together with a marking identifying said knowledge record as idiom-derived and an identifier of said source span; and realizing said knowledge record in said target language using a target-language expression registered in said concept structure for said attested reading.

[330] The method of claim 52, further comprising, where no target-language expression is registered in said concept structure for said attested reading, realizing said attested reading literally together with an indication that said source span expressed said attested reading idiomatically, and refraining from selecting a target-language expression on a basis of surface similarity to said source span. **Linguistic resource registry and derived-artifact-only ingestion (claims 54–57)**

[331] A computer-implemented method for measuring forecast accuracy of a knowledge system, comprising: maintaining a plurality of knowledge records, each said knowledge record encoding a semantic assertion and comprising a temporal anchor identifying an interval over which said semantic assertion holds, an attestation time identifying when said semantic assertion was first recorded by a source of said knowledge record, and a reference into an append-only provenance chain; identifying an event record having an event time and a recorded outcome; selecting a horizon interval preceding said event time; constructing, before any traversal of said knowledge records, a scope description admitting a said knowledge record if and only if said attestation time of said knowledge record is earlier than said event time less said horizon interval, wherein said temporal anchor is not consulted in said admitting and a knowledge record whose attestation time is not established from documented source metadata is excluded from said scope description; executing a forecasting operation against only the knowledge records admitted by said scope description, said forecasting operation producing a predicted outcome and a probability assigned to said predicted outcome; computing an error term from said assigned probability and said recorded outcome; and storing said error term together with said horizon interval, a domain of said event record, and an identifier of each knowledge record admitted by said scope description.

[332] The method of claim 113, wherein a knowledge record excluded by said scope description is not traversable during said forecasting operation rather than being removed from a result of a traversal thereof, such that said excluded knowledge record does not affect a candidate count, a retrieval ordering, a convergence of an iterative stabilization, or a permissible output vocabulary of said forecasting operation.

[333] The method of claim 113, wherein, where a documented creation date of a source of a said knowledge record is contested among a plurality of candidate dates, a latest of said candidate dates is used as said attestation time, such that uncertainty in said attestation time resolves toward exclusion from said scope description.

[334] The method of claim 115, further comprising: maintaining a first candidate hypothesis and a second candidate hypothesis concerning a common subject, each having a confidence value maintained as said first and second parameters; determining that a difference between the confidence values of said first and second candidate hypotheses does not exceed a configured separation floor while an accumulated evidence count of each exceeds a configured minimum; computing, for each of a plurality of candidate observations, a magnitude of a difference between a likelihood of said candidate observation under said first candidate hypothesis and a likelihood of said candidate observation under said second candidate hypothesis; selecting a said candidate observation maximising said magnitude; and emitting a structured request identifying said selected candidate observation, said magnitude, and a confidence value each said candidate hypothesis would take under each of a presence and an absence of said selected candidate observation.

[335] The method of claim 169, wherein said selecting is performed irrespective of a likelihood assigned to said candidate observation by either said candidate hypothesis, such that a candidate

observation predicted equally by both said candidate hypotheses is not selected however probable it is under either.

[336] The method of claim 169, further comprising withholding said emitting, and recording said first and second candidate hypotheses as undiscriminated together with a reason, where said magnitude for every said candidate observation falls below a configured discrimination floor.

[337] The method of claim 169, wherein neither said first candidate hypothesis nor said second candidate hypothesis is advanced or retired by said emitting, each remaining in a proposed state transitionable to a validated state only upon confirmation by an identified human reviewer, and wherein no automated process acts upon said structured request.

[338] The method of claim 169, further comprising, upon a knowledge record corresponding to said selected candidate observation being admitted as a said knowledge record, updating said confidence values of said first and second candidate hypotheses and re-evaluating said difference against said separation floor, and recording an outcome of said selected candidate observation such that said selected candidate observation is not emitted again for said first and second candidate hypotheses. **Deployment topology, module replacement and substrate independence (claims 174–182)**

[339] The method of claim 113, further comprising repeating said selecting, said constructing, said executing and said computing for a plurality of differing horizon intervals preceding a common said event time, and deriving a measure of variation of said error term with said horizon interval within a domain.

[340] The method of claim 113, further comprising determining, over a plurality of said error terms whose assigned probabilities fall within a common band, a proportion of corresponding recorded outcomes that occurred, and reporting a divergence between said proportion and said band as a calibration finding applicable to a confidence interval subsequently rendered by said forecasting operation.

[341] The method of claim 113, wherein a subject of said event record is one of an institution, an organisational process, a system and an aggregate group, and wherein construction of a record storing said error term is refused where said subject resolves to an individual natural person, said refusal occurring at construction rather than by restriction of access to a constructed record. **Event type catalogue, environment vectors and causal validation (claims 119–124)**

— New claims 342–354 (Passes 117–118) —

342. A computer-implemented method as in claim 1, wherein each knowledge record is expressed in a formal, language-independent assertion notation whose vocabulary is a governed predicate catalogue, whose grammar is a fixed record schema, and whose identity function is a semantic fingerprint, such that two statements asserting the same proposition in different natural languages resolve to the same statement, and a candidate not conforming to said notation is refused at admission.

343. The method of claim 1, further comprising maintaining procedural knowledge as provenance-tracked procedural concept models formed from repeated observation, including sensory experience and ingested video, said procedural concept models informing scenario simulation and equipment-control command compilation while never themselves emitting a command.

344. The method of claim 1, further comprising decomposing an uncertainty of a knowledge record into an epistemic component and an aleatory component each carrying explicit sub-parameters, and terminating an evidence-accumulation loop upon a condition defined over said components.

345. The method of claim 1, wherein a knowledge record carries an explicit assertion-polarity field, and said polarity participates in the semantic fingerprint such that an assertion and its negation are distinguished.

346. The method of claim 1, further comprising representing an assertion of more than two participants as a native n-ary structure under a single identifier, a single semantic fingerprint, and a single evidence pair.

347. The method of claim 1, wherein a knowledge record carries an applicability-scope field constituting a record-side complement of a context lock, and a contradiction between two records is evaluated within scope-aware conflict slots defined by said field.

348. The method of claim 1, further comprising maintaining an immutable adjustment-history layer recording each adjustment to a confidence value, whereby said confidence value is deterministically replayable from the record content and an observation stream.

349. The method of claim 1, further comprising hash-linking each superseding version of a knowledge record to its predecessor in a cryptographic integrity chain, without a distributed ledger, token, or ownership-identity node.

350. The method of claim 1, further comprising serializing a knowledge record in a canonical compact encoding for micro-embedded and offline deployment, said canonical compact encoding yielding an identifier and a semantic fingerprint identical to those of a full form of the record.

351. The method of claim 350, wherein said canonical compact encoding constitutes the record's at-rest and wire form fused to the record and not to any store, said encoding being self-describing.

352. The method of claim 351, further comprising storing the encoded record in a host store as an opaque value that the host store does not parse, index, or interpret, an obligation of the host store being to accept and return bytes.

353. The method of claim 352, further comprising expressing a query internally over the record and translating said query at an adapter into a native query language of the host store, whereby the host store never receives the internal query form.

354. The method of claim 351, further comprising carrying an integrity hash chain, an in-record safety classification and access-control list, and an elected zero-knowledge encryption with the encoded record across a storage boundary, whereby the host store can neither read, reorder, nor leak a meaning of the record.

355. The method of claim 1, further comprising providing, to an external service interposed between an artificial-intelligence agent and a network, a read-only verification interface that receives a candidate knowledge record or its semantic fingerprint and returns a signed verdict of verified, contradicted or unknown, said verdict being determined by fingerprint lookup among certified records and, absent a match, by a query for certified records having the same subject and predicate and a conflicting value.

356. The method of claim 355, further comprising reserving a range of frame message types and a header flag of the framed transport for said external service, and accepting from said service, under a registered identity, appends to the hash-chained security record, from which a Merkle root over the entries of one session is computed and sealed as a session certificate verifiable by any party.

357. The method of claim 355, wherein said external service is registered through the agent gateway with read-only verification rights and append-only record rights, receives signed policy bundles, and observes tool calls and hashed tool result records through hooks of the tool server, and wherein said service is never permitted to write knowledge records.

[Drafting note: claims 355–357 are added for support of external agent-wrapper services (FIG. 56); the claim count in the Claim Set heading is to be read as 357.]

358. The method of claim 1, further comprising receiving a question about a capability of the system itself; resolving the question to one or more numbered sections of a specification of the system;

retrieving, for said capability, evidence pairs each comprising a source file of the system and a symbol required to appear therein; checking each evidence pair against the source files of the running system at the time of the question; and returning an answer comprising a status label selected from built, partly built, specified and not yet built, not planned, and not present, together with the section numbers and the evidence pairs as sources.

359. The method of claim 358, wherein a capability is labelled built only when every evidence pair for it is present, and wherein a capability declared built or partly built for which an evidence pair is absent is reported as a contradiction and its description is withheld from the answer.

360. The method of claim 358, further comprising searching the source files for implementation names of each capability declared specified and not yet built, and reporting a contradiction when such a name is found, whereby the self-description is corrected when the implementation advances beyond it.

361. The method of claim 358, wherein the answer is composed by fixed rules without a language model such that the same question about the same program yields the same answer; wherein the source files are read as text without being executed; and wherein any portion of the answer resembling a key or credential is removed before the answer is returned.

362. The method of claim 358, further comprising describing reasoning aspects of the system, including deterministic reasoning, atomic knowledge records, a world model, predictive cognition, a neuromorphic layer, hypothesis generation, simulation, deduction, feedback loops and self-improvement, each with its status label and sources; listing the sections specified but not built as a development agenda; and reporting any self-modification of the system without an administrator's approval as not provided.

[Drafting note: claims 358–362 are added for the self-knowledge layer (FIG. 57); the claim count in the Claim Set heading is to be read as 362.]

ABSTRACT

QueryBook represents a new model of AI - a synthetic cortex and deterministic world-model AI engine built on a new primitive—the Fact Unit, an atomic sub-language semantic unit. QueryBook's engine provides neuromorphic reasoning, multimodal cognition, and an open, flexible architecture. It represents a new paradigm in AI—one that is structured, explainable, multimodal, and universally deployable across virtually every current AI application. QueryBook's ability can be micro-encapsulated for Q&A of single or multiple books and documents in smart devices, used for data analysis, and provide SaaS, ChatBot, Agent, deterministic hybrid additions to current LLMs, specialized LLM-like applications including private AI computational and analysis systems (within option for live data inputs) for those with specialized knowledge, plus unique Fact Unit compressed data search engine capability that can operate alone for specialized fields or as adjunct to current search engines to off load centralized data computation, storage and transmission costs -- bringing to each application a distinct and measurable set of advantages.

These deployment models have modular capability editions (light, medium, and full) selectable at launch, and with a version-independent, provenance-continuous data store that keeps collected Fact Units in one fixed location so that software upgrades never lose or orphan previously harvested facts. QueryBook transforms digital content into a coherent, navigable, intelligent world-model, enabling users to understand, explore, and apply knowledge with unprecedented clarity and precision with option for speech recognition and natural language response that includes emotional inflection as if hearing a human voice-- deterministic, non-hallucinogenic, with citations. Across all applications, internal controls ensure humanistic action and prohibition of rogue actions, and its innovative multi- layer security infrastructure provides a model for AI and computer systems offering an anti-intrusion shield against viral attacks

including Agent Worms. Further additions strengthen this security and integrity posture: an independent Universal Integrity Audit Service (UIAS) produces pre-dispute, tamper-evident proof over the content-addressed Fact Unit ledger for a third party's system, returning only a factual determination (verified, contradicted, or unknown) and never a legal conclusion; and a key-safe external-service mediation layer holds any external-service credential on the server side of the query/presentation boundary so it is never exposed to a client. This protection is extended to AI agents: every agent call passes a single gateway with traps and an AI-aware firewall, a hostile caller is isolated in a decoy with no path to real data, security decisions are sealed in a tamper-evident log, and Fact Units may optionally be certified on a public blockchain for independent verification.

QueryBook Fact Unit graph structure simultaneously permits a (a) new level of enhanced multi-language provision and translation that includes deterministic, dictionary-grounded multilingual acquisition and offline speech with no large language model in the grounding path, regional Dialect Parameter Clusters, a Lecture Query mode that ingests a live or recorded talk as attributed utterances (never promoted to world-truth), (b) Scene Director and Scene Reconstructor that assemble grounded video prompts and reconstruct scenes from prose with all generated media clearly marked as reconstruction, (c) simplified colorization capability, (d) video and audio compression benefits for security applications, video expansion (not yet tested) and, (e) a new Fact Unit based, enhanced TCP/IP hybrid protocol carrying Fact Unit identifiers and provenance references alongside network payload data. Files of any format can be ingested as data, never as instructions, and external AI agents such as OpenClaw connect through a standard tool interface limited to their allow-list. QueryBooks Fact Unit graph architecture also reduces data storage by orders of magnitude, works on commodity server or product hardware, and in larger complex application significantly reduce power consumption, cooling costs and deployment real estate.